Cusp Spectral Degeneration and Local Toric Kummer Extension

Conditional results from the reconstructed candidate S^6 complex-torus fibration

A specialist-verification draft

based on the candidate S^6 construction of Levent Alpöge

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Abstract

Use the period, integral-monodromy, and toric-cusp data from a recent manuscript proposing a complex structure on S^6. Those inputs have been reconstructed and separately verified in the accompanying workbench, but they are imported, rather than reproved, in this standalone note. The regular fibres are marked complex two-tori

$$F_z = \mathbb{C}^2/\mathsf{P}_z\mathbb{Z}^4$$

with an explicit real period matrix P_z. We equip the covering space $\mathbb{R}^4 \cong \mathbb{C}^2$ with its coordinate Euclidean metric and study the resulting scalar fibre Laplacians. Every fixed fibre has a positive first eigenvalue. Nevertheless, on the actual manuscript cusp, where $T = \operatorname{Im}\tau \to \infty$, μ and $\beta + \tau$ remain bounded, one has the sharp asymptotics

$$\lambda_1(F_z) = \frac{4\pi^2}{T^2}\left(1 + \frac{2\min\{b_{\mathrm{I},0}, 0\}}{T} + O(T^{-2})\right), \qquad \operatorname{Vol}(F_z) = T^2 - b_{\mathrm{I},0}T + O(1),$$

where $b_{\mathrm{I},0} = \operatorname{Im}(\beta + \tau)(0)$. The two possible shortest signed Fourier pairs, their exact comparison, and their first corrections are determined below. After rescaling each four-torus to unit volume, the first eigenvalue still tends to zero, now as $4\pi^2/T$. The decomposable vertical operator obtained by integrating the mean-zero fibre Laplacians along a cusp ray is injective but has zero at the bottom of its essential spectrum.

The degeneration is not represented by only one soft Fourier mode. If $\mathcal{D}_T := |\det \mathsf{P}_z| = -TD$ and $N_T(E)$ counts scalar eigenvalues with Fourier multiplicity, then throughout the first transverse band $0 < E < 4\pi^2$ one has the fixed-energy law

$$N_T(E) = \frac{E\mathcal{D}_T}{4\pi} + O_E(T).$$

Thus a two-dimensional density of low modes grows on the order of the fibre covolume. We also give the unit-volume counting law and the exact finite count at the collapsing T^{-2} scale.

The native marked four-tori do not converge to $\mathbb{R}^4$: their pointed smooth limit is $\mathbb{T}^2 \times \mathbb{R}^2$. Replacing the two fixed unit periods by M times those periods gives degree-M^2 local fibre covers. They converge pointed smoothly to $\mathbb{R}^4$, along sequences with $T \to \infty$, exactly when $M \to \infty$ for the unscaled metric. For the pullback of the base-fibre unit-volume metric the condition is $M/T^{1/2} \to \infty$; if every cover is instead normalized to have its own volume one, no choice of M has pointed smooth limit $\mathbb{R}^4$. After the cyclic base change $t_c = v^M$ the cover monodromy is integral, and the complete cover spectrum separates its deck-invariant pullback sector from its new momentum sectors. These finite-cover limits are different from both the native $\mathbb{T}^2 \times \mathbb{R}^2$ limit and the quotient constructions below.

The full monodromy has, up to sign, a unique primitive invariant integral character. It gives a real, non-holomorphic circle quotient of the regular group family. In the convention

above, with $m = \operatorname{Im}\mu$, $D = \operatorname{Im}\beta - 6m^2/T < 0$, and $b_{\mathrm{I},0} = \operatorname{Im}(\beta+\tau)(0)$, its exact quotient circumference and radial asymptotic are

$$\ell_\gamma = \frac{-TD}{\sqrt{T^2+m^2}} = T - b_{\mathrm{I},0} + O(T^{-1}).$$

Four synchronized copies, with no base change or fibre cover, give a real four-torus quotient over the full regular base whose radial fibres converge pointed locally isometrically to $\mathbb{R}^4$. Four copies are minimal within the class of fixed primitive full-monodromy-invariant kernels equipped with the induced product quotient metric. The literal map to this fixed product $\mathbb{T}^4$ target does not extend continuously through the asserted toric cusp and does not factor through the fourfold smash collapse; these statements concern that target and that regular-fibre restriction only.

The included local toric theorem gives the exact Kummer extension after the base change $t = v^N$. If $Z(s) = sB_0 + C(t)$, the source action is not the naive target action with v substituted for t: it is

$$\Psi^{(N)}_{\bar\lambda}(\xi, v) = \left(\exp\left(\frac{2\pi i}{N} C(v^N)\bar\lambda\right) v^{B_0\bar\lambda}\xi, v\right).$$

With this action, $(\xi_1, \xi_2, v) \mapsto (\xi_1^N, \xi_2^N, v^N)$ descends to a finite flat map of degree N^3 to the original completed cusp family. Its factor to the base-changed target has generic degree N^2; for $N > 1$ that relative map is quasi-etale but not flat along the preimages of the double and triple central-fibre strata. This is a completed local cusp morphism, not a cover over the full base.

For the classical Yang–Mills functional at the product connection, Coulomb reduction gives an exact Hessian gap

$$\gamma_T = \frac{4\pi^2}{g_{\mathrm{YM}}^2 T^2}\left(1 + \frac{2\min\{b_{\mathrm{I},0}, 0\}}{T} + O(T^{-2})\right).$$

The corresponding finite cubical cochain model has exact gap $\rho_N/(4\pi^2)$ times this value, where $\rho_N = 4N^2\sin^2(\pi/N)$; it closes as T^{-2} uniformly for every $N \geq 2$, although resolving the varying continuum metric requires $N/T \to \infty$. These are linear classical fluctuation operators, not a nonlinear lattice gauge theory or a quantum Hamiltonian.

An independent compact-link and Gaussian construction keeps the cover index M_{cov} separate from the cubical subdivision count N_{mesh}. It proves the exact holonomy and plaquette formulas for an explicit periodic SU(2) flux and the complete finite cochain spectrum. The associated finite-dimensional Gaussian oscillator has a unique ground state and gap $\sqrt{\lambda_{\min}}$. Along the explicit sequence

$$R_j = j^2, \qquad M_{\mathrm{cov},j} = j, \qquad N_{\mathrm{mesh},j} = j^3,$$

the covering tori converge pointed locally isometrically to $\mathbb{R}^4$, the mesh resolves the varying metric, and both the Gaussian gap and the displayed logarithmic flux action tend to zero. This Gaussian Hilbert space is an L^2-space on a linear Coulomb slice; it is not the compact-link Haar Hilbert space and does not define an interacting Hamiltonian.

For every fixed nondegenerate integral alternating form E on the marked four-torus, the associated constant-curvature magnetic Laplacian has oscillator frequencies $b_1, b_2 > 0$ satisfying

$$b_1 b_2 = \frac{4\pi^2|\operatorname{Pf}(\mathsf{E})|}{\mathcal{D}_T}, \qquad \text{gap} \leq 4\pi\sqrt{\frac{|\operatorname{Pf}(\mathsf{E})|}{\mathcal{D}_T}} = O(T^{-1}).$$

This is a fibrewise statement in the marked cusp family and does not by itself globalize the flux through monodromy. The actual lattice nevertheless has the explicit invariant full-flux class

$$\eta = u \wedge w + 6\gamma \wedge \delta, \qquad \operatorname{Pf}(\eta) = 6.$$

Writing R for logarithmic cusp height, its two charge-one frequencies obey $\mathcal{B}_{R,+}\mathcal{B}_{R,-} = 24\pi^2/D_R$ and have leading terms $12\pi/R$ and $2\pi/R$. Every joint Landau level has complex

multiplicity six, and the first distinct spacing is $4\pi/R + O(R^{-2})$. The corresponding maximal multiplication operators on both the fixed and varying measurable radial Hilbert fields are self-adjoint and have purely continuous spectrum $[0, \infty)$, empty point spectrum, and zero kernel. The radial variable in these operators is only a direct-integral parameter: there is no radial derivative or boundary condition.

The split connection on $\mathscr{L}_{\eta,R} \oplus \mathscr{L}_{\eta,R}^{-1}$ is a non-flat reducible SU(2) Yang–Mills critical point. Its closed Coulomb Jacobi operator has exactly three negative bands on a sufficiently deep cusp tail, each of real multiplicity 48, and hence real Morse index 144. For every nonzero trace-zero integral weight vector $\mathbf{m} \in \mathbb{Z}^r$, the split-Cartan SU($r$) construction has exact rootwise negative bands and deep-tail index $36r \sum_j m_j^2$. These are spectra of classical fibre Hessians at explicit reducible backgrounds, not the spectrum of interacting four-dimensional quantum Yang–Mills theory.

By contrast, the specific degree-six magnetic construction pulled back from the holomorphic first-coordinate elliptic quotient requires $6\mu = a\tau + b$ for integers a, b. We prove, from the stated transformation and cusp laws, that this exceptional locus has no cusp-accumulating sequence and is finite on the completed orbifold base. Consequently that first-coordinate Landau spectrum cannot supply the cusp limit. Finally, all fibre and finite Gaussian degenerations are distinguished from the scalar Laplacian of any smooth metric on the compact total space, which necessarily has compact resolvent and a positive first eigenvalue.

Status and logical scope. All statements involving the global S^6 fibration use the unpublished construction in Alpöge (2026). Its topology, singular-fibre maps, and direct-image argument have been reconstructed and separately verified in the accompanying proof-verification workbench; this standalone note imports that geometric package as stated input instead of reproducing those proofs. A contemporaneous downstream paper treats that construction as an assumption rather than supplying an independent proof (Liang et al., 2026). The linear algebra, Fourier spectra, cusp estimates, and direct-integral argument below are independent deductions from the displayed period and cusp formulae. The regular-locus quotient theorem additionally uses the asserted integral monodromy matrices, which are restated and checked below; its nonextension statement additionally uses the asserted toric cusp chart, and is only a nonextension result for the literal fixed product-circle target with the stated regular-fibre map. The quotient map is real. Its radial circumference and pointed-limit statements use the separately chosen Euclidean metrics on marked cusp fibres; no deck-invariant global fibre metric is asserted. It is neither the holomorphic first-coordinate quotient nor the separate degree-M^2 cover construction. The counting theorem and the arbitrary-flux magnetic theorem are likewise statements along the marked cusp. A general alternating form E is fixed only in the marked lattice; a global magnetic line bundle over the regular family would require additional descent data. Separately, the displayed class $\eta = u \wedge w + 6\gamma \wedge \delta$ is proved invariant under the asserted monodromy generators and is realized by an explicit factor of automorphy on every marked fibre. The radial direct-integral theorem uses Lebesgue measure in logarithmic height and a multiplication operator with the exact maximal domain; it includes no radial derivative. The local toric Kummer theorem included below is proved for a completed cusp disc. It uses the modified source factor $\exp(2\pi i C(v^N)\bar{\lambda}/N)$ and distinguishes the degree-N^3 map to the original cusp family from the generic degree-N^2 relative map to the base-changed target. It supplies no full-base cover. The separate cone targets, simultaneous-root construction, and Seifert filling proofs in the canonical workbench are not imported or claimed here. This paper does prove statements about scalar and magnetic fibre Laplacians, their specified decomposable radial multiplication operators, classical Yang–Mills Hessians at the product and explicit reducible split backgrounds, an honest compact-link flux configuration, and explicitly defined finite cochain and Gaussian models. It constructs neither nonlinear quantum Yang–Mills theory nor a four-dimensional physical Hamiltonian, and makes no physical mass-gap claim.

1 Conditional period data

Let $\mathfrak{h}_z$ be the marked parameter upper half-plane used in the proposed construction. The asserted holomorphic period functions are

$$\tau : \mathfrak{h}_z \longrightarrow \mathfrak{h}, \qquad \mu, \beta : \mathfrak{h}_z \longrightarrow \mathbb{C},$$

and the complex period matrix and its real form are

$$Z(z) = \begin{pmatrix} 6\mu(z) & \tau(z) \\ \beta(z) & \mu(z) \end{pmatrix}, \qquad \mathsf{P}_z = \begin{pmatrix} 6\Re\mu & \Re\tau & 1 & 0 \\ 6\Im\mu & \Im\tau & 0 & 0 \\ \Re\beta & \Re\mu & 0 & 1 \\ \Im\beta & \Im\mu & 0 & 0 \end{pmatrix}. \tag{1.1}$$

Write

$$T = \Im\tau > 0, \qquad m = \Im\mu, \qquad p = \Im\beta, \qquad D = p - \frac{6m^2}{T}. \tag{1.2}$$

The manuscript asserts that an additive normalization of β can be chosen so that $D < 0$ everywhere and proves the determinant identity

$$\det \mathsf{P}_z = TD. \tag{1.3}$$

Thus $\mathsf{P}_z\mathbb{Z}^4$ is a full lattice in $\mathbb{R}^4$ at each regular parameter.

At the distinguished cusp, let t_c be a local coordinate with $t_c = 0$ at the added point. The manuscript gives

$$q = e^{2\pi i\tau} = t_c u_1(t_c), \qquad u_1(0) \neq 0, \tag{1.4}$$

while μ and $b := \beta + \tau$ extend holomorphically across $t_c = 0$. On a sufficiently small cusp disc choose a holomorphic logarithm of u_1 and put

$$h = \frac{1}{2\pi i} \log u_1, \qquad s = \tau - h, \qquad e^{2\pi i s} = t_c. \tag{1.5}$$

The positive radial lift is $s = iR$, where $R \to \infty$. We reserve $T = \operatorname{Im}\tau = R + \operatorname{Im} h$ for the convention in (1.2); the radial height R is therefore not denoted by T in this paper. Along that lift put

$$B_R := \operatorname{Im} b(e^{-2\pi R}), \qquad b_{\mathrm{I},0} := \operatorname{Im} b(0), \qquad m_0 := \operatorname{Im} \mu(0). \tag{1.6}$$

Holomorphy at the cusp gives

$$B_R = b_{\mathrm{I},0} + O(e^{-2\pi R}), \qquad m = m_0 + O(e^{-2\pi R}), \tag{1.7}$$

and, because $b = \beta + \tau$,

$$p = B_R - T, \qquad \mathcal{D}_T := |\det \mathsf{P}_z| = -TD = T(T - B_R) + 6m^2 > 0. \tag{1.8}$$

In particular, along every path in a fixed cusp sector on which $T \to \infty$, one has

$$m = O(1), \qquad p = -T + O(1), \qquad D = -T + O(1). \tag{1.9}$$

The coarse asymptotics suffice for the leading spectral term; the exact radial identities above determine its first correction.

The generator laws needed later are

$$\tau \circ g_0 = \tau - 1, \qquad \mu \circ g_0 = \mu, \qquad \tau \circ g_2 = -\frac{1}{\tau}, \qquad \mu \circ g_2 = 1 + \frac{\mu}{\tau}. \tag{1.10}$$

For the regular-locus quotient, we also use the asserted integral lattice

$$\Lambda = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle \tag{1.11}$$

in the column order of (1.1). Let $\Delta = \langle g_1, g_2 \mid g_1^3 = g_2^4 = 1 \rangle$ denote the manuscript's triangle-group deck action. It must not be identified with the ordinary fundamental group of $B^\circ = B \setminus \{p_0, p_1, p_2\}$. Put $g_0 = (g_1 g_2)^{-1}$. Its asserted action on Λ is generated by

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}, \tag{1.12}$$

and the cusp monodromy is

$$M_0 = (A_1 A_2)^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}. \tag{1.13}$$

Let $\mathfrak{h}_z^\circ$ be the inverse image of B° in the marked parameter cover. We use here the manuscript's assertion that the elliptic fixed-point orbits have been removed, that the deck action on $\mathfrak{h}_z^\circ$ is free, and that its lift to the complex torus family is holomorphic and complex-linear. The fibre over the marked cover is $\mathbb{C}^2/\mathsf{P}_z\Lambda$, and in flat angular coordinates the lifted action is

$$(z, x + \Lambda) \longmapsto (gz, A_g x + \Lambda). \tag{1.14}$$

It is linear and fixes the zero section. Consequently its quotient $J = X^\circ = f^{-1}(B^\circ)$ is, conditionally on this input, a holomorphic family of complex Lie groups rather than merely an affine torus bundle.

2 The fixed-fibre flat spectrum

Equip $\mathbb{R}^4$ with the Euclidean metric in the coordinates used in (1.1), and put

$$F_z^{\mathrm{flat}} = \mathbb{R}^4/\mathsf{P}_z\mathbb{Z}^4. \tag{2.1}$$

This metric is a chosen auxiliary metric determined by the displayed coordinate realization. It is not asserted to be canonical for the complex torus or to extend across a singular fibre of the completed family.

Proposition 2.1 (Fourier spectrum on a marked flat fibre)**.** *If* $\det \mathsf{P}_z \neq 0$*, let* Δ_z *be the nonnegative self-adjoint operator associated with the closed Dirichlet form*

$$q_z[f] = \int_{F_z^{\mathrm{flat}}} |\mathrm{d}f|^2 \, \mathrm{d}\,\mathrm{vol}_{F_z}, \qquad \mathrm{Dom}(q_z) = H^1(F_z^{\mathrm{flat}}).$$

Then $\mathrm{Dom}(\Delta_z) = H^2(F_z^{\mathrm{flat}})$ *and* Δ_z *has spectrum*

$$\mathrm{Spec}(\Delta_z) = \left\{ 4\pi^2 \left\| \mathsf{P}_z^{-\mathsf{T}} k \right\|^2 : k \in \mathbb{Z}^4 \right\}, \tag{2.2}$$

with multiplicities. Its kernel is the constants and

$$\lambda_1(z) = 4\pi^2 \min_{k \in \mathbb{Z}^4 \setminus \{0\}} \left\| \mathsf{P}_z^{-\mathsf{T}} k \right\|^2 > 0. \tag{2.3}$$

For an arbitrary $k = (r, s, n_1, n_2)^{\mathsf{T}}$, set

$$\alpha = \Re\mu, \qquad \eta = \Re\tau, \qquad c = \Re\beta,$$

$$R_z(k) = r - 6\alpha n_1 - cn_2, \qquad S_z(k) = s - \eta n_1 - \alpha n_2. \tag{2.4}$$

The complete dual-vector formula is

$$\mathsf{P}_z^{-\mathsf{T}} k = \left(n_1, \frac{mR_z(k) - pS_z(k)}{-TD}, n_2, \frac{-TR_z(k) + 6mS_z(k)}{-TD} \right). \tag{2.5}$$

Proof. The characters

$$e_k([x]) = \exp\bigl(2\pi i \langle \mathsf{P}_z^{-\mathsf{T}} k, x \rangle\bigr), \qquad k \in \mathbb{Z}^4,$$

form the Fourier basis for the compact abelian group F_z^{flat}. Applying $-\sum_j \partial_{x_j}^2$ gives the eigenvalue in (2.2). The standard closed-form construction and elliptic regularity give the stated form and operator domains. The dual lattice is discrete, so its shortest nonzero vector has positive length. Standard flat-torus spectral theory gives completeness (Chavel, 1984). Finally, solving $\mathsf{P}_z^{\mathsf{T}} x = k$ gives $x_1 = n_1$ and $x_3 = n_2$; the remaining system is

$$\begin{pmatrix} 6m & p \\ T & m \end{pmatrix} \begin{pmatrix} x_2 \\ x_4 \end{pmatrix} = \begin{pmatrix} R_z(k) \\ S_z(k) \end{pmatrix}.$$

Its determinant is $-TD$, and direct inversion proves (2.5). □

3 Sharp degeneration on the actual cusp

The following result uses the actual cusp behavior (1.9); it is not based on an arbitrary family in the ambient period-matrix space.

Theorem 3.1 (Shortest cusp modes and their first corrections)**.** *Assume the conditional period data in Section 1 and restrict to the positive radial lift $s = iR$. Put*

$$e_\gamma = (1,0,0,0)^{\mathsf{T}}, \qquad e_u = (0,1,0,0)^{\mathsf{T}}, \qquad \ell_{\gamma,T} = \mathsf{P}_z^{-\mathsf{T}} e_\gamma, \qquad \ell_{u,T} = \mathsf{P}_z^{-\mathsf{T}} e_u.$$

For all sufficiently large $T = \operatorname{Im}\tau$, every shortest nonzero vector of the dual lattice belongs to

$$\{\pm\ell_{\gamma,T},\ \pm\ell_{u,T}\}.$$

Their exact squared lengths and exact comparison are

$$\|\ell_{\gamma,T}\|^2 = \frac{T^2 + m^2}{\mathcal{D}_T^2}, \qquad \|\ell_{u,T}\|^2 = \frac{(T - B_R)^2 + 36m^2}{\mathcal{D}_T^2}, \tag{3.1}$$

$$\mathcal{D}_T^2\bigl(\|\ell_{\gamma,T}\|^2 - \|\ell_{u,T}\|^2\bigr) = B_R(2T - B_R) - 35m^2. \tag{3.2}$$

Thus the shortest indices are $\pm e_u$ when the right-hand side of (3.2) is positive, $\pm e_\gamma$ when it is negative, and both signed pairs when it vanishes.

The two candidate eigenvalues have the expansions

$$4\pi^2\|\ell_{\gamma,T}\|^2 = \frac{4\pi^2}{T^2}\left(1 + \frac{2b_{\mathrm{I},0}}{T} + O(T^{-2})\right), \tag{3.3}$$

$$4\pi^2\|\ell_{u,T}\|^2 = \frac{4\pi^2}{T^2}\left(1 + O(T^{-2})\right). \tag{3.4}$$

More generally, for each fixed $(r,s)\in\mathbb{Z}^2$,

$$\frac{T^2}{4\pi^2}\lambda_T(r,s,0,0)=r^2+s^2+\frac{2b_{\mathrm{I},0}r^2-10m_0rs}{T}+O_{r,s}(T^{-2}). \tag{3.5}$$

Consequently

$$\boxed{\lambda_1(z)=\frac{4\pi^2}{T^2}\left(1+\frac{2\min\{b_{\mathrm{I},0},0\}}{T}+O(T^{-2})\right).} \tag{3.6}$$

If $b_{\mathrm{I},0}>0$*, the bottom indices are eventually* $\pm e_u$*; if* $b_{\mathrm{I},0}<0$*, they are eventually* $\pm e_\gamma$*. If* $b_{\mathrm{I},0}=0$*, the exact comparison* (3.2) *decides; in particular, if* $m_0\neq 0$ *then the bottom indices are eventually* $\pm e_\gamma$*. The first positive eigenvalue has complex multiplicity two when exactly one signed pair minimizes and multiplicity four when both pairs minimize. All statements are independent of the real parts of* τ,μ,β*.*

Proof. Write $k=(r,s,n_1,n_2)\in\mathbb{Z}^4$ and solve $\mathsf{P}_z^{\mathsf{T}}x=k$. The third and fourth equations are

$$x_1=n_1,\qquad x_3=n_2. \tag{3.7}$$

Hence every mode with $(n_1,n_2)\neq(0,0)$ has dual squared norm at least one. For $k=(r,s,0,0)$, the remaining system is

$$\begin{pmatrix}6m & p\\ T & m\end{pmatrix}\begin{pmatrix}x_2\\ x_4\end{pmatrix}=\begin{pmatrix}r\\ s\end{pmatrix}.$$

Its determinant is $-TD=\mathcal{D}_T$, and $p=B_R-T$, so

$$\mathsf{P}_z^{-\mathsf{T}}(r,s,0,0)=\left(0,\frac{mr+(T-B_R)s}{\mathcal{D}_T},0,\frac{-Tr+6ms}{\mathcal{D}_T}\right). \tag{3.8}$$

Define

$$M_T:=\frac{T}{\mathcal{D}_T}\begin{pmatrix}m & T-B_R\\ -T & 6m\end{pmatrix}. \tag{3.9}$$

Equations (1.7) and (1.8) give

$$M_T=J+\frac{1}{T}\begin{pmatrix}m_0 & 0\\ -b_{\mathrm{I},0} & 6m_0\end{pmatrix}+O_{\mathrm{op}}(T^{-2}),\qquad J=\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}. \tag{3.10}$$

In particular, because J is orthogonal, a constant C exists such that for every $v\in\mathbb{R}^2$ and every sufficiently large T,

$$\left(1-\frac{C}{T}\right)\|v\|\le\|M_Tv\|\le\left(1+\frac{C}{T}\right)\|v\|. \tag{3.11}$$

Thus every plane vector with $r^2+s^2\ge 2$ is eventually longer than all four plane vectors with $r^2+s^2=1$, while the vectors excluded by (3.7) have norm at least one and the four candidates have norm $T^{-1}(1+O(T^{-1}))$. This proves the asserted exhaustive minimizing list.

Putting $(r,s)=(1,0)$ and $(0,1)$ in (3.8) proves (3.1). Subtracting their numerators gives

$$T^2+m^2-(T-B_R)^2-36m^2=B_R(2T-B_R)-35m^2,$$

which proves (3.2). Finally, squaring (3.10) on a fixed vector (r,s) gives

$$\|M_T(r,s)^{\mathsf{T}}\|^2=r^2+s^2+\frac{2b_{\mathrm{I},0}r^2-10m_0rs}{T}+O_{r,s}(T^{-2}).$$

Since $T^2\|\mathsf{P}_z^{-\mathsf{T}}(r,s,0,0)\|^2=\|M_T(r,s)^{\mathsf{T}}\|^2$, this proves (3.5) and the two special expansions. The smaller of the relative corrections $2b_{\mathrm{I},0}$ and 0 is $2\min\{b_{\mathrm{I},0},0\}$, proving (3.6); the exact comparison and Fourier signed-pair counting prove the remaining assertions. □

Corollary 3.2 (Volume and unit-volume degeneration)**.** *Let V_z be the Euclidean volume of F_z^{flat}. Then*

$$V_z = |\det \mathsf{P}_z| = \mathcal{D}_T = T(T - B_R) + 6m^2 = T^2 - b_{\mathrm{I},0}T + 6m_0^2 + O(Te^{-2\pi R}). \tag{3.12}$$

If g_z is the flat metric and

$$\widehat{g}_z = V_z^{-1/2} g_z \tag{3.13}$$

so that $\mathrm{Vol}(F_z, \widehat{g}_z) = 1$, then

$$\lambda_1(F_z, \widehat{g}_z) = \frac{4\pi^2}{T}\left(1 + \frac{-b_{\mathrm{I},0}/2 + 2\min\{b_{\mathrm{I},0}, 0\}}{T} + O(T^{-2})\right) \longrightarrow 0. \tag{3.14}$$

In the normalized metric the two fixed periods have length $\mathcal{D}_T^{-1/4} = T^{-1/2}(1+O(T^{-1}))$, whereas the two long periods have length $T^{1/2}(1 + O(T^{-1}))$.

Proof. Equation (1.8) gives the exact volume and (1.7) gives its expansion. In real dimension four, replacing g by αg multiplies volume by α^2 and Laplace eigenvalues by α^{-1}. Therefore (3.13) multiplies eigenvalues by $V_z^{1/2}$. Combine this with Theorem 3.1. The period-length assertions follow directly from the columns of P_z: the last two have length one, while the first two have length $T(1 + O(T^{-1}))$; all lengths are multiplied by $\mathcal{D}_T^{-1/4}$. □

Proposition 3.3 (Fixed-energy counting law)**.** *Along the radial cusp of Theorem 3.1, retain the positive covolume*

$$\mathcal{D}_T = |\det \mathsf{P}_z| = -TD = \mathrm{Vol}(F_z^{\mathrm{flat}}) \tag{3.15}$$

from (1.8). For $E \geq 0$, count eigenvalues with their Fourier multiplicities by

$$\lambda_T(k) := 4\pi^2 \|\mathsf{P}_z^{-\mathsf{T}} k\|^2, \qquad N_T(E) := \#\{k \in \mathbb{Z}^4 : \lambda_T(k) \leq E\}. \tag{3.16}$$

Then, for every fixed E,

$$N_T(E) = \pi \mathcal{D}_T \sum_{n \in \mathbb{Z}^2} \left(\frac{E}{4\pi^2} - \|n\|^2\right)_+ + O_E(T), \qquad (x)_+ := \max\{x, 0\}. \tag{3.17}$$

The sum is finite. In particular, throughout the first transverse band $0 < E < 4\pi^2$,

$$\boxed{N_T(E) = \frac{E}{4\pi}\mathcal{D}_T + O_E(T).} \tag{3.18}$$

Since $\mathcal{D}_T = T^2(1 + O(T^{-1}))$, the number of modes below any fixed energy in this band grows as $ET^2/(4\pi) + O_E(T)$.

For the unit-volume metric $\widehat{g}_z$ of (3.13), let

$$\widehat{N}_T(E) := \#\{k \in \mathbb{Z}^4 : \lambda_T(k; \widehat{g}_z) \leq E\}. \tag{3.19}$$

For every fixed $E > 0$,

$$\widehat{N}_T(E) = \frac{E}{4\pi}\mathcal{D}_T^{1/2} + O_E(\mathcal{D}_T^{1/4} + 1), \qquad \lim_{T\to\infty} \frac{\widehat{N}_T(E)}{\mathcal{D}_T^{1/2}} = \frac{E}{4\pi}. \tag{3.20}$$

At the collapsing spectral scale define

$$N_T^{\mathrm{coll}}(E) := \#\left\{k \in \mathbb{Z}^4 : \lambda_T(k) \leq \frac{4\pi^2 E}{T^2}\right\}. \tag{3.21}$$

If E is not a sum of two integer squares, then for all sufficiently large T,

$$N_T^{\mathrm{coll}}(E) = \#\{(r,s) \in \mathbb{Z}^2 : r^2 + s^2 \le E\}. \tag{3.22}$$

If E is a sum of two integer squares, then

$$\begin{aligned} \#\{(r,s) : r^2 + s^2 < E\} &\le \liminf_{T\to\infty} N_T^{\mathrm{coll}}(E) \\ &\le \limsup_{T\to\infty} N_T^{\mathrm{coll}}(E) \le \#\{(r,s) : r^2 + s^2 \le E\}. \end{aligned} \tag{3.23}$$

For a threshold vector (r,s), the coefficient $2b_{\mathrm{I},0}r^2 - 10m_0rs$ in (3.5) decides membership whenever it is nonzero; the exact formula (3.25) and higher Taylor coefficients decide the remaining cases.

Proof. Write

$$\alpha = \Re\mu, \qquad \eta = \Re\tau, \qquad c = \Re\beta,$$

and index $k \in \mathbb{Z}^4$ as $k = (r, s, n_1, n_2)$. Solving $\mathsf{P}_z^{\mathsf{T}} x = k$ first gives $x_1 = n_1$ and $x_3 = n_2$. The remaining two equations are

$$\mathcal{A}_T \begin{pmatrix} x_2 \\ x_4 \end{pmatrix} = \begin{pmatrix} r \\ s \end{pmatrix} - \theta_T(n), \quad \mathcal{A}_T := \begin{pmatrix} 6m & p \\ T & m \end{pmatrix}, \quad \theta_T(n) := \begin{pmatrix} 6\alpha n_1 + cn_2 \\ \eta n_1 + \alpha n_2 \end{pmatrix}. \tag{3.24}$$

Here

$$\det \mathcal{A}_T = 6m^2 - pT = -TD = \mathcal{D}_T > 0.$$

Consequently the exact eigenvalue formula is

$$\frac{\lambda_T(r,s,n_1,n_2)}{4\pi^2} = \|n\|^2 + \left\| \mathcal{A}_T^{-1} \left(\begin{pmatrix} r \\ s \end{pmatrix} - \theta_T(n) \right) \right\|^2, \qquad n = (n_1, n_2). \tag{3.25}$$

Put $R_E^2 = E/(4\pi^2)$. A fixed n contributes nothing when $\|n\|^2 > R_E^2$. Otherwise, with $\rho_n = (R_E^2 - \|n\|^2)^{1/2}$, its admissible pairs (r,s) are precisely the integer points of the translated ellipse

$$\theta_T(n) + \mathcal{A}_T \overline{B}_{\mathbb{R}^2}(0, \rho_n). \tag{3.26}$$

When $\rho_n > 0$, this ellipse has area $\pi\mathcal{D}_T\rho_n^2$ and perimeter $O_E(T)$, because $m = O(1)$, $p = -T + O(1)$, and hence $\|\mathcal{A}_T\|_{\mathrm{op}} = O(T)$. When $\rho_n = 0$, its contribution is zero or one and is absorbed by $O_E(1)$.

For completeness, every translated compact convex set $K \subset \mathbb{R}^2$ satisfies

$$\#(K \cap \mathbb{Z}^2) = \mathrm{area}(K) + O(\mathrm{length}(\partial K) + 1), \tag{3.27}$$

with an absolute implied constant. Indeed, center a unit square at every point of $K \cap \mathbb{Z}^2$ and let U be their union. The squares have disjoint interiors and $\mathrm{area}(U) = \#(K \cap \mathbb{Z}^2)$. Every point of $U \triangle K$ lies within distance $\sqrt{2}$ of ∂K: this is immediate for $U \setminus K$, while a point of $K \setminus U$ farther than $\sqrt{2}$ from the boundary would have its nearest lattice point in K and hence lie in the corresponding square. The inner and outer boundary strips of a planar convex set of fixed width have area bounded by an absolute constant times its perimeter plus one (first for polygons, by summing edge rectangles and vertex sectors, and then by polygonal approximation). This proves (3.27).

Only finitely many n occur for fixed E. Applying (3.27) to (3.26), and summing over those n, proves (3.17). If $0 < E < 4\pi^2$, only $n = 0$ occurs, and its area is $\pi\mathcal{D}_T E/(4\pi^2) = E\mathcal{D}_T/(4\pi)$, proving (3.18).

Under $\widehat{g}_z = \mathcal{D}_T^{-1/2} g_z$, every eigenvalue is multiplied by $\mathcal{D}_T^{1/2}$, so $\widehat{N}_T(E) = N_T(E/\mathcal{D}_T^{1/2})$. For large T only $n = 0$ can occur in (3.25). Its ellipse has radius parameter

$$\rho = \left(\frac{E}{4\pi^2 \mathcal{D}_T^{1/2}} \right)^{1/2},$$

area $E\mathcal{D}_T^{1/2}/(4\pi)$, and perimeter $O_E(T\mathcal{D}_T^{-1/4} + 1) = O_E(\mathcal{D}_T^{1/4} + 1)$. The same lattice estimate proves (3.20).

At the collapsing scale, every index with $(n_1, n_2) \neq (0, 0)$ contributes at least $4\pi^2$ and hence contributes nothing for large T. For $n_1 = n_2 = 0$, equations (3.9)–(3.10) give uniformly

$$\frac{T^2 \lambda_T(r, s, 0, 0)}{4\pi^2} = \|M_T(r, s)^{\mathsf{T}}\|^2 \longrightarrow r^2 + s^2.$$

The lower singular-value bound (3.11) restricts a fixed rescaled window to a fixed finite set of pairs (r, s). Termwise comparison proves (3.22) away from thresholds and (3.23) at thresholds. The final refinement follows from (3.5) and the exact eigenvalue formula. □

Remark 3.4 (What the determinant condition does and does not imply). The pointwise condition $\det \mathsf{P}_z \neq 0$ proves that every individual quotient is a compact torus and hence individually gapped. It supplies no uniform lower bound over a noncompact parameter family. The cusp gives an explicit failure of the implication

$$\text{pointwise nondegeneracy} \quad \Longrightarrow \quad \inf_z \lambda_1(z) > 0.$$

The failure persists after volume normalization, so it is anisotropic rather than a mere global dilation.

4 The native pointed limit and finite-cover unspooling

Let e_1, e_2, e_3, e_4 be the standard basis in the real coordinate order of (1.1), and on the positive radial lift put

$$v_{\gamma,T} := \mathsf{P}_z e_\gamma, \qquad v_{u,T} := \mathsf{P}_z e_u, \qquad \mathsf{P}_z \mathbb{Z}^4 = \mathbb{Z} v_{\gamma,T} \oplus \mathbb{Z} v_{u,T} \oplus \mathbb{Z} e_1 \oplus \mathbb{Z} e_3. \tag{4.1}$$

Define the fixed real two-torus

$$\mathbb{T}^2_{\mathrm{re}} := (\mathbb{R} e_1 \oplus \mathbb{R} e_3)/(\mathbb{Z} e_1 \oplus \mathbb{Z} e_3). \tag{4.2}$$

Theorem 4.1 (Native limit and expanding marked covers)**.** *Let o_T be the origin of F_z^{flat}. With the unscaled Euclidean quotient metric,*

$$(F_z^{\mathrm{flat}}, o_T) \longrightarrow (\mathbb{T}^2_{\mathrm{re}} \times (\mathbb{R} e_2 \oplus \mathbb{R} e_4), (0, 0)) \cong (\mathbb{T}^2 \times \mathbb{R}^2, (0, 0)) \tag{4.3}$$

in pointed smooth and pointed Gromov–Hausdorff topology. For every fixed $r < \infty$, the radius-r balls are exactly isometric for all sufficiently large T. Moreover, eventually

$$\mathrm{sys}(\mathsf{P}_z \mathbb{Z}^4) = 1, \qquad \mathrm{inj}(F_z^{\mathrm{flat}}) = \frac{1}{2}. \tag{4.4}$$

For $M \in \mathbb{Z}_{\geq 1}$ define

$$\Lambda_{T,M} := \mathbb{Z} v_{\gamma,T} \oplus \mathbb{Z} v_{u,T} \oplus \mathbb{Z}(M e_1) \oplus \mathbb{Z}(M e_3), \qquad \widetilde{F}_{T,M} := \mathbb{R}^4/\Lambda_{T,M}. \tag{4.5}$$

The identity on $\mathbb{R}^4$ induces a regular holomorphic local-isometric cover

$$c_{T,M} : \widetilde{F}_{T,M} \longrightarrow F_z^{\mathrm{flat}} \tag{4.6}$$

of degree M^2, with deck group $(\mathbb{Z}/M\mathbb{Z})^2$. There are constants $C, C' > 0$, independent of large T and M, for which

$$\min\{T - C, M\} \le \mathrm{sys}(\Lambda_{T,M}) \le \min\{T + C', M\}. \tag{4.7}$$

Consequently, for an integer-valued $M = M(T)$,

$$(\widetilde{F}_{T,M(T)}, \widetilde{o}_{T,M(T)}) \longrightarrow (\mathbb{R}^4, 0) \tag{4.8}$$

in pointed smooth and pointed Gromov–Hausdorff topology if and only if $M(T) \to \infty$.

If the cover instead carries the pullback of the base-fibre unit-volume metric $\widehat{g}_z = \mathcal{D}_T^{-1/2} g_z$, then the same two pointed limits are $\mathbb{R}^4$ if and only if

$$\frac{M(T)}{\mathcal{D}_T^{1/4}} \longrightarrow \infty, \qquad \textit{equivalently} \qquad \frac{M(T)}{\sqrt{T}} \longrightarrow \infty. \tag{4.9}$$

By contrast, if each covering torus is rescaled by

$$g_{T,M}^{\mathrm{cov},1} := (M^2 \mathcal{D}_T)^{-1/2} g_z \tag{4.10}$$

so that the cover itself has volume one, no choice of $M(T)$ gives pointed smooth limit $\mathbb{R}^4$.

Proof. Project the two long periods in (4.1) to the (e_2, e_4)-plane. Their projected column matrix is

$$Y_T = \begin{pmatrix} 6m & T \\ B_R - T & m \end{pmatrix} = T \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \begin{pmatrix} 6m & 0 \\ B_R & m \end{pmatrix}. \tag{4.11}$$

The second matrix is uniformly bounded. Hence, after increasing C,

$$\left\| Y_T \begin{pmatrix} a \\ b \end{pmatrix} \right\| \ge (T - C)\sqrt{a^2 + b^2} \qquad (a, b \in \mathbb{R}). \tag{4.12}$$

First quotient $\mathbb{R}^4$ by $\mathbb{Z}e_1 \oplus \mathbb{Z}e_3$. The result is the product $\mathbb{T}^2_{\mathrm{re}} \times (\mathbb{R}e_2 \oplus \mathbb{R}e_4)$. Every nonidentity remaining deck translation has displacement at least $T - C$ by (4.12). If $T - C > 4r$, no such translate can identify two points of the radius-r ball or shorten their distance, while every minimizing quotient geodesic of length at most r lifts from the origin. The two radius-r balls are therefore exactly isometric. This proves (4.3). A nonzero period with a nonzero long coefficient has length at least $T - C$, whereas the shortest nonzero vector of $\mathbb{Z}e_1 \oplus \mathbb{Z}e_3$ has length one; this proves (4.4).

The four vectors in (4.1) are a lattice basis, so (4.5) has index M^2 and quotient $(\mathbb{Z}/M\mathbb{Z})^2$. Finite-index inclusion of complex lattices makes (4.6) holomorphic and local-isometric. A nonzero vector

$$a v_{\gamma,T} + b v_{u,T} + M(p e_1 + q e_3)$$

has length at least $T - C$ if $(a, b) \neq (0, 0)$, by its plane projection, and at least M otherwise. Conversely, $M e_1$ has length M, while either long generator has length at most $T + C'$ for a uniform C'. This proves (4.7). A flat quotient has radius-r balls exactly Euclidean once its lattice systole exceeds $4r$; hence $M(T) \to \infty$ proves (4.8). If $M(T)$ does not tend to infinity, integrality gives a subsequence on which $M(T) = M_*$ is constant. Quotienting first by $M_*\mathbb{Z}e_1 \oplus M_*\mathbb{Z}e_3$ and repeating the preceding exact-ball argument shows that this subsequence

converges to $(\mathbb{R}/M_*\mathbb{Z})^2 \times \mathbb{R}^2$, not to $\mathbb{R}^4$. This excludes both asserted $\mathbb{R}^4$ topologies and proves necessity.

Under the pullback of $\widehat{g}_z$, all lengths are multiplied by $\mathcal{D}_T^{-1/4}$. The long lower bound becomes $(T-C)\mathcal{D}_T^{-1/4} \sim \sqrt{T}$ and diverges automatically, while Me_1 has length $M\mathcal{D}_T^{-1/4}$. This proves the equivalence (4.9) in the sufficient direction. For necessity, if $\ell_T = M(T)\mathcal{D}_T^{-1/4}$ does not tend to infinity, pass to a subsequence with $\ell_T \to \ell < \infty$. Once the remaining long translations leave every fixed ball, the balls are those of $(\mathbb{R}/\ell_T\mathbb{Z})^2 \times \mathbb{R}^2$. If $\ell > 0$ the pointed limit is $(\mathbb{R}/\ell\mathbb{Z})^2 \times \mathbb{R}^2$; if $\ell = 0$ the pointed Gromov–Hausdorff limit is $\mathbb{R}^2$ and there is collapse, so there is no pointed smooth four-dimensional limit. In neither case is the limit $\mathbb{R}^4$. Finally, under (4.10), the explicit periods Me_1 and $v_{\gamma,T}$ have respective lengths

$$\sqrt{\frac{M}{T}}\,(1+o(1)), \qquad \sqrt{\frac{T}{M}}\,(1+o(1)). \tag{4.13}$$

Their leading factors have product one, so at least one stays bounded. The injectivity radius cannot tend to infinity, which excludes a pointed smooth $\mathbb{R}^4$ limit. □

Theorem 4.2 (Base-changed monodromy and exact cover momenta)**.** *Put*

$$\mathsf{S}_M = \operatorname{diag}(1,1,M,M), \qquad \Lambda^{(M)} = \mathsf{S}_M\mathbb{Z}^4. \tag{4.14}$$

Then $\mathsf{P}_z\Lambda^{(M)} = \Lambda_{T,M}$*. Writing* $M_0 = I + R_0$ *for the cusp monodromy in* (1.13)*, one has*

$$R_0^2 = 0, \qquad \ker R_0 = \operatorname{im} R_0 = \mathbb{Z}\langle \widehat{w}, \widehat{\delta}\rangle, \qquad \mathsf{S}_M^{-1}M_0^M\mathsf{S}_M = M_0. \tag{4.15}$$

Thus, after the cyclic base change $t_c = v^M$*, the covers form a regular relative degree-*M^2 *cover over* $v \neq 0$*. This is punctured descent; the completed local extension and its different absolute and relative degrees are treated in Section 6.1.*

With normalized Haar measure, the complete orthonormal character basis of $\widetilde{F}_{T,M}$ *is*

$$\widetilde{\chi}_{T,M,k}([x]) = \exp\Big(2\pi i\,\Big\langle \mathsf{P}_z^{-\mathsf{T}}\mathsf{S}_M^{-1}k, x\Big\rangle\Big), \qquad k \in \mathbb{Z}^4, \tag{4.16}$$

and its scalar spectrum, with multiplicity, is

$$\operatorname{Spec}(\widetilde{\Delta}_{T,M}) = \left\{4\pi^2\left\|\mathsf{P}_z^{-\mathsf{T}}\mathsf{S}_M^{-1}k\right\|^2 : k \in \mathbb{Z}^4\right\}. \tag{4.17}$$

The pullback, equivalently deck-invariant, sector consists exactly of

$$k \in \mathsf{S}_M\mathbb{Z}^4 \quad\Longleftrightarrow\quad k_3 \equiv k_4 \equiv 0 \pmod{M}. \tag{4.18}$$

There is a constant C_{mom}*, independent of large* T *and* $M \geq 1$*, such that*

$$\sup_{p\in\mathbb{R}^4} \operatorname{dist}\Big(p, \mathsf{P}_z^{-\mathsf{T}}\mathsf{S}_M^{-1}\mathbb{Z}^4\Big) \leq C_{\mathrm{mom}}\left(\frac{1}{T}+\frac{1}{M}\right). \tag{4.19}$$

Hence if $T_j, M_j \to \infty$*, the cover momentum lattices become locally dense in* $\mathbb{R}^4$*, and every* $E \geq 0$ *is a limit of eigenvalues in* (4.17)*.*

Proof. From (1.13), $R_0(a,b,c,d)^{\mathsf{T}} = (0,0,b,-a)^{\mathsf{T}}$. This proves the kernel, image, and square-zero assertions. The binomial theorem gives $M_0^M = I + MR_0$, and direct evaluation on $(a,b,c,d)^{\mathsf{T}}$ gives

$$\mathsf{S}_M^{-1}M_0^M\mathsf{S}_M(a,b,c,d)^{\mathsf{T}} = (a,b,c+b,d-a)^{\mathsf{T}} = M_0(a,b,c,d)^{\mathsf{T}}.$$

Thus M_0^M preserves $\Lambda^{(M)}$, which is exactly descent after $t_c = v^M$.

The dual of $\mathsf{P}_z\mathsf{S}_M\mathbb{Z}^4$ is $\mathsf{P}_z^{-\mathsf{T}}\mathsf{S}_M^{-1}\mathbb{Z}^4$. Fourier theory proves (4.16)–(4.17). Moreover, $c^*_{T,M}\chi_{T,\ell} = \widetilde{\chi}_{T,M,\mathsf{S}_M\ell}$, so a cover character descends exactly under the congruence in (4.18).

Let $b_{j,T,M}$ be the columns of $\mathsf{P}_z^{-\mathsf{T}}\mathsf{S}_M^{-1}$. The exact system (3.24), boundedness of all holomorphic cusp data, and $\mathcal{D}_T \asymp T^2$ give a uniform constant C_0 with

$$\|b_{1,T,M}\| + \|b_{2,T,M}\| \le \frac{C_0}{T}, \qquad \|b_{3,T,M}\| + \|b_{4,T,M}\| \le \frac{C_0}{M}.$$

Express any $p \in \mathbb{R}^4$ uniquely as $p = \sum_j t_j b_{j,T,M}$ and choose $n_j \in \mathbb{Z}$ with $|t_j - n_j| \le 1/2$. Then

$$\left\| p - \sum_j n_j b_{j,T,M} \right\| \le \frac{1}{2}\sum_j \|b_{j,T,M}\| \le \frac{C_0}{2}\left(\frac{1}{T} + \frac{1}{M}\right),$$

which proves (4.19). For $E > 0$, approximate a vector p with $4\pi^2\|p\|^2 = E$; continuity gives eigenvalues tending to E. For $E = 0$, take $k = e_\gamma$, whose momentum is $O(T^{-1})$. □

5 The invariant real quotient and its four-copy limit

Define the short-plane lattice, the primitive integral character, and its kernel lattice by

$$\begin{aligned} K_\Lambda &= \mathbb{Z}\langle \widehat{w}, \widehat{\delta}\rangle, \\ \varepsilon_\gamma(a\widehat{\gamma} + b\widehat{u} + c\widehat{w} + d\widehat{\delta}) = a, \qquad & L_\Lambda = \ker \varepsilon_\gamma = \mathbb{Z}\langle \widehat{u}, \widehat{w}, \widehat{\delta}\rangle. \end{aligned} \tag{5.1}$$

Theorem 5.1 (Regular-locus quotient and exact radial metric)**.** *Assume the conditional global period family and the integral monodromy data in Section 1. Then the following statements hold.*

(i) The formula

$$\chi_\gamma([z, \mathsf{P}_z x]) = \big([z], \varepsilon_\gamma(x) \bmod \mathbb{Z}\big) \tag{5.2}$$

defines a smooth real-analytic fibrewise group epimorphism

$$\chi_\gamma : J \longrightarrow B^\circ \times (\mathbb{R}/\mathbb{Z}). \tag{5.3}$$

Its kernel is a connected real-three-torus family with fibre $(L_\Lambda \otimes \mathbb{R})/L_\Lambda$. The kernel family need not be a product. The quotient is real; it is not a holomorphic quotient of complex tori. Moreover,

$$\mathrm{Hom}(\Lambda, \mathbb{Z})^\Delta = \mathbb{Z}\varepsilon_\gamma, \tag{5.4}$$

so this primitive invariant character is unique up to sign.

(ii) Give each marked fibre the Euclidean quotient metric already used in Section 2 and give the target circle its Riemannian quotient metric. On a radial cusp fibre,

$$\chi_{\gamma,z}([y]) = \left\langle \mathsf{P}_z^{-\mathsf{T}} e_\gamma, y \right\rangle \bmod \mathbb{Z}, \qquad e_\gamma = (1,0,0,0)^{\mathsf{T}}, \tag{5.5}$$

and its circumference is exactly

$$\ell_\gamma(z) = \frac{1}{\|\mathsf{P}_z^{-\mathsf{T}} e_\gamma\|} = \frac{-TD}{\sqrt{T^2 + m^2}}. \tag{5.6}$$

If $b_{\mathrm{I},0} = \operatorname{Im} b(0)$*, then on the positive radial lift*

$$\ell_\gamma(z) = T - b_{\mathrm{I},0} + O(T^{-1}) \longrightarrow \infty. \tag{5.7}$$

The pullback of the first circle Fourier mode is the fibre character with index e_γ*, whose eigenvalue is* $4\pi^2/\ell_\gamma(z)^2$*. The character is global, but the chosen Euclidean fibre metrics need not be invariant under the deck transitions. On the positive radial marked lift, this mode gap-closes with the same leading rate as Theorem 3.1; it is not asserted to be the exact shortest mode at every radial parameter.*

(iii) Let

$$J^{[4]} = J \times_{B^\circ} J \times_{B^\circ} J \times_{B^\circ} J. \tag{5.8}$$

The fourfold product of (5.2) is a quotient

$$\chi_\gamma^{[4]} : J^{[4]} \longrightarrow B^\circ \times (\mathbb{R}/\mathbb{Z})^4 \tag{5.9}$$

with connected twelve-torus kernel. Along the synchronized positive radial cusp, its quotient fibre with the induced product metric is

$$G_z = (\mathbb{R}/\ell_\gamma(z)\mathbb{Z})^4. \tag{5.10}$$

Its systole is $\ell_\gamma(z)$*. Consequently*

$$(G_z, 0) \longrightarrow (\mathbb{R}^4, 0) \tag{5.11}$$

in pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff topology. For every fixed $r > 0$*, the radius-*r *ball is exactly Euclidean once* $\ell_\gamma(z) > 4r$*.*

Equivalently, the external Cartesian product defines

$$J^4 \longrightarrow (B^\circ)^4 \times (\mathbb{R}/\mathbb{Z})^4, \tag{5.12}$$

and $J^{[4]}$ *is its pullback along the diagonal* $B^\circ \to (B^\circ)^4$*. At four independent positive radial cusp parameters* $z_1, \dots, z_4$*, its quotient lattice is*

$$\operatorname{diag}(\ell_\gamma(z_1), \dots, \ell_\gamma(z_4))\mathbb{Z}^4,$$

so the same three pointed limits hold whenever $\min_i \ell_\gamma(z_i) \to \infty$*.*

Among connected torus quotients of n *synchronized copies defined by a fixed primitive full-monodromy-invariant kernel lattice and equipped with the Riemannian quotient of the product Euclidean metric, any quotient whose systole tends to infinity has rank at most* n*. Hence four copies are minimal for a rank-four all-expanding quotient in this class, and (5.9) attains the bound.*

Proof. For a row covector (a, b, c, d), invariance under the first matrix in (1.12) gives $b = 2c$ and $d = 3c$, whereas invariance under the second gives $b = c$ and $d = 2c$. Thus $b = c = d = 0$, which proves (5.4). In particular $\varepsilon_\gamma A_g = \varepsilon_\gamma$ for every $g \in \Delta$. Equation (1.14) therefore proves that (5.2) is well defined. Primitivity of ε_γ proves surjectivity and connectedness of its kernel, and the induced action on $\Lambda/L_\Lambda \simeq \mathbb{Z}$ is trivial. Hence the target circle family is the displayed product. Since the kernel has odd real dimension, it is not a complex subtorus, proving (i).

Putting $(r, s) = (1, 0)$ in (3.8) gives

$$\mathsf{P}_z^{-\mathsf{T}} e_\gamma = \left(0, \frac{m}{-TD}, 0, \frac{-T}{-TD}\right)^{\mathsf{T}}. \tag{5.13}$$

The length of a Riemannian quotient circle defined by a primitive dual character is the reciprocal norm of that character, which proves (5.6). Write $b_{\mathrm{I}}(t_c) = \operatorname{Im} b(t_c)$. Since $p = b_{\mathrm{I}}(t_c) - T$,

$$-TD = T^2 - Tb_{\mathrm{I}}(t_c) + 6m^2. \tag{5.14}$$

Holomorphy at $t_c = 0$ gives $b_{\mathrm{I}}(t_c) = b_{\mathrm{I},0} + O(e^{-2\pi R})$ and $m = \operatorname{Im}\mu(0) + O(e^{-2\pi R})$ on the positive radial lift. Because $T = R + O(1)$, expansion of (5.6) proves (5.7). Formula (5.5) is the standard character formula from Proposition 2.1, and its exponential is precisely the index-e_γ eigenfunction. This proves (ii).

Taking four products proves the quotient and kernel statements in (iii). After multiplying each angular target coordinate by $\ell_\gamma(z)$, its fibre is $\mathbb{R}^4/\ell_\gamma(z)\mathbb{Z}^4$. If $x, y \in B_r(0)$ and $0 \neq \lambda \in \ell_\gamma(z)\mathbb{Z}^4$, then

$$\|x - y + \lambda\| \geq \|\lambda\| - \|x - y\| > \ell_\gamma(z) - 2r > 2r \geq \|x - y\|$$

when $\ell_\gamma(z) > 4r$. The covering map is therefore an exact isometry on the radius-r ball, proving all three pointed convergences. Taking the four quotient maps over independent base parameters gives (5.12); the identical estimate with $\ell_\gamma(z)$ replaced by $\min_i \ell_\gamma(z_i)$ proves its pointed limit and pulling back along the diagonal recovers $J^{[4]}$.

For minimality, equation (1.1) gives $\mathsf{P}_z\widehat{w} = (1,0,0,0)^{\mathsf{T}}$ and $\mathsf{P}_z\widehat{\delta} = (0,0,1,0)^{\mathsf{T}}$. Thus each fixed element of $K_\Lambda^{\oplus n}$ has length bounded independently of the radial parameter. Let $N \subset \Lambda^{\oplus n}$ be the fixed primitive invariant kernel lattice of an all-expanding quotient. If a fixed $k \in K_\Lambda^{\oplus n}$ did not lie in N, primitivity would give $N_{\mathbb{R}} \cap \Lambda^{\oplus n} = N$, so k would determine a nonzero quotient period of bounded length, contradicting divergence of the quotient systole. Therefore $K_\Lambda^{\oplus n} \subset N$.

The first matrix in (1.12) satisfies $A_1\widehat{w} = \widehat{u} - \widehat{w}$. Invariance of N consequently gives $L_\Lambda^{\oplus n} \subset N$, and hence

$$\operatorname{rank}(\Lambda^{\oplus n}/N) \leq 4n - 3n = n.$$

Rank four requires $n \geq 4$, while $N = L_\Lambda^{\oplus 4}$ gives the quotient already constructed. □

Proposition 5.2 (The expanding quotient of the native cusp fibre)**.** *On a radial cusp fibre define*

$$\rho : \mathbb{R}^4 \longrightarrow \mathbb{R}^2, \qquad \rho(x_1, x_2, x_3, x_4) = (x_2, x_4),$$

and

$$Y_z = \begin{pmatrix} 6m & T \\ p & m \end{pmatrix}. \tag{5.15}$$

Then ρ induces an exact sequence with

$$\mathbb{T}^2_{\mathrm{re}} := \frac{\mathbb{R}\langle e_1, e_3\rangle}{\mathbb{Z}\langle e_1, e_3\rangle},$$

namely

$$0 \longrightarrow \mathbb{T}^2_{\mathrm{re}} \longrightarrow F_z^{\mathrm{flat}} \longrightarrow Q_z := \mathbb{R}^2/Y_z\mathbb{Z}^2 \longrightarrow 0, \tag{5.16}$$

and $\operatorname{sys}(Q_z) \geq T - C$ for large T. Thus $(Q_z, 0)$ has pointed local-isometric limit $(\mathbb{R}^2, 0)$, and two synchronized copies satisfy

$$(Q_z \times Q_z, 0) \longrightarrow (\mathbb{R}^4, 0). \tag{5.17}$$

The native fibre itself still has the different pointed limit $\mathbb{T}^2_{\mathrm{re}} \times \mathbb{R}^2$ from Theorem 4.1. The one-copy quotient has kernel lattice K_Λ, and the displayed two-copy quotient has kernel lattice $K_\Lambda^{\oplus 2}$. This is not the global quotient of four copies in Theorem 5.1. No finite cover of B° makes either of these particular kernels full-monodromy invariant.

Proof. The last two columns of (1.1) are killed by ρ, and the projections of the first two columns are the columns of Y_z. Moreover,

$$\det Y_z = 6m^2 - Tp = -TD > 0.$$

If $\rho(x) = Y_z(a,b)^{\mathsf{T}}$, then $x - \mathsf{P}_z(a,b,0,0)^{\mathsf{T}}$ lies in $\ker\rho = \mathbb{R}\langle e_1, e_3\rangle$. Conversely, if $\mathsf{P}_z k \in \ker\rho$ for $k \in \mathbb{Z}^4$, then $Y_z(k_1,k_2)^{\mathsf{T}} = 0$; invertibility of Y_z gives $k_1 = k_2 = 0$. Thus

$$\ker(F_z^{\mathrm{flat}} \longrightarrow Q_z) \simeq \frac{\mathbb{R}\langle e_1, e_3\rangle}{\mathbb{Z}\langle e_1, e_3\rangle} = \mathbb{T}^2_{\mathrm{re}},$$

which proves exactness without identifying this kernel with the invariant kernel L_Λ. By (1.9),

$$T^{-1}Y_z = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + O_{\mathrm{op}}(T^{-1}).$$

The least singular value of Y_z is therefore at least $T - C$, and the same is true of the shortest nonzero vector of $Y_z\mathbb{Z}^2$. The pointed-limit argument in the proof of Theorem 4.1 now applies first to Q_z and then to Q_z^2.

For the finite-cover assertion, direct multiplication gives

$$H_* = A_2A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ -5 & 0 & -1 & 1 \end{pmatrix} = I + N_*, \qquad N_*^2 = 0. \tag{5.18}$$

Consequently, for every nonzero integer n,

$$H_*^n\widehat{w} = \widehat{w} + n(\widehat{u} - \widehat{\delta}), \qquad H_*^n\widehat{\delta} = \widehat{\delta}, \tag{5.19}$$

so $H_*^n K_\Lambda \neq K_\Lambda$. Choose a loop in B° whose monodromy is H_*. Every finite-index subgroup of the ordinary $\pi_1(B^\circ)$ contains a positive power of this loop, and that power still fails to preserve K_Λ. Hence no finite base cover globalizes this particular one-copy marked quotient. Under diagonal monodromy, $(\widehat{w}, 0)$ proves the same assertion for $K_\Lambda^{\oplus 2}$ and thus for its two-copy product. □

Remark 5.3 (Copies, quotients, and covers). The native fibre limit in Theorem 4.1 is $\mathbb{T}^2 \times \mathbb{R}^2$. The same theorem obtains $\mathbb{R}^4$ only after passing to degree-$M(T)^2$ fibre covers with $M(T) \to \infty$. Neither statement is a quotient construction. By contrast, the four-circle limit in Theorem 5.1 changes neither the base nor the fibre lattice: it takes a quotient of four synchronized copies. Proposition 5.2 instead uses a rank-two marked-fibre quotient whose all-expanding behavior is radial cusp behavior and whose kernel does not descend over the full regular base. These quotient constructions are distinct from the degree-M^2 fibre covers, whose punctured descent requires the base change $t_c = v^M$; no such finite-cover construction is used in this paper's globalization theorem.

Proposition 5.4 (Literal fixed-target nonextension through the toric cusp)**.** *Assume the toric cusp chart asserted for the proposed completion. Then neither χ_γ to the fixed product-circle target nor $\chi_\gamma^{[4]}$ to the fixed product $\mathbb{T}^4$ target extends continuously through the toric cusp fibre with its stated regular-fibre restriction.*

Proof. On the positive radial lift $s = iR$, equations (1.1) and (1.5) give

$$Z_R = iRB_0 + C(t_c), \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C(t_c) = \begin{pmatrix} 6\mu & h \\ b-h & \mu \end{pmatrix}, \qquad t_c = e^{-2\pi R}.$$

For $a \in \mathbb{R}^2$, let $x_R(a) = \exp(2\pi i Z_R a) \in (\mathbb{C}^*)^2$ in the dense torus of the cusp model. Its normalized logarithmic position is

$$\frac{(\log|x_{R,1}(a)|, \log|x_{R,2}(a)|)}{\log|t_c|} = B_0 a + \frac{\operatorname{Im} C(t_c) a}{R}. \tag{5.20}$$

In the affine toric chart belonging to $\sigma = \operatorname{conv}(0, e_1, e_2)$, the coordinates are

$$z_0 = \frac{t_c}{x_1 x_2}, \qquad z_1 = x_1, \qquad z_2 = x_2.$$

Writing the left side of (5.20) as $y_R = (y_{R,1}, y_{R,2})$, one has exactly

$$|z_0| = |t_c|^{1-y_{R,1}-y_{R,2}}, \qquad |z_1| = |t_c|^{y_{R,1}}, \qquad |z_2| = |t_c|^{y_{R,2}}.$$

If the limit of y_R lies in the interior of σ, all three exponents are eventually positive, so all three coordinates tend to zero and the path approaches the same toric fixed point.

Take

$$a = \left(-\frac{1}{4}, \frac{1}{4}\right), \qquad a' = \left(-\frac{1}{2}, \frac{1}{4}\right).$$

Then

$$B_0 a = \left(\frac{1}{4}, \frac{1}{4}\right), \qquad B_0 a' = \left(\frac{1}{4}, \frac{1}{2}\right)$$

are both interior to σ, whereas

$$Z_R a = \mathsf{P}_z(a_1, a_2, 0, 0)^{\mathsf{T}}, \qquad Z_R a' = \mathsf{P}_z(a_1', a_2', 0, 0)^{\mathsf{T}},$$

in the real coordinate identification, and hence

$$\chi_{\gamma,R}(x_R(a)) = \frac{3}{4}, \qquad \chi_{\gamma,R}(x_R(a')) = \frac{1}{2} \quad \text{in } \mathbb{R}/\mathbb{Z}.$$

Thus two paths converge to the same cusp point while their character values have different limits. This rules out continuous extension of the literal map to the fixed circle target. Holding the other three factors on identical radial paths proves the assertion for the fixed product $\mathbb{T}^4$ target. □

Corollary 5.5 (The fixed $\mathbb{T}^4$ target and the smash collapse)**.** *Assume in addition the claimed recognition $X \cong_{\text{diff}} S^6$. Then*

$$X^4 \cong_{\text{diff}} (S^6)^4 \not\simeq S^{24}, \qquad H_6(X^4; \mathbb{Z}) \cong \mathbb{Z}^4.$$

After choices of base point and of the recognition, one has $X^{\wedge 4} \cong S^{24}$, but the literal fourfold quotient to the fixed $\mathbb{T}^4$ target has no extension that factors through that smash sphere. More precisely, no map $\Phi : X^4 \to (\mathbb{R}/\mathbb{Z})^4$ restricts on a regular fourfold fibre to $(\chi_{\gamma,z})^4$, and no map $\Psi : X^{\wedge 4} \to (\mathbb{R}/\mathbb{Z})^4$ has that restriction after precomposition with the smash-collapse map $X^4 \to X^{\wedge 4}$. These statements obstruct only the displayed character quotient with the fixed target $(\mathbb{R}/\mathbb{Z})^4$ and the stated regular-fibre restriction; they do not rule out codomain-changing cone targets or any other real or singular globalization mechanism.

Proof. The product homology statement follows from the integral Kunneth theorem, so X^4 is not a homotopy sphere. The smash identification follows from a based recognition $X \cong S^6$. On $F_z^{\text{flat}} \times \{0\}^3$, the fourfold quotient is

$$x \longmapsto (\chi_{\gamma,z}(x), 0, 0, 0),$$

which is nonconstant. It is therefore not constant on the fat wedge and does not factor through the fibre smash product.

For the stronger extension statement, $H^1(X^4;\mathbb{Z}) = 0$. If a map

$$\Phi : X^4 \longrightarrow (\mathbb{R}/\mathbb{Z})^4$$

restricted on one regular fourfold fibre to $(\chi_{\gamma,z})^4$, the pullback of the first target H^1-generator would restrict to the nonzero class ε_γ on the first fibre factor, while every class pulled back to X^4 is zero. This is impossible. A map from $X^{\wedge 4}$ with the corresponding restriction would give such a Φ after precomposition with the smash-collapse map $X^4 \to X^{\wedge 4}$, so it is impossible as well. □

6 The completed local toric Kummer morphism

The spectral degeneration and the invariant-character quotient do not by themselves extend a fibre cover across the cusp. The following local theorem performs that separate task. Its source quotient uses the modified action with $C_N(v) = C(v^N)/N$; consequently the toric power map is equivariant. The degree to the original cusp family is N^3, whereas N^2 is the generic degree only after factoring through the base-changed target. The construction is local at the cusp and makes no full-base claim.

6.1 The Kummer cover extends across the toric cusp

This section determines exactly what is true in the assertion that the degree-N^2 cover of the regular fibres extends by a toric power map. The answer is positive for the completed *local* cusp family, but the source action has to be changed, and two different degrees must not be conflated.

Retain the cusp data

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Z(s) = sB_0 + C(t), \qquad t = e^{2\pi i s},$$

where C is holomorphic on a disc Δ_t about zero. Put

$$c_{\bar\lambda}(t) := \exp\bigl(2\pi i C(t)\bar\lambda\bigr) \in (\mathbb{C}^*)^2, \qquad \bar\lambda \in \bar\Lambda \simeq \mathbb{Z}^2.$$

Let Σ be the fan in $N' = \mathbb{Z}^2 \oplus \mathbb{Z}$ consisting of the cones over the cells of the translation-invariant A_2 triangulation at height one, and write $Y = Y_\Sigma$. On the dense torus, with coordinates (x_1, x_2, t), the target action is

$$\Psi_{\bar\lambda}(x,t) = \bigl(c_{\bar\lambda}(t)t^{B_0\bar\lambda}x, t\bigr). \tag{6.1}$$

Its quotient is the completed cusp family $f_0 \colon N_0 = Y/\bar\Lambda \to \Delta_t$.

Fix an integer $N \geq 1$. Let v be a coordinate on a second disc and make the base change

$$\rho_N \colon \Delta_v \longrightarrow \Delta_t, \qquad \rho_N(v) = v^N, \qquad t = v^N. \tag{6.2}$$

On a copy $Y^{(N)}$ of the same toric variety, with dense-torus coordinates (ξ_1, ξ_2, v), define

$$C_N(v) := \frac{1}{N} C(v^N), \qquad c^{(N)}_{\bar\lambda}(v) := \exp\bigl(2\pi i C_N(v)\bar\lambda\bigr) \tag{6.3}$$

and

$$\Psi^{(N)}_{\bar\lambda}(\xi, v) := \bigl(c^{(N)}_{\bar\lambda}(v) v^{B_0\bar\lambda}\xi, v\bigr). \tag{6.4}$$

Theorem 6.1 (Exact Kummer extension through the cusp)**.** ***Verified.*** *After shrinking the v-disc, the following statements hold.*

1. *The maps in (6.4) extend to a free, properly discontinuous action on $Y^{(N)}$. The quotient*
$$f_N\colon N_0^{\langle N\rangle} := Y^{(N)}/\bar{\Lambda} \longrightarrow \Delta_v$$
is a proper holomorphic map from a smooth threefold. Its central fibre W_N is reduced, irreducible, and nonnormal, and its normalization is a toric del Pezzo surface of degree six.

2. *Multiplication by N on the cocharacter lattice defines a finite flat toric morphism*
$$P_N\colon Y^{(N)} \longrightarrow Y, \qquad P_N(\xi_1,\xi_2,v) = (\xi_1^N,\xi_2^N,v^N), \tag{6.5}$$
of degree N^3. It is equivariant for (6.4) and (6.1), hence descends to a finite flat map
$$\kappa_N\colon N_0^{\langle N\rangle} \longrightarrow N_0, \qquad f_0\circ\kappa_N = \rho_N\circ f_N, \tag{6.6}$$
of degree N^3.

3. *The map κ_N is etale over $v\neq 0$, and*
$$\kappa_N^* W = NW_N, \qquad \mathrm{Ram}(\kappa_N) = (N-1)W_N. \tag{6.7}$$
In particular, the branch support of the map to the original cusp family is the whole central fibre.

4. *Let*
$$N_0[N] := N_0 \times_{\Delta_t} \Delta_v, \qquad t = v^N. \tag{6.8}$$
Then $N_0[N]$ is normal and κ_N factors uniquely as
$$N_0^{\langle N\rangle} \xrightarrow{Q_N} N_0[N] \longrightarrow N_0. \tag{6.9}$$
The map Q_N has generic degree N^2. For $N>1$ it is quasi-etale but not flat: it is etale over the smooth locus of $N_0[N]$, and its non-etale and nonflat loci are exactly the inverse images of the double and triple strata of the base-changed central fibre. Thus Q_N is not a normalization map.

5. *On the reduced central fibres, the induced map $W_N \to W$ has generic degree N^2. It lifts to a commutative diagram*
$$\begin{array}{ccc} E_N \simeq \mathrm{dP}_6 & \xrightarrow{[N]} & E \simeq \mathrm{dP}_6 \\ {\scriptstyle \nu_N}\downarrow & & \downarrow{\scriptstyle \nu} \\ W_N & \longrightarrow & W, \end{array} \tag{6.10}$$
where $[N]$ is toric multiplication by N, of degree N^2, ramified with index N along each of the six toric boundary curves. The diagram respects the paired conductor curves and the two triple-point classes.

6. *Over $v\neq 0$, Q_N is exactly the cover defined by*
$$\mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, N\widehat{w}, N\widehat{\delta}\rangle \subset \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta}\rangle.$$
Its deck group is $(\mathbb{Z}/N\mathbb{Z})^2$. The pulled-back cusp monodromy is M_0^N, and in the displayed cover basis it is M_0:
$$S_N^{-1} M_0^N S_N = M_0, \qquad S_N = \mathrm{diag}(1,1,N,N). \tag{6.11}$$

Proof. We prove the assertions in the order in which the constructions occur.

The source action. For $\bar{\lambda} \in \bar{\Lambda}$, the integral linear map

$$\varphi_{\bar{\lambda}}(y, y_3) = (y + y_3 B_0 \bar{\lambda}, y_3)$$

translates the height-one slice by the integral vector $B_0\bar{\lambda}$. It preserves the A_2 triangulation and hence induces a toric automorphism $\Phi_{\bar{\lambda}}$ of $Y^{(N)}$. On the dense torus this automorphism is

$$\Phi_{\bar{\lambda}}(\xi, v) = (v^{B_0\bar{\lambda}}\xi, v).$$

Multiplication by the nowhere-zero holomorphic function $c_{\bar{\lambda}}^{(N)}(v)$ extends across every toric chart. Since $C_N(v)\bar{\lambda}$ is additive in $\bar{\lambda}$, the resulting maps satisfy $\Psi_{\bar{\lambda}}^{(N)}\Psi_{\bar{\mu}}^{(N)} = \Psi_{\bar{\lambda}+\bar{\mu}}^{(N)}$.

For completeness, freeness and proper discontinuity can be checked directly. Put

$$R_N(v) := -2\pi \operatorname{Im} C_N(v).$$

It is bounded at $v = 0$. At a dense-torus point fixed by $\bar{\lambda}$, logarithmic absolute values give

$$\log|v| B_0\bar{\lambda} + R_N(v)\bar{\lambda} = 0.$$

Choose the disc so small that $|\log|v|| > 2\|R_N\|_\infty$. Because B_0 is orthogonal, the last equality is impossible for $\bar{\lambda} \neq 0$. On the central fibre, $\Psi_{\bar{\lambda}}^{(N)}$ translates every bounded cell by the nonzero vector $B_0\bar{\lambda}$, so it fixes no toric orbit.

For a dense-torus point set

$$y(\xi, v) = \frac{(\log|\xi_1|, \log|\xi_2|)}{\log|v|}, \qquad B_{N,v} = B_0 + \frac{R_N(v)}{\log|v|}.$$

Then

$$y(\Psi_{\bar{\lambda}}^{(N)}(\xi, v)) = y(\xi, v) + B_{N,v}\bar{\lambda}, \qquad |B_{N,v}\bar{\lambda}| \geq \tfrac{1}{2}|\bar{\lambda}|.$$

The logarithmic-position regions associated with any two fixed toric charts are bounded. Hence only finitely many $\bar{\lambda}$ can translate one of them to meet the other. Local finiteness of the fan reduces the same assertion for compact sets to finitely many chart pairs, proving proper discontinuity. Finally, reducing $B_{N,v}^{-1}y(\xi, v)$ modulo $\mathbb{Z}^2$ places every orbit meeting a closed subdisc in finitely many closed unit toric charts. Their union is compact. This proves properness of f_N. The central-fibre and normalization assertions follow exactly from the height-one fan: locally v is the product of the coordinates belonging to the rays through a cell, every component occurs with multiplicity one, the lattice action is transitive on vertices, and the star of a vertex is the complete smooth hexagonal fan.

The toric map and equivariance. Multiplication by N on N' sends every cone $\sigma \in \Sigma$ onto itself as a real cone. It therefore defines the toric morphism P_N. For a maximal cone, let z_0, z_1, z_2 be the target characters dual to its three primitive height-one ray generators, and let w_0, w_1, w_2 be the corresponding source characters. The map on convergent local rings is

$$\mathbb{C}\{z_0, z_1, z_2\} \longrightarrow \mathbb{C}\{w_0, w_1, w_2\}, \qquad z_i \longmapsto w_i^N. \tag{6.12}$$

The target-chart preimage is the corresponding source chart, and the right-hand ring is free of rank N^3 over the left-hand ring, with basis

$$\{w_0^{a_0} w_1^{a_1} w_2^{a_2} : 0 \leq a_i < N\}.$$

Thus P_N is finite flat of degree N^3 on every maximal chart, hence globally.

The choice (6.3) is forced by equivariance. Indeed, writing powers coordinatewise,

$$P_N \Psi_{\bar{\lambda}}^{(N)}(\xi, v) = ((c_{\bar{\lambda}}^{(N)}(v))^N v^{NB_0\bar{\lambda}} \xi^N, v^N) \tag{6.13}$$

$$= (c_{\bar{\lambda}}(v^N)(v^N)^{B_0\bar{\lambda}} \xi^N, v^N) \tag{6.14}$$

$$= \Psi_{\bar{\lambda}} P_N(\xi, v). \tag{6.15}$$

Consequently P_N descends to (6.6). To see finiteness and flatness after the infinite-lattice quotient without any implicit descent theorem, take an evenly covered open set in N_0 and one of its lifts $V \subset Y$. Equivariance gives a unique representative of every source orbit above that open set whose image lies in V; two such representatives cannot differ by a nontrivial lattice translate. The descended map is therefore locally identified with $P_N^{-1}(V) \to V$. It is finite flat of degree N^3 by (6.12).

In the same chart the base characters are

$$t = z_0 z_1 z_2, \qquad v = w_0 w_1 w_2, \qquad P_N^* t = v^N.$$

The Jacobian determinant of (6.12) is

$$N^3 (w_0 w_1 w_2)^{N-1} = N^3 v^{N-1}.$$

This proves (6.7); the local quotient maps are biholomorphisms, so the calculation is unchanged downstairs.

The relative degree-N^2 factor. The equality $(f_0 \circ \kappa_N)([\xi, v]) = v^N = (\rho_N \circ f_N)([\xi, v])$ gives the factorization (6.9). At a triple point the exact local-ring map is

$$\begin{aligned} A_N &:= \frac{\mathbb{C}\{z_0, z_1, z_2, v\}}{(z_0 z_1 z_2 - v^N)}, \qquad B := \mathbb{C}\{w_0, w_1, w_2\}, \\ A_N &\longrightarrow B, \qquad z_i \longmapsto w_i^N, \qquad v \longmapsto w_0 w_1 w_2. \end{aligned} \tag{6.16}$$

Let

$$G_N = \{(\varepsilon_0, \varepsilon_1, \varepsilon_2) \in \mu_N^3 : \varepsilon_0 \varepsilon_1 \varepsilon_2 = 1\} \simeq \mu_N^2$$

act diagonally on B. A monomial $w_0^{a_0} w_1^{a_1} w_2^{a_2}$ is G_N-invariant if and only if

$$a_0 \equiv a_1 \equiv a_2 \pmod{N}.$$

Writing their common residue as r, with $0 \le r < N$, expresses that monomial uniquely in the form

$$(w_0 w_1 w_2)^r (w_0^N)^{b_0} (w_1^N)^{b_1} (w_2^N)^{b_2}.$$

The same coefficientwise argument applies to convergent power series. Hence

$$A_N = B^{G_N}. \tag{6.17}$$

It follows that A_N is normal, that the map has generic degree $|G_N| = N^2$, and that it is a finite quotient rather than a normalization. The corresponding face-chart calculations prove normality along the double and smooth strata as well, so $N_0[N]$ is normal everywhere.

The stabilizer in G_N is trivial when none or exactly one of the w_i vanishes, is isomorphic to μ_N when exactly two vanish, and is all of G_N when all three vanish. Thus Q_N is etale in codimension one and has no ramification divisor, but it is not etale above the double and

triple strata. It is also nonflat at every such point when $N > 1$. At a triple point the closed fibre of (6.16) has length

$$\dim_{\mathbb{C}} \frac{\mathbb{C}\{w_0, w_1, w_2\}}{(w_0^N, w_1^N, w_2^N, w_0 w_1 w_2)} = N^3 - (N-1)^3 \tag{6.18}$$

$$= 3N^2 - 3N + 1, \tag{6.19}$$

which exceeds the generic rank N^2. At a point of a double stratum, localizing at the nonzero third coordinate and choosing its holomorphic Nth root reduces the ramified factor to

$$\frac{\mathbb{C}\{z_0, z_1, v\}}{(z_0 z_1 - v^N)} \longrightarrow \mathbb{C}\{w_0, w_1\}, \qquad z_i \longmapsto w_i^N, \quad v \longmapsto w_0 w_1.$$

Its local generic rank is N, whereas its closed fibre has length

$$\dim_{\mathbb{C}} \frac{\mathbb{C}\{w_0, w_1\}}{(w_0^N, w_1^N, w_0 w_1)} = 2N - 1 > N. \tag{6.20}$$

The remaining transverse nonzero torus coordinate is etale locally. Equations (6.19) and (6.20), together with etaleness elsewhere, identify the nonflat locus exactly.

The regular fibres and monodromy. Choose a logarithm r with $v = e^{2\pi i r}$. The target period matrix at $t = v^N$ is

$$Z(Nr) = NrB_0 + C(v^N),$$

whereas the source residual action is determined by

$$Z_N(r) = rB_0 + \frac{1}{N} C(v^N) = \frac{1}{N} Z(Nr).$$

The source fibre is therefore

$$\frac{\mathbb{C}^2}{\mathbb{Z}^2 + Z_N(r)\mathbb{Z}^2},$$

and multiplication by N on $\mathbb{C}^2$ maps it to the target fibre. Equivalently, in the unscaled additive coordinate it is the identity map

$$\frac{\mathbb{C}^2}{N\mathbb{Z}^2 + Z(Nr)\mathbb{Z}^2} \longrightarrow \frac{\mathbb{C}^2}{\mathbb{Z}^2 + Z(Nr)\mathbb{Z}^2}. \tag{6.21}$$

The imaginary part of $Z(Nr)$ is invertible on a regular fibre, so its period lattice meets $\mathbb{Z}^2$ trivially in the required real sense; the index in (6.21) is N^2. Its deck group is $(\mathbb{Z}/N\mathbb{Z})^2$. In the ordered lattice basis $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$, this is exactly the sublattice obtained by S_N. Since $M_0 = I + R$ with $R^2 = 0$ and $R(a, b, c, d) = (0, 0, b, -a)$, one has

$$M_0^N = I + NR, \qquad S_N^{-1} M_0^N S_N = M_0,$$

proving (6.11).

The central normalization and conductor. Multiplication by N fixes every ray of Σ. It therefore maps a chosen source component of the toric central fibre to the corresponding target component. The star of that ray is the hexagonal fan, so the restriction is the toric power map $[N]\colon \mathrm{dP}_6 \to \mathrm{dP}_6$. On its dense two-torus it has degree N^2; in each toric boundary chart it is an Nth-power map in the normal coordinate, hence has ramification index N along all six boundary curves. Equation (6.15) says that the source and target lattice identifications commute with this map. Consequently opposite boundary curves, their conductor identifications, and the two translation-orbits of triangle vertices descend compatibly. This proves (6.10) and completes the proof. □

Remark 6.2 (Why the unmodified action does not work). If one reuses the target action with v in place of t at the source, then the two sides of equivariance have multiplicative factors

$$c_{\bar{\lambda}}(v)^N v^{NB_0\bar{\lambda}} \quad \text{and} \quad c_{\bar{\lambda}}(v^N)v^{NB_0\bar{\lambda}},$$

respectively. In general $c_{\bar{\lambda}}(v)^N \neq c_{\bar{\lambda}}(v^N)$. Thus the bare formula $(\xi_1, \xi_2, v) \mapsto (\xi_1^N, \xi_2^N, v^N)$ extends on the ambient toric variety but does not descend through the manuscript's quotient unless the source factor (6.3) is inserted. The explicit logarithm $2\pi i C(v^N)\bar{\lambda}/N$ makes this root holomorphic; no unjustified choice of an Nth root is involved.

Remark 6.3 (Exact scope). The theorem constructs a finite morphism of completed families over the local cusp discs. It is a degree-N^3 map to the original family and a degree-N^2 map only after the target is base changed. It is neither a degree-N^2 self-map over the unchanged disc nor the normalization of the base change.

Nothing here supplies a cover over the entire base B. In fact the displayed sublattice is not preserved by the original full monodromy when $N > 1$: for example, the printed matrix A_1 sends $\widehat{u}$ to a vector whose $\widehat{w}$- and $\widehat{\delta}$-coefficients are -1 and 1. A further global base cover and compatible extensions through both finite fillings would have to be constructed separately. The local Kummer morphism also supplies no map to a product sphere, smash sphere, S^{12}, or S^{24}.

Context from the separate cone-target construction. The canonical workbench contains a different, codomain-changing construction whose proof is not reproduced or claimed in this standalone paper. In that construction, collapsing the appropriate cusp boundary pieces gives based maps

$$q_2^{\wedge 2} : X^{\wedge 2} \cong S^{12} \longrightarrow \mathcal{A}^{\wedge 2}, \qquad q_1^{\wedge 4} : X^{\wedge 4} \cong S^{24} \longrightarrow \mathcal{C}^{\wedge 4}.$$

Their synchronized radial target slices have intrinsic varying flat metrics with pointed local-isometric $\mathbb{R}^4$ limits. In every one fixed compatible ambient metric on the compact cone targets, however, those same slices collapse to the relevant cone point. These sphere-source maps have different codomains from the fixed $(\mathbb{R}/\mathbb{Z})^4$ target in Corollary 5.5, so they are outside that literal nonextension statement.

7 A gapless vertical direct-integral operator

Choose a measurable cusp ray $T \mapsto z(T)$ for $T \geq T_0$ and normalize Haar measure on every fibre. The marking identifies the fibre characters measurably. Let

$$\mathcal{H}_0 = \int_{T_0}^{\infty} {}^{\oplus} L_0^2(F_{z(T)}^{\mathrm{flat}})\, w(T)\, \mathrm{d}T, \tag{7.1}$$

where L_0^2 denotes the orthogonal complement of the constants and w is a positive locally integrable weight whose measure is nonzero on arbitrarily distant bounded intervals. Define the decomposable operator

$$H_{\mathrm{v}} = \int_{T_0}^{\infty} {}^{\oplus} \Delta_{z(T)}\big|_{L_0^2}\, w(T)\, \mathrm{d}T. \tag{7.2}$$

Fourier expansion in the marked angular coordinates gives the concrete unitary identification

$$\mathcal{H}_0 \cong \bigoplus_{k \in \mathbb{Z}^4 \setminus \{0\}} L^2([T_0, \infty), w(T)\, \mathrm{d}T), \tag{7.3}$$

under which

$$(H_{\mathrm{v}}u)_k(T) = \lambda_T(k)u_k(T),$$
$$\mathrm{Dom}(H_{\mathrm{v}}) = \left\{ u \in \mathcal{H}_0 : \sum_{k\neq 0} \int_{T_0}^{\infty} \lambda_T(k)^2 |u_k(T)|^2 w(T)\,\mathrm{d}T < \infty \right\}. \tag{7.4}$$

Theorem 7.1 (Continuous spectral threshold at zero)**.** *The operator H_{v} is nonnegative and self-adjoint,*

$$\inf \mathrm{Spec}(H_{\mathrm{v}}) = 0, \qquad \ker H_{\mathrm{v}} = \{0\}. \tag{7.5}$$

Moreover,

$$0 \in \sigma_{\mathrm{ess}}(H_{\mathrm{v}}) \cap \sigma_{\mathrm{cont}}(H_{\mathrm{v}}), \tag{7.6}$$

so there is no positive spectral gap. If the fibrewise constants are retained instead, the kernel contains the entire parameter-space factor and is infinite-dimensional.

Proof. For each $k \neq 0$, multiplication by the positive measurable real function $\lambda_T(k)$ on $L^2([T_0,\infty), w(T)\,\mathrm{d}T)$ is self-adjoint on its maximal multiplication domain. The orthogonal sum of these operators has exactly the domain (7.4), proving self-adjointness and nonnegativity; this is also the standard direct-integral construction (Reed and Simon, 1978, Chapter XIII). Let $e_{\gamma,T}$ be the normalized fixed-fibre character corresponding to $k = e_\gamma$. Proposition 2.1, formula (3.8), and the cusp estimates imply

$$\langle e_{\gamma,T}, \Delta_{z(T)} e_{\gamma,T}\rangle \leq \frac{C}{T^2} \tag{7.7}$$

for large T. Theorem 5.1 additionally identifies this already-defined character as the pullback of the first Fourier mode on the global real circle quotient, but the direct-integral argument does not depend on that identification and does not require it to be the exact shortest fibre mode. Choose pairwise disjoint intervals $I_j = [S_j, S_j + 1]$ with $S_j \to \infty$ and positive $w(T)\mathrm{d}T$ measure. The sections

$$\Psi_j(T) = \frac{\mathbf{1}_{I_j}(T) e_{\gamma,T}}{\left(\int_{I_j} w(T)\,\mathrm{d}T\right)^{1/2}}$$

belong to $\mathrm{Dom}(H_{\mathrm{v}})$, have unit norm, are mutually orthogonal, hence converge weakly to zero, and satisfy

$$\|H_{\mathrm{v}}\Psi_j\|^2 \leq C^2/S_j^4 \longrightarrow 0.$$

Weyl's criterion proves $0 \in \sigma_{\mathrm{ess}}(H_{\mathrm{v}})$, and in particular the spectral bottom is zero.

If $H_{\mathrm{v}}\Psi = 0$, then nonnegativity and decomposability give

$$0 = \int \langle \Psi(T), \Delta_{z(T)}\Psi(T)\rangle w(T)\,\mathrm{d}T.$$

The integrand is nonnegative, hence vanishes almost everywhere. Every fibre vector then lies in the constants, but those were removed in (7.1); therefore $\Psi = 0$. A self-adjoint operator has no residual spectrum, so an injective operator with $0 \in \mathrm{Spec}$ has 0 in its continuous spectrum. Retaining constants instead gives all sections $T \mapsto c(T)\mathbf{1}_T$ as zero modes. □

Remark 7.2 (Vertical, not total-space). The operator (7.2) differentiates only along the torus fibres. It has no derivative in the cusp parameter, no boundary coupling, and no claim to be the Laplacian of a smooth metric on the completed six-dimensional total space. Its gaplessness is exact for the operator defined here, but it must not be silently transferred to a different operator.

8 Classical gauge fluctuation operators on the radial cusp

Let G be a compact Lie group with real Lie algebra $\mathfrak{g}$, equipped with a positive-definite $\mathrm{Ad}(G)$-invariant inner product, and put

$$d_G = \dim_{\mathbb{R}} \mathfrak{g}, \qquad \mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}.$$

Fix a coupling $g_{\mathrm{YM}} > 0$. Every operator in this section acts on the trivial bundle over one of the auxiliary flat tori F_z^{flat} along the positive radial cusp. No physical spacetime identification is made.

8.1 Continuum Hessian and Coulomb spectrum

On the trivial principal bundle $\mathcal{P}_T = F_z^{\mathrm{flat}} \times G$, let A_0 be the product connection and normalize

$$\mathrm{YM}_T(A) = \frac{1}{2g_{\mathrm{YM}}^2} \int_{F_z^{\mathrm{flat}}} |F_A|^2 \, \mathrm{d}\,\mathrm{vol}_{F_z} . \tag{8.1}$$

Proposition 8.1 (Hessian at the product connection)**.** *For smooth $a, b \in \Omega^1(F_z^{\mathrm{flat}}; \mathfrak{g})$,*

$$\mathrm{Hess}_{A_0} \mathrm{YM}_T(a, b) = g_{\mathrm{YM}}^{-2} \langle \mathrm{d}a, \mathrm{d}b \rangle_{L^2}. \tag{8.2}$$

Let d_1 be the maximal closed exterior derivative on $L^2\Omega^1(F_z^{\mathrm{flat}}; \mathfrak{g})$,

$$\mathrm{Dom}(\mathrm{d}_1) = \{a \in L^2\Omega^1 : \mathrm{d}a \in L^2\Omega^2 \ \textit{distributionally}\}.$$

The closed Hessian form and its nonnegative self-adjoint operator are

$$\mathfrak{t}_T[a] = g_{\mathrm{YM}}^{-2} \|\mathrm{d}_1 a\|_{L^2}^2, \qquad \mathrm{Dom}(\mathfrak{t}_T) = \mathrm{Dom}(\mathrm{d}_1), \tag{8.3}$$

$$\mathcal{L}_T = g_{\mathrm{YM}}^{-2} \mathrm{d}_1^* \mathrm{d}_1, \qquad \mathrm{Dom}(\mathcal{L}_T) = \{a \in \mathrm{Dom}(\mathrm{d}_1) : \mathrm{d}_1 a \in \mathrm{Dom}(\mathrm{d}_1^*)\}. \tag{8.4}$$

Its differential expression on smooth one-forms is $g_{\mathrm{YM}}^{-2} \mathrm{d}^* \mathrm{d}$.

Proof. With $F_{A_0 + a} = \mathrm{d}a + \frac{1}{2}[a \wedge a]$,

$$F_{A_0 + sa + tb} = s\,\mathrm{d}a + t\,\mathrm{d}b + O((|s| + |t|)^2).$$

The mixed derivative of (8.1) is therefore (8.2). The maximal distributional exterior derivative is closed and densely defined, so the first representation theorem for closed nonnegative forms gives (8.3) and (8.4), including the displayed operator domain. □

Gauge-longitudinal one-forms and harmonic one-forms are Hessian-zero directions. Remove both by defining

$$\mathscr{C}_T = \{a \in L^2\Omega^1(F_z^{\mathrm{flat}}; \mathfrak{g}) : \mathrm{d}^* a = 0 \text{ distributionally}, \ a \perp \mathcal{H}^1(F_z^{\mathrm{flat}}; \mathfrak{g})\}. \tag{8.5}$$

This closed subspace reduces $\mathcal{L}_T$. On it the form domain is $H^1\Omega^1 \cap \mathscr{C}_T$ and the operator domain is $H^2\Omega^1 \cap \mathscr{C}_T$: on the coclosed slice the graph norm of d_1, together with the L^2 norm, is the Hodge graph norm, so the claims follow from the Gaffney estimate and elliptic regularity on the compact flat torus.

For $k \in \mathbb{Z}^4$ put

$$\xi_{T,k} = \mathsf{P}_z^{-\mathsf{T}} k, \qquad \chi_{T,k}([x]) = e^{2\pi i \langle \xi_{T,k}, x \rangle}.$$

If $\varepsilon(\xi)$ denotes exterior multiplication by a covector and $\iota_{\xi^\sharp}$ contraction by its Euclidean dual vector, direct differentiation gives

$$\mathrm{d}(\chi_{T,k}\alpha) = 2\pi i \chi_{T,k}\varepsilon(\xi_{T,k})\alpha, \tag{8.6}$$

$$\mathrm{d}^*(\chi_{T,k}\alpha) = -2\pi i \chi_{T,k}\iota_{\xi^\sharp_{T,k}}\alpha, \tag{8.7}$$

$$\mathrm{d}^*\mathrm{d}(\chi_{T,k}\alpha) = 4\pi^2\chi_{T,k}\big(|\xi_{T,k}|^2\alpha - \varepsilon(\xi_{T,k})\iota_{\xi^\sharp_{T,k}}\alpha\big). \tag{8.8}$$

Theorem 8.2 (Exact reduced continuum Hessian gap)**.** *For $k \neq 0$, the Coulomb polarization space*

$$\mathcal{T}_{T,k} = \ker \iota_{\xi^\sharp_{T,k}} \subset (\mathbb{R}^4)^*_{\mathbb{C}} \tag{8.9}$$

has complex dimension three. On $\mathcal{T}_{T,k}\otimes\mathfrak{g}_{\mathbb{C}}$ the Hessian eigenvalue is

$$\lambda_T^{\mathrm{YM}}(k) = \frac{4\pi^2}{g_{\mathrm{YM}}^2}|\mathsf{P}_z^{-\mathsf{T}}k|^2 \tag{8.10}$$

with complex multiplicity $3d_G$ on that signed Fourier fibre. If $K_T(\lambda) = \{k \neq 0 : \lambda_T^{\mathrm{YM}}(k) = \lambda\}$, then the total complex multiplicity, and the dimension of the real eigenspace, is $3d_G\#K_T(\lambda)$.

For all sufficiently large T, the first eigenvalue on $\mathscr{C}_T$ is

$$\gamma_T = \frac{4\pi^2}{g_{\mathrm{YM}}^2}\min\left\{\frac{T^2+m^2}{\mathcal{D}_T^2}, \frac{(T-B_R)^2+36m^2}{\mathcal{D}_T^2}\right\}, \tag{8.11}$$

and

$$\boxed{\gamma_T = \frac{4\pi^2}{g_{\mathrm{YM}}^2T^2}\left(1 + \frac{2\min\{b_{\mathrm{I},0},0\}}{T} + O(T^{-2})\right).} \tag{8.12}$$

If $I_T \subset \{\gamma, u\}$ records which of the two exact quantities in (8.11) attain the minimum, the real multiplicity of γ_T is exactly $6d_G\#I_T$.

Proof. For $k \neq 0$, invertibility of P_z gives $\xi_{T,k} \neq 0$. Contraction by $\xi^\sharp_{T,k}$ has rank one on complexified one-forms, so (8.9) has dimension three. On its kernel the second term of (8.8) vanishes, proving (8.10). Fourier completeness gives the whole reduced spectrum. Complex conjugation exchanges k and $-k$; the reality condition on their amplitudes shows that each conjugate pair contributes $6d_G$ real dimensions, proving the multiplicity statement. The zero Fourier mode is precisely the harmonic sector already removed in (8.5).

Minimizing (8.10) is exactly the shortest-dual-vector problem solved by Theorem 3.1. Equations (3.1) and (3.6) prove the exact gap, its expansion, and the bottom multiplicity. □

In lattice-coordinate coefficients a_y, the exact Coulomb projection at mode k is

$$k^{\mathsf{T}}(\mathsf{P}_z^{\mathsf{T}}\mathsf{P}_z)^{-1}a_y = 0, \qquad \Pi_{T,k}a_y = a_y - k\frac{k^{\mathsf{T}}(\mathsf{P}_z^{\mathsf{T}}\mathsf{P}_z)^{-1}a_y}{k^{\mathsf{T}}(\mathsf{P}_z^{\mathsf{T}}\mathsf{P}_z)^{-1}k}. \tag{8.13}$$

Thus Coulomb reduction uses the full physical covector metric, including all cross terms; it is not obtained by deleting a coordinate component.

8.2 Finite cubical cochains and simultaneous cusp–mesh limits

Fix $N \geq 2$ and write $\Lambda_N = (\mathbb{Z}/N\mathbb{Z})^4$. Its vertex n represents the point $\mathsf{P}_z(n/N)$ in the flat torus. Let $E = (\mathbb{R}^4)^*$ with its Euclidean inner product, let $\mathbf{e}_j$ be the standard coordinate columns in lattice-coordinate space, and define the physical coframe

$$\vartheta_T^j = \mathsf{P}_z^{-\mathsf{T}} \mathbf{e}_j \in E, \qquad 1 \leq j \leq 4. \tag{8.14}$$

For $0 \leq p \leq 4$, set

$$C_N^p(T; \mathfrak{g}_{\mathbb{C}}) = \mathrm{Map}(\Lambda_N, \bigwedge^p E_{\mathbb{C}} \otimes \mathfrak{g}_{\mathbb{C}}) \tag{8.15}$$

and equip it with

$$\langle a, b \rangle_{N,p,T} = \frac{\mathcal{D}_T}{N^4} \sum_{n \in \Lambda_N} \langle a(n), b(n) \rangle_{\wedge^p E \otimes \mathfrak{g}_{\mathbb{C}}}. \tag{8.16}$$

This includes the exact Riemann-sum weight. If $G_T = \mathsf{P}_z^{\mathsf{T}} \mathsf{P}_z$, then

$$\left\langle \vartheta_T^{i_1} \wedge \cdots \wedge \vartheta_T^{i_p}, \vartheta_T^{j_1} \wedge \cdots \wedge \vartheta_T^{j_p} \right\rangle = \det\big((G_T^{-1})_{i_a j_b}\big)_{a,b=1}^p. \tag{8.17}$$

Indeed, $\langle \vartheta_T^i, \vartheta_T^j \rangle = \mathbf{e}_i^{\mathsf{T}} \mathsf{P}_z^{-1} \mathsf{P}_z^{-\mathsf{T}} \mathbf{e}_j = (G_T^{-1})_{ij}$, and the inner product induced on an exterior power is the determinant of the matrix of pairings.

Let $(\mathsf{T}_j a)(n) = a(n + \mathbf{e}_j)$ and put

$$D_{N,j}^+ = N(\mathsf{T}_j - \mathrm{id}), \qquad (D_{N,j}^- a)(n) = N(a(n) - a(n - \mathbf{e}_j)). \tag{8.18}$$

Define

$$\mathrm{d}_N a = \sum_{j=1}^4 \varepsilon(\vartheta_T^j) D_{N,j}^+ a. \tag{8.19}$$

This convention stores differential-form coefficients. If $c_I(n)$ instead stores the integral over a positively oriented p-cell, then $a_I(n) = N^p c_I(n)$ carries the unscaled cellular coboundary $\mathsf{T}_j - \mathrm{id}$, together with the inner product

$$\langle c, c' \rangle_{N,p,T}^{\mathrm{int}} = \mathcal{D}_T N^{2p-4} \sum_{n \in \Lambda_N} \langle c(n), c'(n) \rangle, \tag{8.20}$$

isometrically to (8.19) and (8.16). Thus both conventions give exactly the same Hessian symbols and eigenvalues. Explicitly, $\mathcal{D}_T N^{2p-4} \sum |c|^2 = \mathcal{D}_T N^{-4} \sum |N^p c|^2$, and multiplying the unscaled coboundary of c by N^{p+1} gives exactly $\mathrm{d}_N(N^p c)$. The differences commute while exterior multiplications anticommute, so $\mathrm{d}_N^2 = 0$. Translation is unitary for (8.16), and the adjoint of exterior multiplication by ϑ_T^j is contraction by $(\vartheta_T^j)^\sharp$. Hence

$$\mathrm{d}_N^* = -\sum_{j=1}^4 \iota_{(\vartheta_T^j)^\sharp} D_{N,j}^-. \tag{8.21}$$

For $k \in \Lambda_N$, put

$$\chi_{N,k}(n) = e^{2\pi i k \cdot n/N}, \qquad \mathfrak{d}_N(k)_j = N(e^{2\pi i k_j/N} - 1), \tag{8.22}$$

$$\zeta_{N,T}(k) = \sum_j \mathfrak{d}_N(k)_j \vartheta_T^j = \mathsf{P}_z^{-\mathsf{T}} \mathfrak{d}_N(k). \tag{8.23}$$

Then, for every constant exterior amplitude α,

$$\mathrm{d}_N(\chi_{N,k}\alpha) = \chi_{N,k}\varepsilon(\zeta_{N,T}(k))\alpha, \tag{8.24}$$

$$\mathrm{d}_N^*(\chi_{N,k}\alpha) = \chi_{N,k}\iota_{\overline{\zeta_{N,T}(k)}^\sharp}\alpha, \tag{8.25}$$

$$(\mathrm{d}_N\mathrm{d}_N^* + \mathrm{d}_N^*\mathrm{d}_N)(\chi_{N,k}\alpha) = |\zeta_{N,T}(k)|^2\chi_{N,k}\alpha. \tag{8.26}$$

The last identity is the Hermitian exterior-algebra relation $\iota_{\overline{\zeta}^\sharp}\varepsilon(\zeta) + \varepsilon(\zeta)\iota_{\overline{\zeta}^\sharp} = |\zeta|^2\mathrm{id}$. It also proves that the harmonic p-cochains are exactly the constant cochains, of complex dimension $\binom{4}{p}d_G$. Indeed, $\mathfrak{d}_N(k) = 0$ precisely for $k = 0$ in Λ_N, and P_z is invertible, so the Hodge symbol vanishes precisely on the zero Fourier mode.

Define the finite reduced Coulomb slice and Hessian by

$$\mathscr{C}_{N,T} = \ker(\mathrm{d}_N^* : C_N^1 \to C_N^0) \cap (\ker \Delta_{N,T}^{(1)})^\perp, \qquad \Delta_{N,T}^{(1)} = \mathrm{d}_N\mathrm{d}_N^* + \mathrm{d}_N^*\mathrm{d}_N, \tag{8.27}$$

$$\mathcal{L}_{N,T} = g_{\mathrm{YM}}^{-2}\mathrm{d}_N^*\mathrm{d}_N. \tag{8.28}$$

Equivalently, remove $k = 0$ and at each $k \neq 0$ restrict the one-form amplitude to $\ker \iota_{\overline{\zeta_{N,T}(k)}^\sharp}$, a complex three-dimensional space. The exact projection is

$$\Pi_{N,T,k}\alpha = \alpha - \zeta_{N,T}(k)\frac{\iota_{\overline{\zeta_{N,T}(k)}^\sharp}\alpha}{|\zeta_{N,T}(k)|^2}. \tag{8.29}$$

Theorem 8.3 (Finite Coulomb spectrum and every mesh regime)**.** *On the nonzero k-Fourier fibre of $\mathscr{C}_{N,T}$, $\mathcal{L}_{N,T}$ has the single eigenvalue*

$$\lambda_{N,T}^{\mathrm{YM}}(k) = g_{\mathrm{YM}}^{-2}|\mathsf{P}_z^{-\mathsf{T}}\mathfrak{d}_N(k)|^2 = g_{\mathrm{YM}}^{-2}\mathfrak{d}_N(k)^* G_T^{-1}\mathfrak{d}_N(k), \tag{8.30}$$

with complex multiplicity $3d_G$. Put

$$\rho_N = 4N^2\sin^2(\pi/N). \tag{8.31}$$

For all sufficiently large T, uniformly in $N \geq 2$, the first eigenvalue is

$$\gamma_{N,T} = \min\{\lambda_{N,T}^{\mathrm{YM}}(\mathbf{e}_\gamma), \lambda_{N,T}^{\mathrm{YM}}(\mathbf{e}_u)\} \tag{8.32}$$

$$= \frac{\rho_N}{g_{\mathrm{YM}}^2 T^2}\left(1 + \frac{2\min\{b_{\mathrm{I},0}, 0\}}{T} + O(T^{-2})\right). \tag{8.33}$$

If $I_{N,T} \subset \{\gamma, u\}$ records the minimizing coordinate modes and $\eta_N = 1$ for $N = 2$, $\eta_N = 2$ for $N > 2$, its real multiplicity is

$$3d_G\eta_N \# I_{N,T}. \tag{8.34}$$

The following conclusions hold:

(i) For fixed N, $T^2\gamma_{N,T} \to \rho_N/g_{\mathrm{YM}}^2$ and $\gamma_{N,T} \to 0$.

(ii) For every integer-valued $N(T) \geq 2$, with no growth assumption, there are constants $0 < c < C < \infty$ independent of $N(T)$ such that

$$\frac{c}{T^2} \leq \gamma_{N(T),T} \leq \frac{C}{T^2} \tag{8.35}$$

for all large T.

(iii) If $N(T) \to \infty$, then

$$\gamma_{N(T),T} = \frac{4\pi^2}{g_{\mathrm{YM}}^2 T^2}\big(1 + O(T^{-1}) + O(N(T)^{-2})\big). \tag{8.36}$$

(iv) The largest physical elementary edge is

$$\max_j \frac{\|\mathsf{P}_z \mathbf{e}_j\|}{N(T)} = \frac{T}{N(T)}(1 + O(T^{-1})). \tag{8.37}$$

Thus the varying continuum torus is resolved if and only if $N(T)/T \to \infty$. This refinement condition is not needed for the algebraic soft-mode conclusion in (ii).

Proof. The symbol identities preceding the theorem and the exterior-algebra identity show that on the transverse space $\mathrm{d}_N^* \mathrm{d}_N = |\zeta_{N,T}(k)|^2 \mathrm{id}$, proving (8.30) and its multiplicity. Let $\sigma_N = 2N \sin(\pi/N)$, so $\rho_N = \sigma_N^2$. The least nonzero modulus of a coordinate of $\mathfrak{d}_N(k)$ is σ_N, attained precisely at frequency ± 1 (with the two signs equal when $N = 2$). A frequency supported in the first two coordinates but not equal to $\pm\mathbf{e}_\gamma$ or $\pm\mathbf{e}_u$ has squared coordinate norm at least $2\sigma_N^2$. Indeed, either both coordinates are nonzero, or the single frequency is not ± 1; in the latter case $\sin(2\pi/N)/\sin(\pi/N) = 2\cos(\pi/N) \geq \sqrt{2}$ for $N \geq 4$, and $N = 2, 3$ are immediate.

On complex two-vectors $z = (z_1, z_2)$, the relevant soft block is

$$\frac{1}{\mathcal{D}_T} \begin{pmatrix} m & T - B_R \\ -T & 6m \end{pmatrix} z.$$

After multiplication by T, its singular values are $1 + O(T^{-1})$, uniformly on complex vectors, by (3.10). The factor $\sqrt{2}$ therefore separates every other soft-plane frequency from the two coordinate frequencies uniformly in N for large T. For a general $z \in \mathbb{C}^4$, solving $\mathsf{P}_z^{\mathsf{T}} x = z$ shows that the first and third components of x are z_3, z_4. Thus a frequency with nonzero third or fourth coordinate has image norm at least σ_N, whereas either coordinate candidate is $O(\sigma_N/T)$. This proves the exact minimizing list and (8.32); the signed-frequency reality condition proves (8.34). Multiplying (3.3) and (3.4) by ρ_N/g_{YM}^2 proves (8.33).

The elementary sine bounds give $16 \leq \rho_N \leq 4\pi^2$, proving (i) and (ii); $\rho_N = 4\pi^2(1 + O(N^{-2}))$ proves (iii). Finally, the first two columns of P_z have length $T(1 + O(T^{-1}))$, while the last two have length one. Division by $N(T)$ proves (iv). □

Remark 8.4 (Classical linear scope). For each fixed T, the reduced continuum operator is positively gapped; for each fixed (T, N), so is the reduced finite operator. What fails is a positive bound uniform as $T \to \infty$. Before reduction, gauge-longitudinal directions and constant harmonic one-forms are separate Hessian-zero directions. The spaces (8.15) are linear Lie-algebra-valued cochains, not compact-group link variables, and (8.28) is not a Wilson plaquette action. Neither operator contains the nonlinear interaction, a gauge-invariant quantum measure, renormalization, confinement, or a Hamiltonian reconstruction.

9 Full-torus integral magnetic flux

The elliptic quotient studied in the next section is not required in order to define a magnetic Laplacian on the full marked four-torus. We now retain the original lattice, metric, constants, and flux class and give the exact self-adjoint operator. The result is fibrewise along the marked cusp; it does not assert that the chosen class is fixed by the full monodromy.

Use angular coordinates $u \in \mathbb{R}^4/\mathbb{Z}^4$, so that the Euclidean covering coordinate is $x = \mathsf{P}_z u$ and the metric matrix is

$$G_T = \mathsf{P}_z^{\mathsf{T}} \mathsf{P}_z, \qquad \sqrt{\det G_T} = |\det \mathsf{P}_z| = \mathcal{D}_T. \tag{9.1}$$

Fix an integral alternating form

$$\mathsf{E} = \sum_{1\le j<k\le 4} E_{jk}\,\mathrm{d}u_j \wedge \mathrm{d}u_k \in \bigwedge^2 (\mathbb{Z}^4)^*, \qquad E_{jk} \in \mathbb{Z}, \tag{9.2}$$

and denote its skew matrix by the same letter. For $n \in \mathbb{Z}^4$, set

$$J_n^{\mathsf{E}}(u) = \exp\left(-2\pi i \sum_{j<k} E_{jk} n_j u_k\right). \tag{9.3}$$

Since $\sum_{j<k} E_{jk} m_j n_k \in \mathbb{Z}$,

$$J_m^{\mathsf{E}}(u+n) J_n^{\mathsf{E}}(u) = J_{m+n}^{\mathsf{E}}(u).$$

Thus (9.3) defines a Hermitian line bundle $\mathcal{L}_{\mathsf{E},T}$ on F_z^{flat}. The connection

$$\nabla^{\mathsf{E}} = \mathrm{d} + 2\pi i \sum_{j<k} E_{jk} u_j\,\mathrm{d}u_k \tag{9.4}$$

descends because $J_n^{-1}\mathrm{d}J_n = -2\pi i \sum_{j<k} E_{jk} n_j \mathrm{d}u_k$, and its curvature is

$$F_{\nabla^{\mathsf{E}}} = 2\pi i\,\mathsf{E}. \tag{9.5}$$

The Hilbert space is concretely

$$\mathcal{H}_{\mathsf{E},T} = \{f \in L^2_{\mathrm{loc}}(\mathbb{R}^4) : f(u+n) = J_n^{\mathsf{E}}(u) f(u) \text{ for every } n \in \mathbb{Z}^4\}, \tag{9.6}$$

$$\|f\|^2 = \int_{[0,1]^4} |f(u)|^2 \mathcal{D}_T\,\mathrm{d}^4 u.$$

On smooth functions with the same automorphy, define

$$q_{\mathsf{E},T}[f] = \int_{[0,1]^4} (G_T^{-1})^{jk} \overline{\nabla_j^{\mathsf{E}} f}\, \nabla_k^{\mathsf{E}} f\, \mathcal{D}_T\,\mathrm{d}^4 u. \tag{9.7}$$

Theorem 9.1 (Full-torus magnetic oscillator spectrum)**.** *The form (9.7) is nonnegative and closable, with closed form domain $H^1(F_z^{\mathrm{flat}}; \mathcal{L}_{\mathsf{E},T})$. Its Friedrichs operator is*

$$\Delta_{\mathsf{E},T} = (\nabla^{\mathsf{E}})^* \nabla^{\mathsf{E}}, \qquad \mathrm{Dom}(\Delta_{\mathsf{E},T}) = H^2(F_z^{\mathrm{flat}}; \mathcal{L}_{\mathsf{E},T}),$$

and has compact resolvent.

Assume E is nondegenerate. Define $b_1, b_2 > 0$ by

$$\mathrm{Spec}\Big(G_T^{-1/2}(2\pi\mathsf{E})G_T^{-1/2}\Big) = \{ib_1, -ib_1, ib_2, -ib_2\}. \tag{9.8}$$

Then, as a set with finite multiplicities,

$$\mathrm{Spec}(\Delta_{\mathsf{E},T}) = \{(2n_1+1)b_1 + (2n_2+1)b_2 : n_1, n_2 \in \mathbb{Z}_{\ge 0}\}. \tag{9.9}$$

In particular, the spacing from the ground level to the first distinct level is

$$\mathrm{gap}(\Delta_{\mathsf{E},T}) = 2\min\{b_1, b_2\}. \tag{9.10}$$

No multiplicity formula is needed for, or asserted in, these identities.

Proof. In local unitary trivializations, the integrand in (9.7) is the squared norm of a first-order unitary covariant derivative. A finite trivializing cover and a partition of unity show that $\|f\|^2 + q_{\mathsf{E},T}[f]$ is equivalent to the usual local H^1 norm. Its completion is therefore $H^1(F_z^{\mathrm{flat}}; \mathcal{L}_{\mathsf{E},T})$. The representation theorem for closed nonnegative forms gives the Friedrichs operator. Its principal symbol is $G_T^{-1}(\xi, \xi)$, so it is elliptic; elliptic regularity on the compact boundaryless torus identifies its domain with H^2, and the compact inclusion $H^1 \hookrightarrow L^2$ gives compact resolvent.

The real skew-normal-form theorem gives a constant orthonormal frame $v_1, \dots, v_4$ on the Euclidean cover in which $2\pi\mathsf{E}$ is

$$b_1 v^1 \wedge v^2 + b_2 v^3 \wedge v^4,$$

after reversing v^2 or v^4 if necessary. Let ∇_a be covariant differentiation along v_a. On the common smooth invariant core,

$$[\nabla_1, \nabla_2] = i b_1, \qquad [\nabla_3, \nabla_4] = i b_2,$$

and all other commutators vanish. Put

$$C_1 = \nabla_1 - i\nabla_2, \qquad C_2 = \nabla_3 - i\nabla_4, \qquad \Delta_j = -(\nabla_{2j-1}^2 + \nabla_{2j}^2).$$

Direct multiplication, retaining the commutator signs, gives

$$C_j^* C_j = \Delta_j - b_j, \qquad C_j C_j^* = \Delta_j + b_j, \qquad [\Delta_{\mathsf{E},T}, C_j] = -2 b_j C_j. \tag{9.11}$$

Compact resolvent gives a smooth lowest eigenfunction $\psi_0 \neq 0$. If $C_j \psi_0 \neq 0$, the last identity in (9.11) would produce an eigenfunction at a strictly lower eigenvalue. Hence $C_1 \psi_0 = C_2 \psi_0 = 0$, and the first identity gives

$$\Delta_{\mathsf{E},T} \psi_0 = (b_1 + b_2) \psi_0.$$

Repeated application of C_1^* and C_2^* produces nonzero eigenfunctions at all values displayed in (9.9); their nonvanishing follows from $C_j C_j^* = C_j^* C_j + 2 b_j$. Conversely, for an eigenfunction ψ of eigenvalue $\lambda > b_1 + b_2$,

$$(\lambda - b_1 - b_2)\psi = C_1^* C_1 \psi + C_2^* C_2 \psi.$$

Each nonzero $C_j \psi$ lies at the strictly smaller eigenvalue $\lambda - 2 b_j$. Induction over the finite set of eigenvalues below λ therefore expresses every eigenspace in raised ground states and excludes all other spectral values. Raising once in a least-frequency plane proves (9.10). □

Corollary 9.2 (Pfaffian product and cusp gap bound)**.** *Let*

$$d := |\operatorname{Pf}(\mathsf{E})| = |E_{12}E_{34} - E_{13}E_{24} + E_{14}E_{23}| > 0. \tag{9.12}$$

Then

$$b_1 b_2 = \frac{4\pi^2 d}{\mathcal{D}_T}, \tag{9.13}$$

and

$$\boxed{\operatorname{gap}(\Delta_{\mathsf{E},T}) \leq 4\pi \sqrt{\frac{d}{\mathcal{D}_T}} = \frac{4\pi\sqrt{d}}{T}\bigl(1 + O(T^{-1})\bigr).} \tag{9.14}$$

Thus the ground-to-first-level spacing tends to zero for every fixed nondegenerate integral flux on the marked cusp fibres.

Proof. The determinant of the skew matrix in (9.8) is both $b_1^2 b_2^2$ and

$$\frac{\det(2\pi \mathsf{E})}{\det G_T} = \frac{(2\pi)^4 \operatorname{Pf}(\mathsf{E})^2}{\mathcal{D}_T^2}.$$

Taking the positive square root proves (9.13). Since $\min(b_1, b_2) \le \sqrt{b_1 b_2}$, Theorem 9.1 gives the inequality. Finally, $\mathcal{D}_T = T^2(1 + O(T^{-1}))$ by (3.12), which gives the final asymptotic. □

Remark 9.3 (Scope of the magnetic result). Theorem 9.1 concerns a scalar magnetic Laplacian on a Hermitian line bundle over each marked flat four-torus. It is not a Yang–Mills mass-gap theorem, and it does not claim that $[\mathsf{E}] \in H^2(F_z; \mathbb{Z})$ is invariant under the monodromy of the global family. It does show that the failure of the particular elliptic quotient below is not an obstruction to a fibrewise full-torus magnetic gap-closing sequence.

10 Canonical radial notation for the current operator modules

The earlier sections use R for the logarithmic cusp height and $T = \operatorname{Im}\tau$. The current workbench operator modules use the logarithmic height as their independent parameter. In their standalone adaptations below that parameter is consistently denoted by R, so no second meaning is assigned to T. Thus

$$t_c = e^{-2\pi R}, \qquad s = iR, \qquad R \ge R_0, \tag{10.1}$$

and we set

$$\begin{aligned} &\mu_R := \mu(e^{-2\pi R}) = \alpha_R + i m_R, \\ &h_R := h(e^{-2\pi R}) = \eta_R + i H_R, \\ &b_R := b(e^{-2\pi R}) = \xi_R + i B_R, \\ &c_R := \xi_R - \eta_R, \qquad L_R := R + H_R, \qquad q_R := L_R - B_R, \\ &D_R := L_R q_R + 6 m_R^2. \end{aligned} \tag{10.2}$$

All quantities on the right are real. In particular $L_R = \operatorname{Im}\tau_R$, $-q_R = \operatorname{Im}\beta_R$, and $D_R > 0$ is the physical covolume, whereas the determinant function D in (1.2) equals $-D_R/L_R < 0$. Within the imported operator modules below, bare m denotes the limiting constant, not the varying function in (1.2):

$$m := \operatorname{Im}\mu(0) = m_0, \qquad H := \operatorname{Im} h(0), \qquad B := \operatorname{Im} b(0) = b_{\mathrm{I},0}. \tag{10.3}$$

Holomorphy at $t_c = 0$ gives, without identifying any of these three constants,

$$m_R = m + O(e^{-2\pi R}), \qquad H_R = H + O(e^{-2\pi R}), \qquad B_R = B + O(e^{-2\pi R}). \tag{10.4}$$

We retain the ordered real coordinates

$$(z_1, z_2) = (x_1 + i y_1, x_2 + i y_2) \longmapsto (x_1, y_1, x_2, y_2). \tag{10.5}$$

They carry the unrescaled Euclidean metric used throughout the operator modules.

Proposition 10.1 (Exact translation to canonical radial variables)**.** *In the ordered marking* $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$*, the real period matrix is*

$$P_R = \begin{pmatrix} 6\alpha_R & \eta_R & 1 & 0 \\ 6 m_R & L_R & 0 & 0 \\ c_R & \alpha_R & 0 & 1 \\ -q_R & m_R & 0 & 0 \end{pmatrix}. \tag{10.6}$$

Its determinant and positive covolume are

$$\det P_R = -D_R, \qquad \operatorname{covol}(P_R\mathbb{Z}^4) = D_R > 0, \qquad \operatorname{Vol}(F_R) = D_R, \tag{10.7}$$

where $F_R = \mathbb{R}^4/P_R\mathbb{Z}^4$. *The complete inverse transpose is*

$$P_R^{-\mathsf{T}} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ \dfrac{m_R}{D_R} & \dfrac{q_R}{D_R} & \dfrac{-6m_R\alpha_R - q_R\eta_R}{D_R} & \dfrac{-m_Rc_R - q_R\alpha_R}{D_R} \\ 0 & 0 & 0 & 1 \\ -\dfrac{L_R}{D_R} & \dfrac{6m_R}{D_R} & \dfrac{6(L_R\alpha_R - m_R\eta_R)}{D_R} & \dfrac{L_Rc_R - 6m_R\alpha_R}{D_R} \end{pmatrix}. \tag{10.8}$$

For $k = (r, s, n_1, n_2)^{\mathsf{T}} \in \mathbb{Z}^4$, *define*

$$\mathcal{R}_R(k) = r - 6\alpha_R n_1 - c_R n_2, \qquad S_R(k) = s - \eta_R n_1 - \alpha_R n_2. \tag{10.9}$$

Then

$$P_R^{-\mathsf{T}}k = \left(n_1, \frac{m_R\mathcal{R}_R(k) + q_R S_R(k)}{D_R}, n_2, \frac{-L_R\mathcal{R}_R(k) + 6m_R S_R(k)}{D_R}\right). \tag{10.10}$$

The scalar Laplacian defined by the H^1 *Dirichlet form has operator domain* $H^2(F_R)$, *compact resolvent, and eigenvalues*

$$\lambda_R(k) = 4\pi^2\|P_R^{-\mathsf{T}}k\|^2, \qquad k \in \mathbb{Z}^4, \tag{10.11}$$

with Fourier multiplicity.

Finally,

$$D_R = R^2 + (2H - B)R + H(H - B) + 6m^2 + O(Re^{-2\pi R}). \tag{10.12}$$

Proof. The identity

$$Z = s\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \begin{pmatrix} 6\mu & h \\ b - h & \mu \end{pmatrix},$$

which follows from (1.5) and $b = \beta + \tau$, gives (10.6) by taking real and imaginary parts in the fixed coordinate order. Expansion along its last two columns gives

$$\det P_R = -L_Rq_R - 6m_R^2 = -D_R.$$

The sign $D_R > 0$ is exactly the earlier condition $D < 0$, since $D = \operatorname{Im}\beta_R - 6m_R^2/L_R = -D_R/L_R$ and $L_R > 0$.

Multiplying the displayed candidate inverse transpose by P_R^{T} gives the identity matrix entry by entry: the two nontrivial 2×2 minors reduce to $L_Rq_R + 6m_R^2 = D_R$, and the real-part terms cancel by the definitions of c_R and $S_R(k)$. This proves (10.8), and multiplication by k gives (10.10).

The characters $x \mapsto \exp(2\pi i\langle P_R^{-\mathsf{T}}k, x\rangle)$ form the complete Fourier basis of the compact quotient. Applying the nonnegative Euclidean Laplacian gives (10.11); closed-form theory and elliptic regularity give the stated form and operator domains, and Rellich compactness gives compact resolvent. Finally, substituting $L_R = R + H_R$, $q_R = R + H_R - B_R$, and the three holomorphic expansions into $D_R = L_Rq_R + 6m_R^2$ gives the last formula with every constant and sign retained. □

11 Cubical holonomy and finite gauge systems

This section records three constructions on the regular cusp fibres while keeping two independent integers separate throughout: M_{cov} is the index used to enlarge two periods of a fibre, whereas N_{mesh} is the number of cubical subdivisions in each coordinate direction. In particular, no mesh refinement is identified with a change of covering space.

Let

$$P_R = \begin{pmatrix} 6\operatorname{Re}\mu_R & \operatorname{Re}\tau_R & 1 & 0 \\ 6\operatorname{Im}\mu_R & \operatorname{Im}\tau_R & 0 & 0 \\ \operatorname{Re}\beta_R & \operatorname{Re}\mu_R & 0 & 1 \\ \operatorname{Im}\beta_R & \operatorname{Im}\mu_R & 0 & 0 \end{pmatrix}, \qquad F_R = \mathbb{R}^4/P_R\mathbb{Z}^4. \tag{11.1}$$

Write

$$L_R = \operatorname{Im}\tau_R, \qquad m_R = \operatorname{Im}\mu_R, \qquad q_R = -\operatorname{Im}\beta_R, \qquad D_R = L_R q_R + 6m_R^2 > 0. \tag{11.2}$$

Expanding the determinant along the last two columns gives

$$\det P_R = L_R \operatorname{Im}\beta_R - 6m_R^2 = -D_R.$$

Along the positive radial cusp the source data give

$$L_R = R + O(1), \qquad q_R = R + O(1), \qquad m_R = O(1), \tag{11.3}$$

and all three real parts occurring in (11.1) are bounded. Equivalently, after ordering physical coordinates as $(x_1, x_3; x_2, x_4)$, the first two columns have block form

$$\begin{pmatrix} \mathsf{A}_R^{\mathrm{blk}} \\ \mathsf{B}_R^{\mathrm{blk}} \end{pmatrix}, \qquad \mathsf{B}_R^{\mathrm{blk}} = \begin{pmatrix} 6m_R & L_R \\ -q_R & m_R \end{pmatrix} = R\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \mathsf{C}_R^{\mathrm{blk}}, \qquad \sup_R \|\mathsf{C}_R^{\mathrm{blk}}\|_{\mathrm{op}} < \infty. \tag{11.4}$$

For $M_{\mathrm{cov}} \geq 1$, put

$$S_{M_{\mathrm{cov}}} = \operatorname{diag}(1, 1, M_{\mathrm{cov}}, M_{\mathrm{cov}}), \qquad P_{R,M_{\mathrm{cov}}} = P_R S_{M_{\mathrm{cov}}}, \tag{11.5}$$

and define the degree-M_{cov}^2 cover

$$\widetilde{F}_{R,M_{\mathrm{cov}}} = \mathbb{R}^4/P_{R,M_{\mathrm{cov}}}\mathbb{Z}^4 \longrightarrow F_R. \tag{11.6}$$

Indeed, $S_{M_{\mathrm{cov}}}\mathbb{Z}^4 \subset \mathbb{Z}^4$ and

$$\mathbb{Z}^4/S_{M_{\mathrm{cov}}}\mathbb{Z}^4 \cong (\mathbb{Z}/M_{\mathrm{cov}}\mathbb{Z})^2,$$

so the identity on $\mathbb{R}^4$ induces the asserted cover and its deck group has order M_{cov}^2. Its Euclidean quotient metric, written on the coordinate torus $\mathbb{R}^4/\mathbb{Z}^4$, has Gram matrix and covolume

$$G_{R,M_{\mathrm{cov}}} = P_{R,M_{\mathrm{cov}}}^{\mathsf{T}} P_{R,M_{\mathrm{cov}}}, \qquad \mathcal{D}_{R,M_{\mathrm{cov}}} = \sqrt{\det G_{R,M_{\mathrm{cov}}}} = M_{\mathrm{cov}}^2 D_R. \tag{11.7}$$

11.1 The transported cubical subdivision and continuum holonomy

Fix an integer $N_{\mathrm{mesh}} \geq 2$, and let $K_{N_{\mathrm{mesh}}}$ be the standard cubical subdivision of $\mathbb{R}^4/\mathbb{Z}^4$ obtained by dividing every coordinate circle into N_{mesh} equal intervals. Its vertex set and positively oriented coordinate edges are

$$V_{N_{\mathrm{mesh}}} = (\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4, \qquad E_{N_{\mathrm{mesh}}} = V_{N_{\mathrm{mesh}}} \times \{1, 2, 3, 4\}. \tag{11.8}$$

Proposition 11.1 (Exact cubical realization and the torsion vertices)**.** *The linear map $P_{R,M_{\mathrm{cov}}}$ induces a diffeomorphism*

$$\Phi_{R,M_{\mathrm{cov}}} : \mathbb{R}^4/\mathbb{Z}^4 \longrightarrow \widetilde{F}_{R,M_{\mathrm{cov}}}, \qquad [y] \longmapsto [P_{R,M_{\mathrm{cov}}} y]. \tag{11.9}$$

Transporting $K_{N_{\mathrm{mesh}}}$ by this diffeomorphism gives an honest cubical cell decomposition of the whole torus. On vertices its restriction is the bijection

$$\begin{aligned} \iota_{R,M_{\mathrm{cov}},N_{\mathrm{mesh}}} : \quad & (\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4 \xrightarrow{\cong} \widetilde{F}_{R,M_{\mathrm{cov}}}[N_{\mathrm{mesh}}], \\ & [n] \longmapsto \left[\frac{1}{N_{\mathrm{mesh}}} P_{R,M_{\mathrm{cov}}} n\right]. \end{aligned} \tag{11.10}$$

Here the codomain is the finite N_{mesh}-torsion subgroup, not the entire torus. The coordinate edge (n, j) is the path

$$\gamma_{n,j}(t) = \left[\frac{1}{N_{\mathrm{mesh}}} P_{R,M_{\mathrm{cov}}}(n + t\mathbf{e}_j)\right], \qquad 0 \le t \le 1. \tag{11.11}$$

More generally, for an ordered subset $I = \{i_1, \dots, i_p\} \subset \{1, 2, 3, 4\}$, the corresponding closed p-cell is parametrized by

$$c_{n,I}(t_1, \dots, t_p) = \left[\frac{1}{N_{\mathrm{mesh}}} P_{R,M_{\mathrm{cov}}} \left(n + \sum_{a=1}^{p} t_a \mathbf{e}_{i_a}\right)\right], \qquad (t_1, \dots, t_p) \in [0, 1]^p. \tag{11.12}$$

Proof. If $y - y' \in \mathbb{Z}^4$, then $P_{R,M_{\mathrm{cov}}}(y - y')$ belongs to the quotient lattice, so (11.9) is well defined. The matrix $P_{R,M_{\mathrm{cov}}}$ is invertible, and its inverse induces the inverse map on the two quotients. Hence it is a diffeomorphism and carries every open cell of the standard subdivision diffeomorphically onto an open cell of the target.

Every point displayed in (11.10) is killed by N_{mesh}. If it is zero, then

$$\frac{P_{R,M_{\mathrm{cov}}} n}{N_{\mathrm{mesh}}} \in P_{R,M_{\mathrm{cov}}} \mathbb{Z}^4.$$

Invertibility gives $n \in N_{\mathrm{mesh}}\mathbb{Z}^4$, proving injectivity. Conversely, if $[x]$ is killed by N_{mesh}, choose a lift $x \in \mathbb{R}^4$. There is $n \in \mathbb{Z}^4$ such that $N_{\mathrm{mesh}} x = P_{R,M_{\mathrm{cov}}} n$, and therefore $[x] = [P_{R,M_{\mathrm{cov}}} n / N_{\mathrm{mesh}}]$. This proves surjectivity. Formula (11.11) is simply the transport of the corresponding coordinate edge; at a wrap-around endpoint the extra vector $P_{R,M_{\mathrm{cov}}} \mathbf{e}_j$ is a period. □

Let G be a compact matrix Lie group and use the trivial principal bundle over $\widetilde{F}_{R,M_{\mathrm{cov}}}$. A connection is represented by $A \in \Omega^1(\widetilde{F}_{R,M_{\mathrm{cov}}}; \mathfrak{g})$. For an oriented path γ, define $U_A(\gamma)$ by

$$\frac{dU}{dt} = -A_{\gamma(t)}(\dot{\gamma}(t)) U(t), \qquad U(0) = 1, \qquad U_A(\gamma) = U(1). \tag{11.13}$$

Proposition 11.2 (Typed continuum-to-link holonomy map)**.** *The formula*

$$\mathrm{Hol}_{R,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(A)_{n,j} = U_A(\gamma_{n,j}) \tag{11.14}$$

defines a map

$$\mathrm{Hol}_{R,M_{\mathrm{cov}},N_{\mathrm{mesh}}} : \Omega^1(\widetilde{F}_{R,M_{\mathrm{cov}}}; \mathfrak{g}) \longrightarrow G^{E_{N_{\mathrm{mesh}}}}. \tag{11.15}$$

For the convention

$$A^g = g^{-1} A g + g^{-1} dg, \qquad g \in C^\infty(\widetilde{F}_{R,M_{\mathrm{cov}}}, G), \tag{11.16}$$

it is equivariant for evaluation of g at the vertices:

$$\mathrm{Hol}(A^g)_{n,j} = g(\gamma_{n,j}(1))^{-1} \,\mathrm{Hol}(A)_{n,j}\, g(\gamma_{n,j}(0)). \tag{11.17}$$

Proof. The smooth matrix ODE (11.13) has a unique solution on every edge, which proves that (11.15) is defined. Let U solve that ODE for A, set $g_t = g(\gamma_{n,j}(t))$, and put $V(t) = g_t^{-1}U(t)g_0$. Direct differentiation gives

$$V'(t) = -(g_t^{-1}A(\dot\gamma_{n,j})g_t + g_t^{-1}g_t')V(t), \qquad V(0) = 1.$$

This is exactly the parallel-transport equation for A^g. Uniqueness and evaluation at $t = 1$ prove (11.17). Notice that the endpoint inverse in this formula is forced by the joint conventions (11.13) and (11.16). □

11.2 An explicit compact-link SU(2) flux

The ordered plaquette convention used below is most naturally written for the inverse-transport link variable

$$\mathcal{U}(A)_{n,j} = \mathrm{Hol}_{R,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(A)^{-1}_{n,j}. \tag{11.18}$$

Equation (11.17) then becomes

$$\mathcal{U}(A^g)_{n,j} = g(\gamma_{n,j}(0))^{-1}\mathcal{U}(A)_{n,j}g(\gamma_{n,j}(1)). \tag{11.19}$$

Thus the product in (11.25) transforms by conjugation at its base vertex. The symbols $U_j(n)$ in the remainder of this subsection use this inverse-transport convention.

Use

$$\mathfrak{su}(2) = \{X \in M_2(\mathbb{C}) : X^* = -X,\ \mathrm{tr}\, X = 0\}, \qquad \langle X, Y\rangle_{\mathfrak{su}(2)} = -\frac{1}{2}\,\mathrm{tr}(XY), \tag{11.20}$$

on real elements, and fix

$$H_{\mathrm{link}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}. \qquad \|H_{\mathrm{link}}\|^2 = 1, \qquad \exp(2\pi H_{\mathrm{link}}) = 1. \tag{11.21}$$

Write $N = N_{\mathrm{mesh}}$ temporarily inside the next displayed link formulas, and represent n_1, n_2 by integers in $\{0, \dots, N-1\}$.

Theorem 11.3 (A periodic compact-link flux and its exact action)**.** *On the oriented edges of K_N, define*

$$U_1(n) = \exp\left(-\frac{2\pi n_2}{N^2}H_{\mathrm{link}}\right), \tag{11.22}$$

$$U_2(n) = \begin{cases} 1, & 0 \le n_2 \le N-2, \\ \exp(2\pi n_1 H_{\mathrm{link}}/N), & n_2 = N-1, \end{cases} \tag{11.23}$$

$$U_3(n) = U_4(n) = 1. \tag{11.24}$$

For

$$U_{ij}(n) = U_i(n)U_j(n+\mathbf{e}_i)U_i(n+\mathbf{e}_j)^{-1}U_j(n)^{-1}, \tag{11.25}$$

one has

$$\begin{aligned} &U_{12}(n) = \exp(2\pi H_{\mathrm{link}}/N^2) && \text{for every } n, \\ &U_{ij}(n) = 1 && \text{for } 1 \le i < j \le 4, \quad (i,j) \ne (1,2). \end{aligned} \tag{11.26}$$

Let Log *be the principal logarithm on the displayed plaquette. It is unambiguous because $0 < 2\pi/N^2 \le \pi/2$. Define the logarithmic curvature two-form and its metric-weighted action*

by

$$\mathcal{F}_N^{\log}(n) = N^2 \operatorname{Log}(U_{12}(n))\, dy^1 \wedge dy^2, \tag{11.27}$$

$$S^{\log}_{R,M_{\mathrm{cov}},N}(U) = \frac{\mathcal{D}_{R,M_{\mathrm{cov}}}}{2g_{\mathrm{YM}}^2 N^4} \sum_{n\in V_N} \|\mathcal{F}_N^{\log}(n)\|^2_{G^{-1}_{R,M_{\mathrm{cov}}}\otimes\mathfrak{su}(2)}. \tag{11.28}$$

Then

$$\mathcal{F}_N^{\log}(n) = 2\pi H_{\mathrm{link}}\, dy^1 \wedge dy^2, \qquad S^{\log}_{R,M_{\mathrm{cov}},N}(U) = \frac{2\pi^2 M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2 D_R}. \tag{11.29}$$

In particular, the native-fibre value $M_{\mathrm{cov}} = 1$ is $2\pi^2/(g_{\mathrm{YM}}^2 D_R)$, independently of the mesh.

There is no canonical scalar Wilson action for a general nonorthogonal metric obtained merely by assigning independent character weights to all coordinate plaquettes. For this configuration, however, the exact continuum-matched $(1,2)$-sector Wilson functional is unambiguous if it is defined explicitly by the following formula. Its ordered plaquette holonomy and real trace are the Wilson lattice-action construction Wilson (1974); the metric coefficient and normalization here are the present definition, and their consequences are proved below rather than imported from that source:

$$S^{W,12}_{R,M_{\mathrm{cov}},N}(U) = \frac{\mathcal{D}_{R,M_{\mathrm{cov}}}\kappa^{12}_{R,M_{\mathrm{cov}}}}{2g_{\mathrm{YM}}^2} \sum_{n\in V_N} \left(2 - \operatorname{Re}\operatorname{tr} U_{12}(n)\right), \tag{11.30}$$

where

$$\kappa^{12}_{R,M_{\mathrm{cov}}} = (G^{-1}_{R,M_{\mathrm{cov}}})_{11}(G^{-1}_{R,M_{\mathrm{cov}}})_{22} - (G^{-1}_{R,M_{\mathrm{cov}}})^2_{12} = D_R^{-2}. \tag{11.31}$$

Its exact value and continuum-mesh limit are

$$S^{W,12}_{R,M_{\mathrm{cov}},N}(U) = \frac{M_{\mathrm{cov}}^2 N^4}{2g_{\mathrm{YM}}^2 D_R}\left(2 - 2\cos\frac{2\pi}{N^2}\right), \tag{11.32}$$

$$= \frac{M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2 D_R}\left(2\pi^2 - \frac{2\pi^4}{3N^4} + \frac{4\pi^6}{45N^8} + O(N^{-12})\right), \tag{11.33}$$

$$\lim_{N\to\infty} S^{W,12}_{R,M_{\mathrm{cov}},N}(U) = \frac{2\pi^2 M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2 D_R}. \tag{11.34}$$

Proof. All links lie in the same diagonal circle subgroup, so they commute. If $0 \le n_2 \le N-2$, the two U_2-terms in (11.25) are trivial and

$$U_{12}(n) = \exp(-2\pi n_2 H_{\mathrm{link}}/N^2)\exp(2\pi(n_2+1)H_{\mathrm{link}}/N^2) = \exp(2\pi H_{\mathrm{link}}/N^2).$$

If $n_2 = N-1$, then $U_1(n+\mathbf{e}_2) = 1$. Moreover,

$$U_2(n+\mathbf{e}_1)U_2(n)^{-1} = \exp(2\pi H_{\mathrm{link}}/N).$$

For $n_1 = N-1$, the exponent computed with representatives differs from $2\pi H_{\mathrm{link}}/N$ by $-2\pi H_{\mathrm{link}}$, whose exponential is one; hence the formula also holds across that wrap boundary. It follows that

$$U_{12}(n) = \exp\left(-\frac{2\pi(N-1)}{N^2}H_{\mathrm{link}}\right)\exp\left(\frac{2\pi}{N}H_{\mathrm{link}}\right) = \exp(2\pi H_{\mathrm{link}}/N^2).$$

The functions U_1, U_2 are independent of n_3, n_4, and $U_3 = U_4 = 1$, proving every remaining assertion in (11.26). Reversing the orientation gives $U_{21}(n) = U_{12}(n)^{-1}$; it is not an additional trivial plaquette.

The cover Gram matrix has lower-right principal block $M_{\mathrm{cov}}^2 I_2$. Jacobi's identity for complementary minors therefore gives

$$\kappa^{12}_{R,M_{\mathrm{cov}}} = \frac{\det(M_{\mathrm{cov}}^2 I_2)}{\det G_{R,M_{\mathrm{cov}}}} = \frac{M_{\mathrm{cov}}^4}{M_{\mathrm{cov}}^4 D_R^2} = D_R^{-2},$$

which proves (11.31). By (11.21) and the principal-log choice, $N^2 \operatorname{Log} U_{12} = 2\pi H_{\mathrm{link}}$. Hence every one of the N^4 summands in (11.28) equals $4\pi^2 D_R^{-2}$. Substitution of $\mathcal{D}_{R,M_{\mathrm{cov}}} = M_{\mathrm{cov}}^2 D_R$ proves (11.29).

Finally, H_{link} has eigenvalues $\pm i$, so

$$\operatorname{Re}\operatorname{tr}\exp(2\pi H_{\mathrm{link}}/N^2) = 2\cos(2\pi/N^2).$$

There are N^4 equal plaquettes. Substitution proves (11.32). Applying

$$1 - \cos x = \frac{x^2}{2} - \frac{x^4}{24} + \frac{x^6}{720} + O(x^8)$$

with $x = 2\pi/N^2$ proves (11.33) and (11.34). □

11.3 Finite Gaussian quantization with independent cover and mesh

Let

$$\vartheta^j_{R,M_{\mathrm{cov}}} = P^{-\mathsf{T}}_{R,M_{\mathrm{cov}}} \mathbf{e}_j \in (\mathbb{R}^4)^*, \qquad 1 \le j \le 4. \tag{11.35}$$

For $0 \le p \le 4$, define the real finite cochain spaces

$$C^p_{N_{\mathrm{mesh}}}(R, M_{\mathrm{cov}}) = \operatorname{Map}\left(V_{N_{\mathrm{mesh}}}, \bigwedge^p (\mathbb{R}^4)^* \otimes \mathfrak{su}(2)\right). \tag{11.36}$$

The exterior powers carry the metric induced by the physical coframe (11.35); explicitly, for increasing multi-indices $I = (i_1, \dots, i_p)$ and $J = (j_1, \dots, j_p)$,

$$\left\langle \vartheta^{i_1} \wedge \cdots \wedge \vartheta^{i_p}, \vartheta^{j_1} \wedge \cdots \wedge \vartheta^{j_p} \right\rangle = \det\left((G^{-1}_{R,M_{\mathrm{cov}}})_{i_a j_b} \right)_{a,b=1}^{p}. \tag{11.37}$$

Together with (11.20), this defines the pointwise tensor-product metric. The cochain inner product is

$$\langle a, b \rangle_p = \frac{\mathcal{D}_{R,M_{\mathrm{cov}}}}{N^4_{\mathrm{mesh}}} \sum_{n \in V_{N_{\mathrm{mesh}}}} \langle a(n), b(n) \rangle. \tag{11.38}$$

Let $\mathsf{T}_j a(n) = a(n + \mathbf{e}_j)$, set

$$D^+_{N_{\mathrm{mesh}},j} = N_{\mathrm{mesh}}(\mathsf{T}_j - 1), \qquad D^-_{N_{\mathrm{mesh}},j} a(n) = N_{\mathrm{mesh}}(a(n) - a(n - \mathbf{e}_j)), \tag{11.39}$$

and define

$$d_{N_{\mathrm{mesh}}} a = \sum_{j=1}^{4} \varepsilon(\vartheta^j_{R,M_{\mathrm{cov}}}) D^+_{N_{\mathrm{mesh}},j} a. \tag{11.40}$$

The commuting differences and anticommuting exterior multiplications give $d^2_{N_{\mathrm{mesh}}} = 0$. Its adjoint is

$$d^*_{N_{\mathrm{mesh}}} = -\sum_{j=1}^{4} \iota_{(\vartheta^j_{R,M_{\mathrm{cov}}})^\sharp} D^-_{N_{\mathrm{mesh}},j}. \tag{11.41}$$

Define the real reduced Coulomb space

$$\mathscr{C}_{R,M_{\mathrm{cov}},N_{\mathrm{mesh}}} = \ker(d^*_{N_{\mathrm{mesh}}} : C^1 \to C^0) \cap \left(\ker \Delta^{(1)}_{N_{\mathrm{mesh}}}\right)^{\perp}, \tag{11.42}$$

$$\Delta^{(1)}_{N_{\mathrm{mesh}}} = d_{N_{\mathrm{mesh}}} d^*_{N_{\mathrm{mesh}}} + d^*_{N_{\mathrm{mesh}}} d_{N_{\mathrm{mesh}}}. \tag{11.43}$$

On this finite-dimensional space put

$$\mathcal{L}_{R,M_{\mathrm{cov}},N_{\mathrm{mesh}}} = g_{\mathrm{YM}}^{-2} d^*_{N_{\mathrm{mesh}}} d_{N_{\mathrm{mesh}}}. \tag{11.44}$$

Proposition 11.4 (Exact finite Hessian spectrum)**.** *For*

$$\mathfrak{d}_{N_{\mathrm{mesh}}}(k)_j = N_{\mathrm{mesh}} \left(e^{2\pi i k_j / N_{\mathrm{mesh}}} - 1\right), \qquad k \in (\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4, \tag{11.45}$$

the positive spectrum of $\mathcal{L}$ is obtained from

$$\lambda_{R,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(k) = \frac{1}{g_{\mathrm{YM}}^2} \left\| P_R^{-\mathsf{T}} S_{M_{\mathrm{cov}}}^{-1} \mathfrak{d}_{N_{\mathrm{mesh}}}(k) \right\|^2, \qquad k \neq 0. \tag{11.46}$$

Each complex Fourier fibre has transverse multiplicity $3 \dim \mathfrak{su}(2) = 9$. In particular,

$$\lambda_{\min}(\mathcal{L}) = \min_{k \neq 0} \lambda_{R,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(k) > 0. \tag{11.47}$$

For $\mathbf{e}_1 = (1,0,0,0)^{\mathsf{T}}$, the exact mode value is

$$\lambda_{R,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(\mathbf{e}_1) = \frac{4N_{\mathrm{mesh}}^2 \sin^2(\pi/N_{\mathrm{mesh}})}{g_{\mathrm{YM}}^2} \frac{L_R^2 + m_R^2}{D_R^2}; \tag{11.48}$$

it is independent of M_{cov}.

Proof. Translation is unitary for (11.38), so $(D_j^+)^* = -D_j^-$. Exterior multiplication by ϑ^j has adjoint contraction by $(\vartheta^j)^\sharp$, proving (11.41). On the Fourier character $\chi_k(n) = e^{2\pi i k\cdot n/N_{\mathrm{mesh}}}$, the derivative symbol is exterior multiplication by

$$\zeta(k) = P_{R,M_{\mathrm{cov}}}^{-\mathsf{T}} \mathfrak{d}_{N_{\mathrm{mesh}}}(k) = P_R^{-\mathsf{T}} S_{M_{\mathrm{cov}}}^{-1} \mathfrak{d}_{N_{\mathrm{mesh}}}(k).$$

The Hermitian exterior-algebra identity

$$\iota_{\overline{\zeta}^\sharp} \varepsilon(\zeta) + \varepsilon(\zeta) \iota_{\overline{\zeta}^\sharp} = \|\zeta\|^2 \,\mathrm{id}$$

shows that on the Coulomb polarization $\ker \iota_{\overline{\zeta}^\sharp} \subset (\mathbb{C}^4)^*$, the operator d^*d is multiplication by $\|\zeta\|^2$. For $k \neq 0$, $\mathfrak{d}_{N_{\mathrm{mesh}}}(k) \neq 0$, so invertibility of $P_{R,M_{\mathrm{cov}}}$ gives $\zeta(k) \neq 0$. The contraction has rank one, and its kernel has dimension three. The zero Fourier fibre consists exactly of constant harmonic cochains and was removed in (11.42). Fourier completeness now proves (11.46)–(11.47).

For $k = \mathbf{e}_1$, the only nonzero component of $\mathfrak{d}_{N_{\mathrm{mesh}}}(k)$ has modulus

$$2N_{\mathrm{mesh}} \sin(\pi/N_{\mathrm{mesh}}),$$

and $S_{M_{\mathrm{cov}}}^{-1} \mathbf{e}_1 = \mathbf{e}_1$. Solving $P_R^{\mathsf{T}} \xi = \mathbf{e}_1$, or directly inverting the lower 2×2 block in (11.1), gives

$$\|P_R^{-\mathsf{T}} \mathbf{e}_1\|^2 = \frac{L_R^2 + m_R^2}{D_R^2}.$$

Substitution proves (11.48). □

Let da denote Lebesgue measure on $\mathscr{C}_{R,M_{\mathrm{cov}},N_{\mathrm{mesh}}}$ induced by (11.38) and define

$$\mathcal{H}^{\mathrm{Gauss}}_{R,M_{\mathrm{cov}},N_{\mathrm{mesh}}} = L^2(\mathscr{C}_{R,M_{\mathrm{cov}},N_{\mathrm{mesh}}}, da). \tag{11.49}$$

Theorem 11.5 (Finite Gaussian oscillator and a resolved cusp sequence)**.** *The Friedrichs realization on* (11.49) *of*

$$\widehat{H}^{\mathrm{Gauss}}_{R,M_{\mathrm{cov}},N_{\mathrm{mesh}}} = -\frac{1}{2}\Delta_{\mathscr{C}} + \frac{1}{2}\langle a, \mathcal{L}_{R,M_{\mathrm{cov}},N_{\mathrm{mesh}}} a\rangle \tag{11.50}$$

has a unique Gaussian ground state and spectral gap

$$\mathrm{gap}(\widehat{H}^{\mathrm{Gauss}}) = \sqrt{\lambda_{\min}(\mathcal{L}_{R,M_{\mathrm{cov}},N_{\mathrm{mesh}}})}. \tag{11.51}$$

Consequently,

$$\mathrm{gap}(\widehat{H}^{\mathrm{Gauss}}) \le \frac{2N_{\mathrm{mesh}}\sin(\pi/N_{\mathrm{mesh}})}{g_{\mathrm{YM}}}\frac{\sqrt{L_R^2+m_R^2}}{D_R}. \tag{11.52}$$

There is an explicit diagonal sequence for which the cover geometry tends locally to $\mathbb{R}^4$, the cubical mesh resolves that geometry, and the Gaussian gap tends to zero. Explicitly, for integers $j \to \infty$, take

$$R_j = j^2, \qquad M_{\mathrm{cov},j} = j, \qquad N_{\mathrm{mesh},j} = j^3. \tag{11.53}$$

Then

$$\begin{aligned} \mathrm{sys}(\widetilde{F}_{R_j,M_{\mathrm{cov},j}}) &\longrightarrow \infty, \\ \max_r \frac{\|P_{R_j,M_{\mathrm{cov},j}}\mathbf{e}_r\|}{N_{\mathrm{mesh},j}} &\longrightarrow 0, \\ \mathrm{gap}(\widehat{H}^{\mathrm{Gauss}}) &\longrightarrow 0, \\ S^{\log}_{R_j,M_{\mathrm{cov},j},N_{\mathrm{mesh},j}}(U) &\longrightarrow 0. \end{aligned} \tag{11.54}$$

Thus the pointed cover tori converge locally isometrically to $\mathbb{R}^4$ while their finite linearized Gaussian systems are simultaneously refined.

Proof. The real finite-dimensional operator $\mathcal{L}$ is positive definite on the reduced Coulomb space. Choose an orthonormal eigenbasis with eigenvalues $0 < \lambda_1 \le \cdots \le \lambda_r$. In the resulting coordinates, (11.50) is the tensor sum

$$\sum_{s=1}^{r}\left(-\frac{1}{2}\frac{\partial^2}{\partial x_s^2} + \frac{1}{2}\lambda_s x_s^2\right).$$

The normalized Hermite basis has eigenvalues $\sum_s (n_s + \frac{1}{2})\sqrt{\lambda_s}$, with $n_s \in \mathbb{Z}_{\ge 0}$. The all-zero multi-index is the unique ground state, and the smallest possible positive excitation is $\sqrt{\lambda_1}$. This proves (11.51); applying the minimum to the explicit $\mathbf{e}_1$-mode in (11.48) proves (11.52).

For completeness, reorder physical coordinates as in (11.4). Every vector of the cover lattice has the form

$$\begin{pmatrix} \mathsf{A}^{\mathrm{blk}}_R n + M_{\mathrm{cov}}\ell \\ \mathsf{B}^{\mathrm{blk}}_R n \end{pmatrix}, \qquad n, \ell \in \mathbb{Z}^2.$$

Let $C_{\mathrm{blk}} = \sup_R \|\mathsf{C}^{\mathrm{blk}}_R\|_{\mathrm{op}}$. If $n \neq 0$, then $\|\mathsf{B}^{\mathrm{blk}}_R n\| \ge (R - C_{\mathrm{blk}})\|n\|$; if $n = 0$ and the lattice vector is nonzero, its norm is at least M_{cov}. Hence

$$\mathrm{sys}(\widetilde{F}_{R,M_{\mathrm{cov}}}) \ge \min\{R - C_{\mathrm{blk}}, M_{\mathrm{cov}}\}. \tag{11.55}$$

This tends to infinity for the first two choices in (11.53). For any fixed $r > 0$, once the systole exceeds $4r$, the quotient map from $\mathbb{R}^4$ is injective and distance-preserving on the radius-r ball: for a nonzero period λ and x, y in that ball,

$$\|x - y + \lambda\| \geq \|\lambda\| - \|x - y\| > 4r - 2r \geq \|x - y\|.$$

This proves the pointed local-isometric $\mathbb{R}^4$ limit.

The first two columns of P_R have norm $O(R)$, while the last two columns of $P_{R,M_{\mathrm{cov}}}$ have norm exactly M_{cov}. Therefore the largest physical edge in the N_{mesh}-subdivision has length $O(\max\{R, M_{\mathrm{cov}}\}/N_{\mathrm{mesh}})$, which is $O(j^{-1})$ for (11.53). Finally, (11.3) gives $\sqrt{L_R^2 + m_R^2}/D_R = R^{-1}(1 + O(R^{-1}))$, while $2N \sin(\pi/N) \leq 2\pi$. The right-hand side of (11.52) is therefore $O(j^{-2})$. Moreover, (11.29) is $O(j^{-2})$, since $M_{\mathrm{cov},j}^2/D_{R_j} = j^2/(j^4 + O(j^2))$. This proves every assertion in (11.54). □

11.4 The exponential map and the exact physical boundary

Let $C^1 = C^1_{N_{\mathrm{mesh}}}(R, M_{\mathrm{cov}})$, regarded only as a finite-dimensional Lie-algebra cochain space, and define componentwise

$$\mathrm{Exp}_{N_{\mathrm{mesh}}} : C^1 \longrightarrow \mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}, \qquad a \longmapsto (\exp(a_e/N_{\mathrm{mesh}}))_e\,. \tag{11.56}$$

Here, if $a(n) = \sum_{j=1}^4 \vartheta^j_{R,M_{\mathrm{cov}}} \otimes a_j(n)$, then $a_e = a_j(n)$ for the oriented edge $e = (n, j)$.

Proposition 11.6 (Local chart, not a global embedding)**.** *The map* (11.56) *is smooth and a local diffeomorphism at zero, but it is not globally injective. Under left trivialization of every tangent space at the identity,*

$$D\,\mathrm{Exp}_{N_{\mathrm{mesh}}}(0)\,a = N_{\mathrm{mesh}}^{-1} a. \tag{11.57}$$

If S_W is any smooth compact-link action for which the identity configuration is critical, then

$$D^2(S_W \circ \mathrm{Exp}_{N_{\mathrm{mesh}}})_0[a, b] = N_{\mathrm{mesh}}^{-2} D^2(S_W)_{\mathbf{1}}[a, b] \tag{11.58}$$

under the same tangent identification. Thus, for a prescribed cochain quadratic form Q,

$$D^2(S_W \circ \mathrm{Exp}_{N_{\mathrm{mesh}}})_0 = Q$$

holds exactly when

$$D^2(S_W)_{\mathbf{1}} = N_{\mathrm{mesh}}^2 Q$$

under the coefficient identification. Both this scaling and the tensor structure must be verified for the chosen compact-link action; neither follows from the exponential map alone.

Proof. For the Lie-group exponential, $D\exp_0$ is the identity. Applying the chain rule to $a_e \mapsto a_e/N_{\mathrm{mesh}}$ gives (11.57), which is invertible, so the inverse function theorem gives a local diffeomorphism. It is not globally injective: a cochain which is zero on every edge except one, where it equals $2\pi N_{\mathrm{mesh}} H_{\mathrm{link}}$, has the same image as the zero cochain by (11.21). Finally, the second-derivative chain rule contains a term involving $DS_W(\mathbf{1})$, which vanishes by criticality; the two remaining first derivatives of Exp each supply the factor N_{mesh}^{-1}. This proves (11.58). □

The Gaussian Hilbert space in (11.49) is not the compact-link Yang–Mills Hilbert space. At this finite lattice, the latter kinematic and gauge-invariant spaces would be

$$\mathcal{H}^{\mathrm{link}}_{N_{\mathrm{mesh}}} = L^2(\mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}, dU), \qquad \mathcal{H}^{\mathrm{inv}}_{N_{\mathrm{mesh}}} = \left(\mathcal{H}^{\mathrm{link}}_{N_{\mathrm{mesh}}}\right)^{\mathrm{SU}(2)^{V_{N_{\mathrm{mesh}}}}}, \tag{11.59}$$

where dU is product normalized Haar measure and the superscript denotes the nonlinear vertex-gauge action. Here "gauge-invariant" refers only to the link-configuration space of this four-dimensional cell complex. Without a chosen time direction or transfer matrix, it is not asserted to be a canonical Hamiltonian physical Hilbert space. Lüscher's construction of a selfadjoint, strictly positive transfer matrix Lüscher (1977) and Menotti–Pelissetto's reflection-positivity theorem for the Wilson action Menotti and Pelissetto (1987) concern the standard Wilson lattice setting after the requisite reflection/time structure has been fixed; they do not supply those omitted choices for the nonorthogonal sector functional (11.30). The Osterwalder–Schrader reconstruction boundary is likewise the one stated in the primary Euclidean axioms papers Osterwalder and Schrader (1973, 1975), not a consequence of a finite Gaussian oscillator calculation. The differences are structural:

1. $\mathcal{H}^{\mathrm{Gauss}}$ is an L^2-space on a noncompact linear Coulomb slice with Lebesgue measure; $\mathcal{H}^{\mathrm{inv}}$ is a gauge-invariant subspace of an L^2-space on compact link variables with Haar measure.

2. The local exponential map is neither a global coordinate chart nor a measure-preserving bijection. No unitary map between the two displayed Hilbert spaces has been constructed.

3. A link configuration U determines the Dirac distribution δ_U, but

$$\delta_U \in \mathcal{D}'(\mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}), \qquad \delta_U \notin L^2(\mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}, dU). \tag{11.60}$$

   Indeed, product Haar measure is nonatomic, so a point mass is singular with respect to it and has no square-integrable density.

4. No nonlinear compact-link Hamiltonian, Gauss-law spectral problem, time direction, reflection-positive transfer matrix, renormalized-scale limit, or Osterwalder–Schrader reconstruction is supplied here.

Accordingly, (11.51) is an exact theorem about the finite quadratic oscillator defined by the Coulomb-restricted operator $g_{\mathrm{YM}}^{-2} d^* d$. It is not an eigenvalue statement about a nonperturbative physical Yang–Mills Hamiltonian, and the small classical action in (11.29) is not by itself a physical-state Rayleigh quotient.

12 Integral magnetic sectors and a reducible $SU(2)$ critical point on the radial cusp

Retain the positive radial flat tori $F_R = \mathbb{R}^4/P_R\mathbb{Z}^4$, their Euclidean quotient metrics, and the real quantities $L_R, q_R, m_R, D_R, H, B, m$ from equations (10.2) and (10.12) and theorem 10.1. In particular,

$$L_R = \operatorname{Im}\tau_R, \qquad q_R = -\operatorname{Im}\beta_R, \qquad D_R = L_R q_R + 6m_R^2 = \mathrm{Vol}(F_R) > 0.$$

The real coordinate order is the already fixed order

$$x = (x_1, y_1, x_2, y_2)^{\mathsf{t}}.$$

12.1 The integral coframe

Let $e_1, \dots, e_4$ be the standard basis of $\mathbb{R}^4$, put $u = P_R^{-1}x$, and define

$$\theta_R^j := \mathrm{d}u_j = (P_R^{-\mathsf{T}} e_j)^{\mathsf{t}} (\mathrm{d}x_1, \mathrm{d}y_1, \mathrm{d}x_2, \mathrm{d}y_2)^{\mathsf{t}}. \tag{12.1}$$

Proposition 12.1 (Integral coframe and the distinguished unit-flux plane)**.** *The forms θ_R^j are the integral coframe dual to the ordered marked lattice basis:*

$$\int_{[0,1] \ni s \mapsto [x + s P_R e_k]} \theta_R^j = \delta_{jk}.$$

The first two members are exactly

$$\theta_R^1 = \frac{m_R}{D_R}\,\mathrm{d}y_1 - \frac{L_R}{D_R}\,\mathrm{d}y_2, \qquad \theta_R^2 = \frac{q_R}{D_R}\,\mathrm{d}y_1 + \frac{6m_R}{D_R}\,\mathrm{d}y_2, \tag{12.2}$$

and hence

$$\theta_R^1 \wedge \theta_R^2 = \frac{1}{D_R}\,\mathrm{d}y_1 \wedge \mathrm{d}y_2, \qquad |\theta_R^1 \wedge \theta_R^2|^2 = \frac{1}{D_R^2}. \tag{12.3}$$

Proof. For $n \in \mathbb{Z}^4$,

$$u(x + P_R n) = u(x) + n,$$

which proves the integral-period assertion. The first two columns of the already-proved matrix $P_R^{-\mathsf{T}}$ in equation (10.8) give equation (12.2). Their wedge coefficient is

$$\frac{6m_R^2 + L_R q_R}{D_R^2} = \frac{1}{D_R}.$$

The coordinate forms $\mathrm{d}y_1, \mathrm{d}y_2$ are orthonormal for the fixed quotient metric, proving equation (12.3). □

12.2 The unit-flux line and the topologically trivial rank-two sum

The following construction is made in the marked coordinates u of (12.1); it therefore applies to the actual skew lattice $P_R\mathbb{Z}^4$, not to a rectangular replacement.

Proposition 12.2 (Explicit factor of automorphy and connection)**.** *For $n \in \mathbb{Z}^4$, set*

$$j_n(u) = \exp(-2\pi i n_1 u_2). \tag{12.4}$$

The quotient

$$\mathscr{L}_R = (\mathbb{R}^4 \times \mathbb{C})/\mathbb{Z}^4, \qquad n \cdot (u, z) = (u + n, j_n(u) z),$$

is a Hermitian line bundle over F_R. The connection on its pullback to $\mathbb{R}^4$,

$$\widetilde{\nabla}^{\mathscr{L}_R} = \mathrm{d} + 2\pi i u_1\,\mathrm{d}u_2, \tag{12.5}$$

descends, and its curvature is

$$F_{\mathscr{L}_R} = 2\pi i\,\mathrm{d}u_1 \wedge \mathrm{d}u_2 = 2\pi i\,\theta_R^1 \wedge \theta_R^2. \tag{12.6}$$

With the unitary Chern–Weil convention $c_1 = [\,iF/(2\pi)\,]$, one has $c_1(\mathscr{L}_R) = -[\theta_R^1 \wedge \theta_R^2]$.

Proof. For $m, n \in \mathbb{Z}^4$,

$$j_m(u+n)j_n(u) = j_{m+n}(u)\exp(-2\pi i m_1 n_2) = j_{m+n}(u).$$

Thus (12.4) is a factor of automorphy. Its values are unitary, so the standard fibre metric descends. If $\mathcal{A} = 2\pi i u_1\,\mathrm{d}u_2$, then

$$\mathcal{A}(u+n) = \mathcal{A}(u) + 2\pi i n_1\,\mathrm{d}u_2, \qquad j_n^{-1}\mathrm{d}j_n = -2\pi i n_1\,\mathrm{d}u_2.$$

Hence $\mathcal{A}(u+n) + j_n^{-1}\mathrm{d}j_n = \mathcal{A}(u)$, which is precisely the descent identity for $\mathrm{d}+\mathcal{A}$ under the stated quotient convention. Taking $\mathrm{d}\mathcal{A}$ gives (12.6). The Chern–Weil statement follows without a sign suppression:

$$\frac{i}{2\pi}F_{\mathscr{L}_R} = -\theta_R^1 \wedge \theta_R^2.$$

□

Let $\mathscr{L}_R^{-1} = \mathscr{L}_R^*$ carry the dual metric and dual connection, and set

$$\mathscr{E}_R = \mathscr{L}_R \oplus \mathscr{L}_R^{-1}. \tag{12.7}$$

Proposition 12.3 (Determinant, c_2, and explicit $SU(2)$-triviality)**.** *The factors of automorphy of $\mathscr{E}_R$ are*

$$G_n(u) = \begin{pmatrix} j_n(u) & 0 \\ 0 & j_n(u)^{-1} \end{pmatrix} \in SU(2).$$

Evaluation gives a canonical parallel determinant trivialization

$$\det \mathscr{E}_R = \mathscr{L}_R \otimes \mathscr{L}_R^{-1} \xrightarrow{\mathrm{ev}} F_R \times \mathbb{C}.$$

Moreover

$$c_1(\mathscr{E}_R) = 0, \qquad c_2(\mathscr{E}_R) = 0 \quad \text{in } H^4(F_R;\mathbb{Z}),$$

and the underlying principal $SU(2)$-bundle is smoothly trivial. The last assertion follows from the explicit nullhomotopy below and does not rely only on the vanishing of a de Rham Chern form.

Proof. The determinant assertion follows pointwise from $j_n j_n^{-1} = 1$. The induced determinant connection has connection form

$$\mathrm{tr}\begin{pmatrix} 2\pi i u_1\,\mathrm{d}u_2 & 0 \\ 0 & -2\pi i u_1\,\mathrm{d}u_2 \end{pmatrix} = 0,$$

so the evaluation trivialization is parallel.

Let $c = c_1(\mathscr{L}_R)$. The integral Whitney identity gives

$$c(\mathscr{E}_R) = (1+c)(1-c) = 1 - c^2.$$

The factor j_n depends only on (u_1, u_2), so $\mathscr{L}_R$ is the pullback of a line bundle L_0 on $\mathbb{R}^2/\mathbb{Z}^2$ under the marked projection $[u_1, u_2, u_3, u_4] \mapsto [u_1, u_2]$. Hence

$$c^2 = \pi_{12}^*(c_1(L_0)^2) = 0$$

integrally because $H^4(\mathbb{T}^2;\mathbb{Z}) = 0$. This proves the two Chern-class identities on the four-torus.

It remains to trivialize the $SU(2)$-bundle itself. On the fundamental square $0 \le u_1, u_2 \le 1$ for the first two marked coordinates, the only nontrivial seam is

$$(0, t, v) \sim (1, t, \gamma(t)v), \qquad \gamma(t) = \begin{pmatrix} e^{-2\pi i t} & 0 \\ 0 & e^{2\pi i t} \end{pmatrix}.$$

For $0 \le r \le 1$, define

$$a(r,t) = (1-r)e^{-2\pi i t} + r, \qquad s(r) = \sin(\pi r), \qquad N(r,t) = \big(|a(r,t)|^2 + s(r)^2\big)^{1/2},$$

and

$$U(r,t) = \frac{1}{N(r,t)} \begin{pmatrix} a(r,t) & -s(r) \\ s(r) & \overline{a(r,t)} \end{pmatrix}. \tag{12.8}$$

If $0 < r < 1$, then $s(r) > 0$; at $r = 0, 1$, one has $|a(r,t)| = 1$. Thus N never vanishes. The two columns in (12.8) are orthonormal and its determinant is one, so $U(r,t) \in SU(2)$. It is smooth and periodic in t, and

$$U(0,t) = \gamma(t), \qquad U(1,t) = I_2.$$

This is an explicit nullhomotopy of the clutching loop. To see directly that it produces a smooth global frame, choose a smooth $\chi : [0,1] \to [0,1]$ which is zero near 0 and one near 1, and put

$$K(\sigma, t) = U(1 - \chi(\sigma), t).$$

Then $K = I_2$ near $\sigma = 0$, $K = \gamma$ near $\sigma = 1$, and $K(\sigma, 0) = K(\sigma, 1)$. The required trivialization, written in the direction from the clutched bundle to the product bundle, is

$$[\sigma, t, v] \longmapsto ([\sigma, t], K(\sigma, t)^{-1} v). \tag{12.9}$$

Indeed, the quotient convention gives $[0, t, v] = [1, t, \gamma(t)v]$, and the two representatives have the same image because

$$K(1,t)^{-1}\gamma(t)v = \gamma(t)^{-1}\gamma(t)v = v = K(0,t)^{-1}v.$$

At the other seam, $K(\sigma,0)^{-1} = K(\sigma,1)^{-1}$. The constant collars used in the definition of χ make the local matrix in equation (12.9) constant after applying the seam transition, so the map and its inverse are smooth across the seam. It is fibrewise unitary and determinant one. Extending it independently of u_3, u_4 and pulling it through the marking gives the required global $SU(2)$-frame of $\mathscr{E}_R$. □

12.3 A non-flat reducible Yang–Mills critical connection

The symbol $H = \operatorname{Im} h(0)$ is already fixed by equation (10.3). For the requested Lie-algebra element, fix instead the unambiguous notation

$$\langle X, Y \rangle_{\mathfrak{su}(2)} = -\frac{1}{2}\operatorname{Tr}(XY), \qquad H_{\mathrm{mag}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}. \tag{12.10}$$

Theorem 12.4 (Reducible critical point and exact action)**.** *The direct-sum connection on $\mathscr{E}_R$ has, in the lifted splitting,*

$$\nabla^R = \mathrm{d} + 2\pi u_1 \, \mathrm{d}u_2 \, H_{\mathrm{mag}}, \qquad F_{\nabla^R} = 2\pi H_{\mathrm{mag}} \, \theta_R^1 \wedge \theta_R^2 = \frac{2\pi}{D_R} H_{\mathrm{mag}} \, \mathrm{d}y_1 \wedge \mathrm{d}y_2. \tag{12.11}$$

After any of the global $SU(2)$-trivializations from theorem 12.3, this is a connection on the trivial principal $SU(2)$-bundle. It is non-flat and satisfies

$$\mathrm{d}^*_{\nabla^R} F_{\nabla^R} = 0.$$

For the normalization

$$\mathrm{YM}_R(\nabla) = \frac{1}{2g^2_{\mathrm{YM}}} \int_{F_R} |F_\nabla|^2 \,\mathrm{d}\,\mathrm{vol}_{F_R}, \tag{12.12}$$

its action is exactly

$$\mathrm{YM}_R(\nabla^R) = \frac{2\pi^2}{g^2_{\mathrm{YM}} D_R}. \tag{12.13}$$

Its first inverse-power correction, in the canonical scalar notation $H = \operatorname{Im} h(0)$, $B = \operatorname{Im} b(0)$, is

$$\mathrm{YM}_R(\nabla^R) = \frac{2\pi^2}{g^2_{\mathrm{YM}} R^2} \left(1 + \frac{B - 2H}{R} + O(R^{-2})\right). \tag{12.14}$$

Proof. The connection matrix is

$$\begin{pmatrix} 2\pi i u_1 \,\mathrm{d}u_2 & 0 \\ 0 & -2\pi i u_1 \,\mathrm{d}u_2 \end{pmatrix} = 2\pi u_1 \,\mathrm{d}u_2 \, H_{\mathrm{mag}},$$

so equation (12.11) follows from theorem 12.2 and equation (12.3). The diagonal endomorphism H_{mag} is a globally defined parallel section of $\operatorname{ad} \mathscr{E}_R$, because every transition G_n commutes with it. The forms θ^j_R are parallel constant forms for the descended flat metric. Thus $\nabla^R F_{\nabla^R} = 0$, which implies $\mathrm{d}^*_{\nabla^R} F_{\nabla^R} = 0$. The curvature is nonzero because $D_R > 0$.

The chosen Lie-algebra normalization gives

$$|H_{\mathrm{mag}}|^2 = -\frac{1}{2} \operatorname{Tr}(H^2_{\mathrm{mag}}) = 1.$$

Equations equations (10.7) and (12.3) therefore give

$$\int_{F_R} |F_{\nabla^R}|^2 \,\mathrm{d}\,\mathrm{vol}_{F_R} = 4\pi^2 D_R \frac{1}{D_R^2} = \frac{4\pi^2}{D_R},$$

which proves equation (12.13). Finally, equation (10.12) and the elementary reciprocal expansion

$$\frac{1}{R^2 + aT + O(1)} = \frac{1}{R^2}\left(1 - \frac{a}{T} + O(R^{-2})\right)$$

with $a = 2H - B$ prove equation (12.14). □

12.4 Constant-curvature magnetic Laplacians for arbitrary integral nondegenerate flux

We next give the Hilbert space, closed form, self-adjoint realization, and oscillator spectrum in a form valid for every nondegenerate integral alternating form on the marked four-torus.

Let

$$\mathsf{E} = \sum_{1 \le j < k \le 4} E_{jk} \,\mathrm{d}u_j \wedge \mathrm{d}u_k \in \bigwedge^2 (\mathbb{Z}^4)^*, \qquad E_{jk} \in \mathbb{Z},$$

and let the same symbol denote its skew matrix. For $n \in \mathbb{Z}^4$, put

$$J_n^{\mathsf{E}}(u) = \exp\left(-2\pi i \sum_{j<k} E_{jk} n_j u_k\right). \tag{12.15}$$

The integer in

$$\sum_{j<k} E_{jk} m_j n_k$$

shows directly that $J_m^{\mathsf{E}}(u+n) J_n^{\mathsf{E}}(u) = J_{m+n}^{\mathsf{E}}(u)$. Thus (12.15) defines a Hermitian line bundle $\mathscr{L}_{\mathsf{E},R}$. On its pullback, set

$$\nabla^{\mathsf{E}} = \mathrm{d} + 2\pi i \sum_{j<k} E_{jk} u_j \, \mathrm{d}u_k. \tag{12.16}$$

Exactly as in theorem 12.2, the connection descends and

$$F_{\nabla^{\mathsf{E}}} = 2\pi i \, \mathsf{E}. \tag{12.17}$$

In u-coordinates the flat metric matrix is

$$G_R = P_R^{\mathsf{T}} P_R, \qquad \sqrt{\det G_R} = D_R. \tag{12.18}$$

Define the Hilbert space concretely by

$$\begin{aligned} \mathcal{H}_{\mathsf{E},R} &= \{\, f \in L^2_{\mathrm{loc}}(\mathbb{R}^4) : f(u+n) = J_n^{\mathsf{E}}(u) f(u) \text{ for every } n \in \mathbb{Z}^4 \}, \\ \|f\|^2 &= \int_{[0,1]^4} |f(u)|^2 \sqrt{\det G_R}\, \mathrm{d}^4 u. \end{aligned} \tag{12.19}$$

Let $\mathcal{C}_{\mathsf{E},R}$ be the smooth functions with the same automorphy. On this dense subspace define

$$q_{\mathsf{E},R}[f] = \int_{[0,1]^4} (G_R^{-1})^{jk} \overline{\nabla_j^{\mathsf{E}} f} \nabla_k^{\mathsf{E}} f \sqrt{\det G_R}\, \mathrm{d}^4 u. \tag{12.20}$$

Theorem 12.5 (Friedrichs realization and oscillator spectrum)**.** *The form* (12.20) *is nonnegative and closable. Its closed form domain is*

$$\mathrm{Dom}\, \overline{q}_{\mathsf{E},R} = H^1(F_R; \mathscr{L}_{\mathsf{E},R}).$$

Its Friedrichs operator is the magnetic Laplacian

$$\Delta_{\mathsf{E},R} = (\nabla^{\mathsf{E}})^* \nabla^{\mathsf{E}}, \qquad \mathrm{Dom}(\Delta_{\mathsf{E},R}) = H^2(F_R; \mathscr{L}_{\mathsf{E},R}),$$

and it has compact resolvent.

Assume now that E *is nondegenerate. Let* $b_1, b_2 > 0$ *be defined by the eigenvalues*

$$\mathrm{Spec}\Big(G_R^{-1/2} (2\pi \mathsf{E}) G_R^{-1/2}\Big) = \{ i b_1, -i b_1, i b_2, -i b_2 \}. \tag{12.21}$$

Then, as a set (with finite multiplicities),

$$\mathrm{Spec}(\Delta_{\mathsf{E},R}) = \{ (2n_1+1) b_1 + (2n_2+1) b_2 : n_1, n_2 \in \mathbb{Z}_{\geq 0} \}. \tag{12.22}$$

In particular, the ground-to-first-distinct-level spacing is

$$\mathrm{gap}(\Delta_{\mathsf{E},R}) = 2 \min(b_1, b_2). \tag{12.23}$$

No multiplicity formula is needed for, or asserted in, these identities.

Proof. In local unitary trivializations, the integrand in (12.20) is the squared norm of a first-order unitary covariant derivative. A finite trivializing cover and a partition of unity show that the form norm $\|f\|^2 + q_{\mathsf{E},R}[f]$ is equivalent to the usual local H^1 norm. Its completion is therefore $H^1(F_R; \mathscr{L}_{\mathsf{E},R})$, proving closability and identifying the closed domain. The representation theorem for closed nonnegative forms gives the Friedrichs operator. Its local principal symbol is $G_R^{-1}(\xi, \xi)$, so it is uniformly elliptic. Standard interior elliptic regularity on the compact manifold without boundary identifies the operator domain with H^2. The compact inclusion $H^1 \hookrightarrow L^2$ implies compact resolvent.

The real skew-normal-form theorem supplies a constant orthonormal frame $v_1, \dots, v_4$ on the Euclidean cover in which the real two-form $2\pi\mathsf{E}$ is

$$b_1 v^1 \wedge v^2 + b_2 v^3 \wedge v^4.$$

Changing the sign of v^2 or v^4, if necessary, makes both frequencies positive. Let ∇_a denote covariant differentiation along v_a. Each ∇_a is formally skew-adjoint on the common invariant core $\mathcal{C}_{\mathsf{E},R}$; the closures used below are the ones determined by the closed magnetic form. On that smooth core one has

$$[\nabla_1, \nabla_2] = i b_1, \qquad [\nabla_3, \nabla_4] = i b_2,$$

and all other commutators vanish. Put

$$C_1 = \nabla_1 - i\nabla_2, \qquad C_2 = \nabla_3 - i\nabla_4, \qquad \Delta_j = -(\nabla_{2j-1}^2 + \nabla_{2j}^2).$$

Direct multiplication, retaining the commutator signs, gives

$$C_j^* C_j = \Delta_j - b_j, \qquad C_j C_j^* = \Delta_j + b_j, \qquad [\Delta_{\mathsf{E},R}, C_j] = -2 b_j C_j. \tag{12.24}$$

Because the resolvent is compact, the lowest eigenvalue has a smooth eigenfunction $\psi_0 \neq 0$. If $C_j \psi_0 \neq 0$, the last identity in (12.24) would make it an eigenfunction with a strictly smaller eigenvalue. Hence $C_1 \psi_0 = C_2 \psi_0 = 0$, and the first identity gives

$$\Delta_{\mathsf{E},R} \psi_0 = (b_1 + b_2) \psi_0.$$

Conversely, repeated application of C_1^*, C_2^* produces nonzero eigenfunctions with eigenvalues $(2n_1 + 1)b_1 + (2n_2 + 1)b_2$. Nonvanishing follows from $C_j C_j^* = C_j^* C_j + 2b_j$.

It remains to exclude other eigenvalues. For an eigenfunction ψ with eigenvalue $\lambda > b_1 + b_2$, the operator identity

$$(\lambda - b_1 - b_2)\psi = C_1^* C_1 \psi + C_2^* C_2 \psi$$

expresses ψ as a sum of raising operators applied to the lower eigenfunctions $C_j \psi$. Induction over the finite set of eigenvalues below λ reduces every eigenfunction to raised ground states. This proves (12.22). Raising once in a least-frequency plane proves (12.23). □

Corollary 12.6 (Frequency product and cusp spacing bound)**.** *Let*

$$d = |\operatorname{Pf}(\mathsf{E})| = |E_{12}E_{34} - E_{13}E_{24} + E_{14}E_{23}| > 0.$$

Then

$$b_1 b_2 = \frac{4\pi^2 d}{D_R}, \tag{12.25}$$

and

$$\begin{aligned} \operatorname{gap}(\Delta_{\mathsf{E},R}) &\leq 4\pi \sqrt{\frac{d}{D_R}} \\ &= \frac{4\pi\sqrt{d}}{R}\left(1 + \frac{B - 2H}{2R} + O(R^{-2})\right). \end{aligned} \tag{12.26}$$

Proof. The determinant of the skew matrix in (12.21) is both $b_1^2 b_2^2$ and

$$\frac{\det(2\pi \mathsf{E})}{\det G_R} = \frac{(2\pi)^4 \operatorname{Pf}(\mathsf{E})^2}{D_R^2}.$$

Taking the positive square root proves (12.25). Since $\min(b_1, b_2) \le \sqrt{b_1 b_2}$, Theorem 12.5 gives (12.26). Expanding the square root of equation (10.12) gives the displayed first correction. □

12.5 A background-gauge negative mode

We finish with the classical background-gauge Jacobi operator at the reducible critical connection. Every sign below uses exactly the inner product and the element H_{mag} fixed in equation (12.10).

Put

$$\mathscr{H}_R^{\mathrm{ad},1} := L^2\Omega^1(F_R; \operatorname{ad}\mathscr{E}_R).$$

Put also

$$\mathscr{H}_R^{\mathrm{ad},0} := L^2\Omega^0(F_R; \operatorname{ad}\mathscr{E}_R), \qquad D_R^{\max} := \mathrm{d}_{\nabla^R}^{*,\max}.$$

For the resolvent argument these real Hilbert spaces are complexified; all operators commute with the real structure, so their restrictions to real sections are unchanged. Here the maximal distributional adjoint is the closed, densely defined operator whose domain and value are specified without an implicit regularity assumption by

$$\begin{aligned} \operatorname{Dom}(D_R^{\max}) := \{a \in \mathscr{H}_R^{\mathrm{ad},1} :\ & \text{there is a } u \in \mathscr{H}_R^{\mathrm{ad},0} \text{ such that} \\ & \langle a, \mathrm{d}_{\nabla^R}\phi\rangle_{L^2} = \langle u, \phi\rangle_{L^2} \text{ for every } \phi \in C^\infty(F_R; \operatorname{ad}\mathscr{E}_R)\}, \\ D_R^{\max} a := u. & \end{aligned} \tag{12.27}$$

The test space is dense, so u is unique. The definition also proves closedness directly: if $a_n \to a$ and $D_R^{\max} a_n \to u$ in L^2, passage to the limit in the displayed testing identity gives $a \in \operatorname{Dom}(D_R^{\max})$ and $D_R^{\max} a = u$. Consequently the background slice

$$\mathscr{S}_R := \ker D_R^{\max} \subset \mathscr{H}_R^{\mathrm{ad},1}. \tag{12.28}$$

is a closed subspace.

The scalar covariant Laplacian used below is fixed as follows. On $C^\infty(F_R; \operatorname{ad}\mathscr{E}_R)$ start with the nonnegative form

$$q_R^{(0)}[\phi, \psi] := \langle \mathrm{d}_{\nabla^R}\phi, \mathrm{d}_{\nabla^R}\psi\rangle_{L^2}.$$

Its closure has domain $H^1\Omega^0(F_R; \operatorname{ad}\mathscr{E}_R)$, and its associated Friedrichs operator is

$$\Delta_{\nabla^R}^{(0)} := \mathrm{d}_{\nabla^R}^* \mathrm{d}_{\nabla^R}, \qquad \operatorname{Dom}(\Delta_{\nabla^R}^{(0)}) = H^2\Omega^0(F_R; \operatorname{ad}\mathscr{E}_R). \tag{12.29}$$

The H^2 domain follows from elliptic regularity on the compact flat torus; in particular, this notation always denotes the nonnegative self-adjoint realization, not merely its differential expression.

For an adjoint-valued one-form a, define the smooth bundle endomorphism

$$(V_R a)_\nu := -2 \sum_{\mu=1}^{4} [F_{\nu\mu}, a_\mu].$$

It is symmetric: after interchanging μ, ν, the identities $F_{\mu\nu} = -F_{\nu\mu}$ and skew-adjointness of $\mathrm{ad}(F_{\nu\mu})$ give $\langle V_R a, b\rangle = \langle a, V_R b\rangle$. On the full one-form Hilbert space define

$$\mathfrak{j}_R^{\mathrm{gf}}[a,b] := \langle \nabla^R a, \nabla^R b\rangle_{L^2} + \langle V_R a, b\rangle_{L^2}, \qquad \mathrm{Dom}(\mathfrak{j}_R^{\mathrm{gf}}) = H^1\Omega^1(F_R; \mathrm{ad}\,\mathscr{E}_R). \tag{12.30}$$

Because V_R is a bounded symmetric endomorphism on the compact torus, this form is closed and bounded below. Its associated self-adjoint operator $\mathcal{J}_R^{\mathrm{gf}}$ has

$$\mathrm{Dom}(\mathcal{J}_R^{\mathrm{gf}}) = H^2\Omega^1(F_R; \mathrm{ad}\,\mathscr{E}_R) \tag{12.31}$$

and differential expression

$$(\mathcal{J}_R^{\mathrm{gf}} a)_\nu = -\sum_{\mu=1}^{4} \nabla_{e_\mu}\nabla_{e_\mu} a_\nu - 2\sum_{\mu=1}^{4}[F_{\nu\mu}, a_\mu].$$

Proposition 12.7 (Intertwining and the reducing Coulomb slice)**.** *At the Yang–Mills connection ∇^R, the following formal identities hold on smooth sections:*

$$\mathcal{J}_R^{\mathrm{gf}}\mathrm{d}_{\nabla^R} = \mathrm{d}_{\nabla^R}\Delta^{(0)}_{\nabla^R}, \qquad \mathrm{d}^*_{\nabla^R}\mathcal{J}_R^{\mathrm{gf}} = \Delta^{(0)}_{\nabla^R}\mathrm{d}^*_{\nabla^R}. \tag{12.32}$$

For every $z \in \mathbb{C}\setminus\mathbb{R}$, the correctly typed closed-operator resolvent identity is

$$D_R^{\max}(\mathcal{J}_R^{\mathrm{gf}} - z)^{-1} f = (\Delta^{(0)}_{\nabla^R} - z)^{-1} D_R^{\max} f, \qquad f \in \mathrm{Dom}(D_R^{\max}). \tag{12.33}$$

In particular, $\mathscr{S}_R$ reduces $\mathcal{J}_R^{\mathrm{gf}}$. The part of the full operator in this subspace and its closed form are therefore

$$\mathcal{J}_R := \mathcal{J}_R^{\mathrm{gf}}\big|_{\mathscr{S}_R}, \quad \mathrm{Dom}(\mathcal{J}_R) = H^2\Omega^1(F_R; \mathrm{ad}\,\mathscr{E}_R)\cap\mathscr{S}_R, \quad \mathrm{Dom}(\mathfrak{j}_R) = H^1\Omega^1(F_R; \mathrm{ad}\,\mathscr{E}_R)\cap\mathscr{S}_R \tag{12.34}$$

with $\mathfrak{j}_R = \mathfrak{j}_R^{\mathrm{gf}}|_{\mathrm{Dom}(\mathfrak{j}_R)}$. The operator $\mathcal{J}_R$ is self-adjoint on $\mathscr{S}_R$, and $\mathfrak{j}_R$ is its closed, lower-bounded form.

Proof. Write $\nabla_\mu = \nabla^R_{e_\mu}$ in the constant orthonormal frame, so $[\nabla_\mu, \nabla_\nu]\phi = [F_{\mu\nu}, \phi]$. For a smooth adjoint-valued scalar ϕ, commuting the derivatives without suppressing either curvature contribution gives

$$\begin{aligned}
&\big(\mathcal{J}_R^{\mathrm{gf}}\mathrm{d}_{\nabla^R}\phi - \mathrm{d}_{\nabla^R}\Delta^{(0)}_{\nabla^R}\phi\big)_\nu \\
&\quad = -\nabla_\mu\nabla_\mu\nabla_\nu\phi + \nabla_\nu\nabla_\mu\nabla_\mu\phi - 2[F_{\nu\mu}, \nabla_\mu\phi] \\
&\quad = -[\nabla_\mu F_{\mu\nu}, \phi] = [(\mathrm{d}^*_{\nabla^R}F_{\nabla^R})_\nu, \phi].
\end{aligned}$$

The last member vanishes by theorem 12.4. This proves the first identity in (12.32). Since both Laplacians and $\mathcal{J}_R^{\mathrm{gf}}$ are formally symmetric, pairing it against arbitrary smooth zero- and one-forms proves the second identity.

Fix $z \notin \mathbb{R}$, take $f \in \mathrm{Dom}(D_R^{\max})$, and put $u = (\mathcal{J}_R^{\mathrm{gf}} - z)^{-1} f$. The full operator domain gives $u \in H^2$, hence $D_R^{\max} u \in H^1$. Moreover $(\mathcal{J}_R^{\mathrm{gf}} - z)u = f$ and the distributional extension of the second formal identity imply

$$(\Delta^{(0)}_{\nabla^R} - z) D_R^{\max} u = D_R^{\max} f \quad \text{in } \mathcal{D}'(F_R; \mathrm{ad}\,\mathscr{E}_R). \tag{12.35}$$

The right side is in L^2. Elliptic regularity for the scalar Laplacian upgrades $D_R^{\max}u$ from H^1 to H^2, so it belongs to $\mathrm{Dom}(\Delta^{(0)}_{\nabla^R})$. Applying the bounded scalar resolvent to (12.35) proves (12.33); no extension of $D_R^{\max}$ to all of L^2 is asserted or needed.

If $f \in \mathscr{S}_R$, the right side of (12.33) is zero, hence both resolvents at z and at $\overline{z}$ preserve $\mathscr{S}_R$. For $g \in \mathscr{S}_R^{\perp}$ and $s \in \mathscr{S}_R$, self-adjointness of the full operator gives

$$\langle (\mathcal{J}_R^{\mathrm{gf}} - z)^{-1} g, s \rangle = \langle g, (\mathcal{J}_R^{\mathrm{gf}} - \overline{z})^{-1} s \rangle = 0.$$

Thus the resolvent preserves both $\mathscr{S}_R$ and its orthogonal complement, which is precisely the reducing property. The standard part of a self-adjoint operator on a reducing subspace is self-adjoint, and its domain is the intersection in (12.34). The reducing orthogonal projection commutes with the spectral calculus of the lower-bounded full operator, including its form square root; this proves the asserted form domain and completes the proof. □

Theorem 12.8 (Exact charged Landau negative mode)**.** *For the self-adjoint Yang–Mills Jacobi operator $\mathcal{J}_R$ on the closed background slice $\mathscr{S}_R$ defined above, the differential expression in a constant orthonormal frame $(e_\mu)_{\mu=1}^4$ is*

$$(\mathcal{J}_R a)_\nu = -\sum_{\mu=1}^{4} \nabla_{e_\mu} \nabla_{e_\mu} a_\nu - 2 \sum_{\mu=1}^{4} [F_{\nu\mu}, a_\mu]. \tag{12.36}$$

There exists a nonzero smooth real $\mathfrak{su}(2)$-valued one-form a such that

$$\|a\|_{L^2} = 1, \qquad \mathrm{d}^*_{\nabla^R} a = 0, \qquad \mathcal{J}_R a = -\frac{4\pi}{D_R} a. \tag{12.37}$$

Equivalently, for the action normalization equation (12.12),

$$\mathrm{YM}''_R\big|_{\nabla^R}(a, a) = -\frac{4\pi}{g_{\mathrm{YM}}^2 D_R}. \tag{12.38}$$

Thus the reducible non-flat critical connection has a strictly negative classical second-variation direction for every finite R.

Proof. First isolate the exact two-dimensional magnetic problem. Let

$$W = \{x_1 = x_2 = 0\} \subset \mathbb{R}^4,$$

so that (y_1, y_2) are oriented orthonormal coordinates on W. By equation (10.6), orthogonal projection to W sends the first two marked lattice generators to

$$v_1 = (6m_R, -q_R), \qquad v_2 = (L_R, m_R),$$

and sends the last two to zero. Since $\det(v_1, v_2) = D_R > 0$, projection descends to

$$\varpi_R : F_R \longrightarrow \Sigma_R := W/(\mathbb{Z}v_1 + \mathbb{Z}v_2).$$

The factors and connection of $\mathscr{L}_R^2$ depend only on (u_1, u_2). They therefore define a Hermitian line bundle with unitary connection $(\mathscr{M}_R, \nabla^{\mathscr{M}_R})$ on Σ_R such that

$$(\mathscr{L}_R^2, \nabla^{\mathscr{L}_R^2}) = \varpi_R^*(\mathscr{M}_R, \nabla^{\mathscr{M}_R}). \tag{12.39}$$

Let $\Delta_+ = (\nabla^{\mathscr{M}_R})^* \nabla^{\mathscr{M}_R}$ be the nonnegative self-adjoint Bochner Laplacian on $L^2(\Sigma_R; \mathscr{M}_R)$, defined by its closed H^1 form and with operator domain $H^2(\Sigma_R; \mathscr{M}_R)$. More precisely, if $c_R > 0$ is the constant Riemannian volume of a fibre of ϖ_R, then

$$U_R : L^2(\Sigma_R; \mathscr{M}_R) \longrightarrow L^2(F_R; \mathscr{L}_R^2), \qquad U_R s = c_R^{-1/2} \varpi_R^* s,$$

is an isometry. Its image

$$\mathscr{B}_R := U_R L^2(\Sigma_R; \mathscr{M}_R)$$

is the closed fibre-basic subspace. The pulled-back realization is, by definition,

$$\Delta_{+,\mathrm{bas}} := U_R \Delta_+ U_R^{-1} \quad \text{on } \mathscr{B}_R, \qquad \mathrm{Dom}(\Delta_{+,\mathrm{bas}}) = U_R H^2(\Sigma_R; \mathscr{M}_R),$$

and its closed form domain is $U_R H^1(\Sigma_R; \mathscr{M}_R)$. Only this operator on $\mathscr{B}_R$ is self-adjoint. In the formulas below we abbreviate $\Delta_{+,\mathrm{bas}}$ again to Δ_+; no self-adjoint operator on the full $L^2(F_R; \mathscr{L}_R^2)$ is thereby defined or asserted. On this root line the curvature is

$$F_{\mathscr{L}_R^2} = 4\pi i\, \theta_R^1 \wedge \theta_R^2 = i\kappa_R \, \mathrm{d}y_1 \wedge \mathrm{d}y_2, \qquad \kappa_R = \frac{4\pi}{D_R} > 0. \tag{12.40}$$

We now construct an exact lowest Landau section on this actual projected lattice. In the marked coordinates put

$$\mathcal{D}_1 := \partial_{u_1}, \qquad \mathcal{D}_2 := \partial_{u_2} + 4\pi i u_1,$$

and define

$$\zeta_R := \frac{q_R - 6im_R}{m_R + iL_R}, \qquad v_R := -\operatorname{Im} \zeta_R = \frac{D_R}{m_R^2 + L_R^2} > 0. \tag{12.41}$$

Indeed,

$$\operatorname{Im}\big((q_R - 6im_R)(m_R - iL_R)\big) = -(L_R q_R + 6m_R^2) = -D_R,$$

which also checks the sign $v_R > 0$. The series

$$\Theta_R(u_1, u_2) = \sum_{\ell \in \mathbb{Z}} \exp\Big(-2\pi i \zeta_R (\ell + u_1)^2 + 4\pi i \ell u_2\Big) \tag{12.42}$$

converges normally with all derivatives, because the absolute value of its ℓ-th summand is $\exp(-2\pi v_R(\ell + u_1)^2)$. Reindexing the series and using $\ell \in \mathbb{Z}$ give

$$\Theta_R(u_1 + 1, u_2) = e^{-4\pi i u_2} \Theta_R(u_1, u_2), \qquad \Theta_R(u_1, u_2 + 1) = \Theta_R(u_1, u_2). \tag{12.43}$$

Thus Θ_R is a smooth section of $\mathscr{L}_R^2$. Termwise differentiation gives

$$(\mathcal{D}_1 + \zeta_R \mathcal{D}_2)\Theta_R = 0. \tag{12.44}$$

The exact period block on the (y_1, y_2)-plane is

$$\begin{pmatrix} 6m_R & L_R \\ -q_R & m_R \end{pmatrix}.$$

Its inverse transpose gives the physical covariant derivatives

$$\nabla_{y_1}^+ = \frac{m_R \mathcal{D}_1 + q_R \mathcal{D}_2}{D_R}, \qquad \nabla_{y_2}^+ = \frac{-L_R \mathcal{D}_1 + 6m_R \mathcal{D}_2}{D_R}. \tag{12.45}$$

Consequently

$$\nabla_{y_1}^+ - i\nabla_{y_2}^+ = \frac{m_R + iL_R}{D_R}(\mathcal{D}_1 + \zeta_R \mathcal{D}_2). \tag{12.46}$$

Orthogonality of the u_2-Fourier modes also gives the nonzero norm

$$\int_0^1 \int_0^1 |\Theta_R|^2 \, \mathrm{d}u_1 \, \mathrm{d}u_2 = \sum_{\ell \in \mathbb{Z}} \int_0^1 e^{-4\pi v_R (\ell + u_1)^2} \, \mathrm{d}u_1 = \frac{1}{2\sqrt{v_R}}.$$

Set

$$\phi_R := \left(\frac{2\sqrt{v_R}}{D_R}\right)^{1/2} \Theta_R$$

and pull it back by ϖ_R. Since the marked-coordinate volume element is $D_R\,\mathrm{d}^4u$, it has unit $L^2(F_R)$-norm. Its covariant derivatives in the x_1, x_2 directions vanish, and

$$(\nabla^+_{y_1} - i\nabla^+_{y_2})\phi_R = 0, \qquad \Delta_+\phi_R = \kappa_R\phi_R, \qquad \Delta_+\phi_R = -((\nabla^+_{y_1})^2 + (\nabla^+_{y_2})^2)\phi_R. \tag{12.47}$$

Indeed, equation (12.45) and $[\mathcal{D}_1, \mathcal{D}_2] = 4\pi i$ give

$$[\nabla^+_{y_1}, \nabla^+_{y_2}] = \frac{m_R(6m_R) - q_R(-L_R)}{D_R^2}\,4\pi i = i\kappa_R.$$

Direct multiplication then gives

$$(\nabla^+_{y_1} - i\nabla^+_{y_2})^*(\nabla^+_{y_1} - i\nabla^+_{y_2}) = \Delta_+ - \kappa_R,$$

and the second equation in (12.47) follows from the first.

In $\mathfrak{sl}_2(\mathbb{C})$, put

$$X_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad X_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then

$$[H_{\mathrm{mag}}, X_+] = 2iX_+, \qquad [H_{\mathrm{mag}}, X_-] = -2iX_-.$$

The X_+-root line in $\mathrm{ad}(\mathscr{L}_R \oplus \mathscr{L}_R^{-1}) \otimes \mathbb{C}$ is $\mathscr{L}_R^2$, so

$$a^+ = \phi_R X_+(\mathrm{d}y_1 - i\,\mathrm{d}y_2)$$

is a globally defined complexified adjoint-valued one-form. The first equation in (12.47) gives

$$\mathrm{d}^*_{\nabla^R} a^+ = -(\nabla^+_{y_1} - i\nabla^+_{y_2})\phi_R\, X_+ = 0.$$

Put $\mathcal{B}_R := 2\pi/D_R$, so that $\kappa_R = 2\mathcal{B}_R$ and $F_{y_1y_2} = \mathcal{B}_R H_{\mathrm{mag}}$. The two nonzero components of a^+ are $a^+_{y_1} = \phi_R X_+$ and $a^+_{y_2} = -i\phi_R X_+$. Substitution into (12.36), with no suppressed charge factor, gives

$$\begin{aligned}(\mathcal{J}_R a^+)_{y_1} &= \kappa_R\phi_R X_+ - 2[\mathcal{B}_R H_{\mathrm{mag}}, -i\phi_R X_+] = -\kappa_R\phi_R X_+,\\ (\mathcal{J}_R a^+)_{y_2} &= -i\kappa_R\phi_R X_+ - 2[-\mathcal{B}_R H_{\mathrm{mag}}, \phi_R X_+] = i\kappa_R\phi_R X_+.\end{aligned}$$

Thus $\mathcal{J}_R a^+ = -\kappa_R a^+$.

For completeness, formula (12.36) follows by expanding $F_{\nabla^R + ta} = F_{\nabla^R} + t\,\mathrm{d}_{\nabla^R}a + \frac{1}{2}t^2[a \wedge a]$, integrating the quadratic form by parts on the background slice, and using the flat-base Weitzenbock identity. The Hodge Laplacian contributes $-[F_{\nu\mu}, a_\mu]$, and the curvature term in the second variation contributes the identical term, giving the coefficient -2.

Finally take the anti-Hermitian real form

$$a := \frac{1}{\sqrt{2}}(a^+ - (a^+)^\dagger) = \frac{1}{\sqrt{2}}\left(\phi_R X_+(\mathrm{d}y_1 - i\,\mathrm{d}y_2) - \overline{\phi_R} X_-(\mathrm{d}y_1 + i\,\mathrm{d}y_2)\right). \tag{12.48}$$

It is nonzero, traceless, and anti-Hermitian. For $X(z) := zX_+ - \bar{z}X_-$, the fixed inner product gives $-\frac{1}{2}\operatorname{Tr}(X(z)^2) = |z|^2$. The two orthonormal one-form components in equation (12.48) therefore have total pointwise norm $|a|^2 = |\phi_R|^2$, so $\|a\|_{L^2} = 1$. Complex conjugation and Hermitian adjunction preserve both the gauge equation and the real Jacobi operator, so (12.37) follows from the calculation for a^+. The Hessian identity follows from $\mathrm{YM}''_R(a,a) = g_{\mathrm{YM}}^{-2}\langle a, \mathcal{J}_R a\rangle_{L^2}$ on the background slice. □

13 The native full-flux magnetic sector

This section retains the canonical radial parameter, period matrix, Euclidean quotient metric, marked lattice, and every sign already fixed in section 10. It treats the monodromy-invariant integral class that uses both elementary alternating blocks. It is therefore distinct from the unit-flux plane treated in section 12. No metric rescaling, finite covering, rectangular replacement of the lattice, or orientation suppression occurs below.

For $R > R_0$, retain

$$\begin{aligned} L_R &= R + H_R = \operatorname{Im} \tau_R, \qquad q_R = L_R - B_R = -\operatorname{Im} \beta_R, \\ m_R &= \operatorname{Im} \mu_R, \qquad D_R = L_R q_R + 6m_R^2 > 0, \end{aligned} \tag{13.1}$$

and write

$$\eta_R^{\mathrm{re}} := \operatorname{Re} h_R.$$

This scalar is denoted by η_R in section 10; the superscript is retained throughout this section so that it cannot be confused with the native integral two-form introduced below. The complete period matrix is

$$P_R = \begin{pmatrix} 6\alpha_R & \eta_R^{\mathrm{re}} & 1 & 0 \\ 6m_R & L_R & 0 & 0 \\ c_R & \alpha_R & 0 & 1 \\ -q_R & m_R & 0 & 0 \end{pmatrix}, \qquad \det P_R = -D_R. \tag{13.2}$$

Thus $F_R = \mathbb{R}^4/P_R\mathbb{Z}^4$ has the unrescaled Euclidean quotient metric and volume D_R. In the marking coordinates $y \in \mathbb{R}^4$, the map $y \mapsto x = P_R y$ pulls the Euclidean metric back to $G_R = P_R^{\mathsf{T}} P_R$.

The limiting cusp constants remain

$$m = \operatorname{Im} \mu(0), \qquad H = \operatorname{Im} h(0), \qquad B = \operatorname{Im} b(0), \tag{13.3}$$

so holomorphy at $e^{-2\pi R} = 0$ gives

$$\begin{aligned} m_R &= m + O(e^{-2\pi R}), \quad H_R = H + O(e^{-2\pi R}), \\ B_R &= B + O(e^{-2\pi R}), \quad D_R = R^2 + (2H - B)R \\ &\qquad + (H^2 - BH + 6m^2) + O(Re^{-2\pi R}). \end{aligned} \tag{13.4}$$

13.1 The invariant class, its cocycle, and both orientations

The matrices A_1, A_2 in (1.12) act on the marked homology, equivalently on the angular lattice used there. On the dual integral cohomology lattice put

$$T_j := A_j^{-\mathsf{T}}, \qquad j = 1, 2. \tag{13.5}$$

Direct inversion of those displayed matrices gives the actions written below. In the resulting ordered cohomological marking $\mathbb{Z}\langle \gamma, u, w, \delta \rangle$, put

$$\eta = u \wedge w + 6\gamma \wedge \delta. \tag{13.6}$$

The two finite monodromy generators act by

$$\begin{aligned} &T_1\gamma = \gamma, \quad T_1 u = -u - w, \quad T_1 w = -6\gamma + u, \quad T_1\delta = 2\gamma + u + w + \delta, \\ &T_2\gamma = \gamma, \quad T_2 u = 6\gamma + w, \quad T_2 w = -u, \quad T_2\delta = -3\gamma + u + \delta. \end{aligned}$$

Directly, without passing to a normalized representative,

$$\begin{aligned}(\textstyle\bigwedge^2 T_1)\eta &= (-u-w)\wedge(-6\gamma+u)+6\gamma\wedge(2\gamma+u+w+\delta)\\ &= u\wedge w+6\gamma\wedge\delta,\\ (\textstyle\bigwedge^2 T_2)\eta &= (6\gamma+w)\wedge(-u)+6\gamma\wedge(-3\gamma+u+\delta)\\ &= u\wedge w+6\gamma\wedge\delta.\end{aligned}$$

Hence η is invariant under both generators.

On $Y=\mathbb{R}^4/\mathbb{Z}^4$, with ordered coordinates y^1,y^2,y^3,y^4, its constant representative and alternating matrix are

$$\eta_R = \mathrm{d}y^2\wedge\mathrm{d}y^3+6\,\mathrm{d}y^1\wedge\mathrm{d}y^4,\qquad E_\eta=\begin{pmatrix}0&0&0&6\\0&0&1&0\\0&-1&0&0\\-6&0&0&0\end{pmatrix}. \tag{13.7}$$

In the paired basis $(e_2,e_3;e_1,e_4)$, the alternating elementary divisors are $(1,6)$; consequently

$$\mathrm{SNF}(E_\eta)=\mathrm{diag}(1,1,6,6),\qquad \mathrm{Pf}(E_\eta)=6,\qquad \det E_\eta=36. \tag{13.8}$$

The actual Hermitian line bundle is the quotient of $\mathbb{R}^4\times\mathbb{C}$ by

$$\begin{aligned}&(y,z)\longmapsto(y+e_1,e^{12\pi i y^4}z), && (y,z)\longmapsto(y+e_2,e^{2\pi i y^3}z),\\ &(y,z)\longmapsto(y+e_3,z), && (y,z)\longmapsto(y+e_4,z).\end{aligned} \tag{13.9}$$

The only possibly nontrivial generator commutators acquire the factors $e^{12\pi i}=1$ and $e^{2\pi i}=1$, so these maps define a $\mathbb{Z}^4$-action. Denote the quotient by $\mathscr{L}_{\eta,R}\to F_R$, using the marking $[y]\mapsto[P_R y]$. The connection

$$\nabla_{\eta,R}=\mathrm{d}-2\pi i(6y^1\,\mathrm{d}y^4+y^2\,\mathrm{d}y^3) \tag{13.10}$$

descends through the displayed cocycle. For instance translation by e_1 changes its connection form by $-12\pi i\,\mathrm{d}y^4$, which equals the negative logarithmic derivative of $e^{12\pi i y^4}$; the e_2 check is identical. Therefore

$$F_{\nabla_{\eta,R}}=-2\pi i\eta_R,\qquad c_1(\mathscr{L}_{\eta,R})=\frac{i}{2\pi}F_{\nabla_{\eta,R}}=\eta_R. \tag{13.11}$$

There are two opposite orientations which must not be conflated. Write

$$\mathrm{or}_{\mathrm{mark}}=\mathrm{d}y^1\wedge\mathrm{d}y^2\wedge\mathrm{d}y^3\wedge\mathrm{d}y^4 \tag{13.12}$$

and let or_x be the native complex orientation of $(z_1,z_2)=(x_1+iy_1,x_2+iy_2)$. Then

$$P_R^*\,\mathrm{or}_x=-D_R\,\mathrm{or}_{\mathrm{mark}}, \tag{13.13}$$

whereas

$$\frac{1}{2}\eta_R^2=6\,\mathrm{or}_{\mathrm{mark}}\,. \tag{13.14}$$

Consequently

$$\frac{1}{2}\langle\eta_R^2,[F_R]_{\mathrm{mark}}\rangle=6,\qquad \frac{1}{2}\langle\eta_R^2,[F_R]_{x\text{-complex}}\rangle=-6. \tag{13.15}$$

13.2 The exact physical frequencies

In the physical Euclidean coordinates the two-form matrix is

$$\Omega_R = P_R^{-\mathsf{T}} E_\eta P_R^{-1} = \frac{1}{D_R} \begin{pmatrix} 0 & -q_R & 0 & -6m_R \\ q_R & 0 & 6m_R & 0 \\ 0 & -6m_R & 0 & 6L_R \\ 6m_R & 0 & -6L_R & 0 \end{pmatrix}. \tag{13.16}$$

After the exact permutation $(x^1, x^2, x^3, x^4) \mapsto (x^1, x^3; x^2, x^4)$, its off-diagonal block is

$$K_R = \begin{pmatrix} -q_R & -6m_R \\ -6m_R & 6L_R \end{pmatrix}, \qquad \det K_R = -6D_R. \tag{13.17}$$

Define the magnetic scalars

$$S_R := 6L_R - q_R, \qquad \Xi_R := \sqrt{(6L_R + q_R)^2 + 144m_R^2}. \tag{13.18}$$

Here S_R has no argument and is distinct from the affine residual $S_R(k)$ used in the dual-lattice calculation of theorem 10.1. The signed eigenvalues of K_R are

$$\kappa_{R,\pm} = \frac{S_R \pm \Xi_R}{2}. \tag{13.19}$$

Direct expansion, with no asymptotic replacement, gives

$$\Xi_R^2 - S_R^2 = 24(L_R q_R + 6m_R^2) = 24D_R > 0. \tag{13.20}$$

Thus $\kappa_{R,+} > 0 > \kappa_{R,-}$. The two positive charge-one magnetic frequencies are

$$\boxed{\mathcal{B}_{R,+} = \frac{\pi(S_R + \Xi_R)}{D_R}, \qquad \mathcal{B}_{R,-} = \frac{\pi(\Xi_R - S_R)}{D_R}.} \tag{13.21}$$

They obey the exact identities

$$\mathcal{B}_{R,+}\mathcal{B}_{R,-} = \frac{24\pi^2}{D_R}, \tag{13.22}$$

$$\mathcal{B}_{R,+} + \mathcal{B}_{R,-} = \frac{2\pi\Xi_R}{D_R}, \qquad \mathcal{B}_{R,+} - \mathcal{B}_{R,-} = \frac{2\pi S_R}{D_R}, \tag{13.23}$$

$$\mathcal{B}_{R,+} = \frac{24\pi}{\Xi_R - S_R}, \qquad \mathcal{B}_{R,-} = \frac{24\pi}{\Xi_R + S_R}. \tag{13.24}$$

The last line follows by multiplying by the conjugate factor and using equation (13.20).

Keeping the three independent limiting coefficients H, B, m, one obtains

$$\mathcal{B}_{R,-} = \frac{2\pi}{R} - \frac{2\pi H}{R^2} + \frac{2\pi}{R^3}\left(H^2 - \frac{6}{7}m^2\right) + O(R^{-4}), \tag{13.25}$$

$$\mathcal{B}_{R,+} = \frac{12\pi}{R} + \frac{12\pi(B-H)}{R^2} + \frac{12\pi}{R^3}\left((B-H)^2 - \frac{36}{7}m^2\right) + O(R^{-4}), \tag{13.26}$$

$$\rho_R := \frac{\mathcal{B}_{R,+}}{\mathcal{B}_{R,-}} = 6 + \frac{6B}{R} + \frac{6B^2 - 6BH - 180m^2/7}{R^2} + O(R^{-3}). \tag{13.27}$$

These follow by substituting equation (13.4) into equation (13.24) and expanding the reciprocals; no sign of H, B, m is assumed. In particular, on a sufficiently deep cusp tail,

$$S_R > 0, \qquad \mathcal{B}_{R,+} > \mathcal{B}_{R,-} > 0, \qquad 5 < \rho_R < 7. \tag{13.28}$$

13.3 Joint Landau levels and their exact multiplicities

Let $\Delta_{\eta,R} = \nabla^*_{\eta,R}\nabla_{\eta,R}$ be the Friedrichs magnetic Laplacian on $L^2(F_R, \mathscr{L}_{\eta,R})$. Its form domain is $H^1(F_R; \mathscr{L}_{\eta,R})$, its operator domain is $H^2(F_R; \mathscr{L}_{\eta,R})$, and compactness of the torus gives compact resolvent. We now prove the multiplicity rather than infer it from the Pfaffian alone.

Define A^{pol}_R on the marked tangent space by

$$G_R(A^{\mathrm{pol}}_R v, w) = 2\pi E_\eta(v, w), \qquad Q^{\mathrm{pol}}_R = \bigl(-(A^{\mathrm{pol}}_R)^2\bigr)^{1/2}, \qquad J_{\eta,R} = A^{\mathrm{pol}}_R (Q^{\mathrm{pol}}_R)^{-1}. \tag{13.29}$$

The endomorphism A^{pol}_R is G_R-skew-adjoint and invertible, so Q^{pol}_R is positive, commutes with A^{pol}_R, and

$$J^2_{\eta,R} = -1, \qquad E_\eta(J_{\eta,R}v, J_{\eta,R}w) = E_\eta(v, w), \qquad 2\pi E_\eta(v, J_{\eta,R}v) = G_R(Q^{\mathrm{pol}}_R v, v) > 0. \tag{13.30}$$

For every signed charge $q \in \mathbb{Z} \setminus \{0\}$, use the convention

$$\mathscr{L}^{\otimes q}_{\eta,R} = \begin{cases} (\mathscr{L}_{\eta,R})^{\otimes q}, & q > 0, \\ (\mathscr{L}^\vee_{\eta,R})^{\otimes(-q)}, & q < 0, \end{cases} \qquad A^{\mathrm{pol}}_{q,R} = qA^{\mathrm{pol}}_R, \qquad Q^{\mathrm{pol}}_{q,R} = \bigl(-(A^{\mathrm{pol}}_{q,R})^2\bigr)^{1/2} = |q|Q^{\mathrm{pol}}_R, \tag{13.31}$$

and define

$$J_{q,R} = A^{\mathrm{pol}}_{q,R}(Q^{\mathrm{pol}}_{q,R})^{-1} = \mathrm{sgn}(q) J_{\eta,R}. \tag{13.32}$$

All these coefficients are constant, hence the Nijenhuis tensor of $J_{q,R}$ vanishes and the marking transports it to a complex structure on F_R. The induced connection $\nabla^{(q)}_{\eta,R}$ has

$$F_{\nabla^{(q)}_{\eta,R}} = -2\pi i q \eta_R, \qquad 2\pi q E_\eta(v, J_{q,R}v) = |q| G_R(Q^{\mathrm{pol}}_R v, v) > 0. \tag{13.33}$$

Thus its curvature is of type $(1,1)$, and $(\nabla^{(q)}_{\eta,R})^{0,1}$ makes $\mathscr{L}^{\otimes q}_{\eta,R}$ a positive Appell–Humbert line bundle on the auxiliary complex torus $(F_R, J_{q,R})$.

The auxiliary orientation is fixed, rather than repaired by an absolute value. For $q > 0$, compatibility identifies the $J_{q,R}$-complex orientation with the E_η-symplectic orientation; for $q < 0$, replacing J by $-J$ preserves complex orientation in complex dimension two. Hence

$$[F_R]_{J_{q,R}} = [F_R]_{\mathrm{mark}} = -[F_R]_{x\text{-complex}}. \tag{13.34}$$

This auxiliary positive structure is not the native complex structure inherited from the S^6 construction. Indeed, with $E_\eta = \mathrm{Im}\, H_{\eta,R}$, the native Appell–Humbert calculation is

$$G_{\eta,R} = \begin{pmatrix} 2L_R & 2m_R \\ 2m_R & -q_R/3 \end{pmatrix}, \qquad \det G_{\eta,R} = -\frac{2}{3} D_R < 0, \qquad H_{\eta,R} = 2G^{-1}_{\eta,R}. \tag{13.35}$$

Consequently $H_{\eta,R}$, and also $qH_{\eta,R}$ for every nonzero q, has signature $(1,1)$. The native line bundle is holomorphic because $q\eta_R$ is of type $(1,1)$, but it is indefinite and is not the positive bundle just constructed.

The tangent and canonical bundles of the auxiliary complex torus are trivial. Kodaira vanishing and Hirzebruch–Riemann–Roch therefore give

$$\begin{aligned} &\dim_{\mathbb{C}} H^0\bigl((F_R, J_{q,R}), \mathscr{L}^{\otimes q}_{\eta,R}\bigr) = \int_{[F_R]_{J_{q,R}}} \frac{(q\eta_R)^2}{2} = \int_{[F_R]_{\mathrm{mark}}} \frac{(q\eta_R)^2}{2} = 6q^2, \\ &\dim_{\mathbb{C}} H^0\bigl((F_R, J_{\eta,R}), \mathscr{L}_{\eta,R}\bigr) = 6, \\ &\dim_{\mathbb{C}} H^0\bigl((F_R, J_{2,R}), \mathscr{L}^{\otimes 2}_{\eta,R}\bigr) = 24. \end{aligned} \tag{13.36}$$

No absolute value occurs in this auxiliary Hirzebruch–Riemann–Roch integral. Equivalently, the positive polarization has elementary divisors $(|q|, 6|q|)$, and the Appell–Humbert theta theorem gives their product $6q^2$; see Beauville, Theorems 2.6 (pp. 107–108) and 3.5 (p. 112). On the native complex torus the same signed Hirzebruch–Riemann–Roch calculation instead gives

$$\chi((F_R, J_{\mathrm{native},R}), \mathscr{L}_{\eta,R}^{\otimes q}) = \int_{[F_R]_{x\text{-complex}}} \frac{(q\eta_R)^2}{2} = -6q^2, \tag{13.37}$$

and positivity vanishing is unavailable there. Thus the positive auxiliary section count and the native signed Euler characteristic concern distinct Dolbeault problems and distinct complex orientations.

Choose a G_R-orthonormal frame $e_+, f_+ = J_{\eta,R}e_+, e_-, f_- = J_{\eta,R}e_-$ which diagonalizes Q_R^{pol}, labelled so its positive eigenvalues are $\mathcal{B}_{R,+}, \mathcal{B}_{R,-}$. Put

$$a_\sigma = \frac{\nabla_{e_\sigma} + i\nabla_{f_\sigma}}{\sqrt{2\mathcal{B}_{R,\sigma}}}, \qquad a_\sigma^* = \frac{-\nabla_{e_\sigma} + i\nabla_{f_\sigma}}{\sqrt{2\mathcal{B}_{R,\sigma}}}, \qquad \sigma \in \{+,-\}. \tag{13.38}$$

The curvature commutators give

$$[a_\sigma, a_\tau^*] = \delta_{\sigma\tau}, \qquad [a_\sigma, a_\tau] = [a_\sigma^*, a_\tau^*] = 0, \tag{13.39}$$

and, for $N_\sigma = a_\sigma^* a_\sigma$,

$$\Delta_{\eta,R} = \mathcal{B}_{R,+}(2N_+ + 1) + \mathcal{B}_{R,-}(2N_- + 1). \tag{13.40}$$

The simultaneous vacuum $\ker a_+ \cap \ker a_- = H^0((F_R, J_{\eta,R}), \mathscr{L}_{\eta,R})$ has complex dimension six by equation (13.36).

Theorem 13.1 (Complete native joint Landau decomposition)**.** *For every $n_+, n_- \in \mathbb{Z}_{\geq 0}$, the joint eigenspace*

$$\mathcal{H}_{R;n_+,n_-} = \ker(N_+ - n_+) \cap \ker(N_- - n_-) \tag{13.41}$$

has complex dimension six and energy

$$E_{R;n_+,n_-} = (2n_+ + 1)\mathcal{B}_{R,+} + (2n_- + 1)\mathcal{B}_{R,-}. \tag{13.42}$$

Moreover

$$L^2(F_R, \mathscr{L}_{\eta,R}) = \widehat{\bigoplus}_{n_+,n_- \geq 0} \mathcal{H}_{R;n_+,n_-}, \tag{13.43}$$

and the map from the vacuum to the joint level is the unitary

$$\frac{(a_+^*)^{n_+}(a_-^*)^{n_-}}{\sqrt{n_+!n_-!}}. \tag{13.44}$$

For a numerical eigenvalue, the multiplicity is six times the number of joint pairs in equation (13.42) producing that number.

Proof. The elliptic operator has compact resolvent, and the two commuting number operators preserve its finite-dimensional eigenspaces; they can therefore be diagonalized simultaneously. If $N_\sigma\psi = \lambda_\sigma\psi$, then $\lambda_\sigma \geq 0$. Repeated application of a_σ lowers the Laplacian energy by $2\mathcal{B}_{R,\sigma} > 0$, so it terminates. If $a_\sigma^r\psi \neq 0$ and $a_\sigma^{r+1}\psi = 0$, then

$$0 = \|a_\sigma^{r+1}\psi\|^2 = (\lambda_\sigma - r)\|a_\sigma^r\psi\|^2,$$

and hence $\lambda_\sigma = r \in \mathbb{Z}_{\geq 0}$. Lowering both indices reaches the simultaneous vacuum.

The commutator identities give, for a vacuum vector v,

$$\|(a_\sigma^*)^n v\|^2 = n!\|v\|^2, \qquad a_\sigma^n (a_\sigma^*)^n v = n! v.$$

On the n-th level they also give $(a_\sigma^*)^n a_\sigma^n = n!$. Operators in the two planes commute, so equation (13.44) is unitary with inverse $a_+^{n_+} a_-^{n_-}/\sqrt{n_+! n_-!}$. This proves the dimension, energy, and completeness assertions. Orthogonal joint eigenspaces with coincident numerical energies must be added, which proves the final sentence. □

The compact-resolvent lowering argument proves that no additional joint eigenspaces occur. Independently, this is exactly the case $m = 2$, $k = 1$ of Gordon–Kirwin–Schueth–Webb, Proposition 4.5, article pp. 9–10 (volume pp. 175–176), doi:10.1090/pspum/084/1355. Their signed tensor-power statement gives every oscillator tuple multiplicity $|k|^m V_{E_\eta} = |k|^2 \operatorname{Pf}(E_\eta) = 6k^2$, so it also supplies the external completeness check for every $k \neq 0$.

The exact ground energy and first distinct spacing are therefore

$$E_0(R) = \mathcal{B}_{R,+} + \mathcal{B}_{R,-} = \frac{2\pi \Xi_R}{D_R} = \frac{14\pi}{R} + \frac{\pi(12B - 14H)}{R^2} + O(R^{-3}), \tag{13.45}$$

$$g_0(R) = 2\min\{\mathcal{B}_{R,+}, \mathcal{B}_{R,-}\} = \frac{2\pi(\Xi_R - |S_R|)}{D_R} = \frac{48\pi}{\Xi_R + |S_R|}. \tag{13.46}$$

On the deep tail equation (13.28), this is $g_0(R) = 2\mathcal{B}_{R,-}$, with expansion

$$g_0(R) = \frac{4\pi}{R} - \frac{4\pi H}{R^2} + \frac{4\pi}{R^3}\left(H^2 - \frac{6}{7}m^2\right) + O(R^{-4}). \tag{13.47}$$

Corollary 13.2 (Heat trace and exact fixed-energy count)**.** *For $t > 0$,*

$$\operatorname{Tr}(e^{-t\Delta_{\eta,R}}) = \frac{3}{2\sinh(t\mathcal{B}_{R,+})\sinh(t\mathcal{B}_{R,-})} = \frac{D_R}{16\pi^2 t^2}(1 + O_t(R^{-2})). \tag{13.48}$$

For $E > 0$, let $N_{\eta,R}(E)$ count complex eigenvalues not exceeding E, with multiplicity. Then

$$N_{\eta,R}(E) = \begin{cases} 0, & E < E_0(R), \\ 6 \displaystyle\sum_{n_+=0}^{\lfloor (E-E_0(R))/(2\mathcal{B}_{R,+}) \rfloor} \left(1 + \left\lfloor \frac{E - E_0(R) - 2\mathcal{B}_{R,+} n_+}{2\mathcal{B}_{R,-}} \right\rfloor\right), & E \geq E_0(R). \end{cases} \tag{13.49}$$

For fixed $E > 0$, as $R \to \infty$,

$$N_{\eta,R}(E) = \frac{D_R}{32\pi^2}(E - E_0(R))^2 + O_E(R) = \frac{E^2 D_R}{32\pi^2} + O_E(R). \tag{13.50}$$

Proof. The heat trace is the product of the two geometric series in equation (13.42), multiplied by six. Expanding both hyperbolic sines and using equation (13.22) proves its asymptotic. The counting formula enumerates the lattice points

$$(n_+, n_-) \in \mathbb{R}^2_{\geq 0}, \qquad 2\mathcal{B}_{R,+} n_+ + 2\mathcal{B}_{R,-} n_- \leq E - E_0(R).$$

Its triangular area is $(E - E_0(R))^2/(8\mathcal{B}_{R,+}\mathcal{B}_{R,-})$. The difference between the lattice count and the area is bounded by a constant times the perimeter plus one, hence is $O_E(R)$. Multiplying by six and using the exact frequency product gives the first asymptotic; the second follows from $E_0(R) = O(R^{-1})$ and $D_R = O(R^2)$. □

13.4 The split $SU(2)$ critical connection

Put

$$\mathscr{E}_{\eta,R} = \mathscr{L}_{\eta,R} \oplus \mathscr{L}_{\eta,R}^{-1}, \qquad A_{\eta,R} = \nabla_{\eta,R} \oplus \nabla_{\eta,R}^{-1}. \tag{13.51}$$

The determinant is canonically trivial and

$$c(\mathscr{E}_{\eta,R}) = (1+\eta_R)(1-\eta_R) = 1-\eta_R^2, \qquad c_2(\mathscr{E}_{\eta,R}) = -\eta_R^2. \tag{13.52}$$

Thus the orientation-sensitive Chern numbers are

$$\langle c_2(\mathscr{E}_{\eta,R}), [F_R]_{\text{mark}} \rangle = -12, \qquad \langle c_2(\mathscr{E}_{\eta,R}), [F_R]_{x\text{-complex}} \rangle = +12. \tag{13.53}$$

The cohomology class $c_2 = -\eta_R^2$ is nonzero, so this split $SU(2)$-bundle is not smoothly trivial. This is another exact distinction from the unit-flux split bundle of theorem 12.3.

With

$$H_{\text{mag}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad \langle X, Y \rangle_{\mathfrak{su}(2)} = -\frac{1}{2}\operatorname{Tr}(XY), \tag{13.54}$$

one has $|H_{\text{mag}}| = 1$ and

$$F_{A_{\eta,R}} = -2\pi\eta_R H_{\text{mag}}. \tag{13.55}$$

This curvature is parallel; hence $d^*_{A_{\eta,R}} F_{A_{\eta,R}} = 0$, and $A_{\eta,R}$ is Yang–Mills. For

$$\mathrm{YM}_R(A) = \frac{1}{2g_{\mathrm{YM}}^2} \int_{F_R} |F_A|^2 \, \mathrm{d}\,\mathrm{vol}_{g_R}, \tag{13.56}$$

diagonalizing equation (13.17) gives the exact action

$$\mathrm{YM}_R(A_{\eta,R}) = \frac{D_R}{2g_{\mathrm{YM}}^2}(\mathcal{B}_{R,+}^2 + \mathcal{B}_{R,-}^2) \tag{13.57}$$

$$= \frac{2\pi^2}{g_{\mathrm{YM}}^2 D_R}(q_R^2 + 36L_R^2 + 72m_R^2) \tag{13.58}$$

$$= \frac{24\pi^2}{g_{\mathrm{YM}}^2} + \frac{2\pi^2 S_R^2}{g_{\mathrm{YM}}^2 D_R}. \tag{13.59}$$

Indeed $q_R^2 + 36L_R^2 + 72m_R^2 = 12D_R + (6L_R - q_R)^2$. Consequently

$$\mathrm{YM}_R(A_{\eta,R}) = \frac{74\pi^2}{g_{\mathrm{YM}}^2} + \frac{70\pi^2 B}{g_{\mathrm{YM}}^2 R} + O(R^{-2}). \tag{13.60}$$

13.5 The closed Jacobi operator and exact Coulomb reduction

We next define the operator before computing its spectrum. Let

$$\mathscr{H}_R^1 = L^2\Omega^1(F_R; \operatorname{ad}\mathscr{E}_{\eta,R}), \qquad \mathscr{H}_R^0 = L^2\Omega^0(F_R; \operatorname{ad}\mathscr{E}_{\eta,R}) \tag{13.61}$$

be real Hilbert spaces. Their complexifications carry the root decomposition used below, and every operator constructed here commutes with the anti-linear real structure.

Define the maximal distributional codifferential by

$$\operatorname{Dom} D_R^{\max} = \{a \in \mathscr{H}_R^1 : \text{there is } u \in \mathscr{H}_R^0 \text{ such that} \\ \langle a, d_{A_{\eta,R}}\phi\rangle_{L^2} = \langle u, \phi\rangle_{L^2} \text{ for every smooth } \phi\}, \tag{13.62}$$

$$D_R^{\max} a = u, \qquad \mathscr{S}_R := \ker D_R^{\max}. \tag{13.63}$$

Uniqueness follows from density of the test space. Passing to the limit in the defining identity proves that $D_R^{\max}$ is closed, hence the Coulomb slice $\mathscr{S}_R$ is a closed subspace.

On adjoint-valued scalars, let $\Delta^{(0)}_{A_{\eta,R}}$ be the nonnegative self-adjoint operator of the closed form

$$\mathfrak{q}_R^{(0)}[\phi,\psi] = \langle d_{A_{\eta,R}}\phi, d_{A_{\eta,R}}\psi\rangle_{L^2}, \qquad \operatorname{Dom}\mathfrak{q}_R^{(0)} = H^1\Omega^0. \tag{13.64}$$

Elliptic regularity gives

$$\Delta^{(0)}_{A_{\eta,R}} = d^*_{A_{\eta,R}} d_{A_{\eta,R}}, \qquad \operatorname{Dom}\Delta^{(0)}_{A_{\eta,R}} = H^2\Omega^0. \tag{13.65}$$

In a constant orthonormal frame, define the symmetric zeroth-order endomorphism

$$(V_R a)_\nu = -2\sum_{\mu=1}^{4}[F_{A_{\eta,R},\nu\mu}, a_\mu]. \tag{13.66}$$

Symmetry follows from invariance of the compact Lie-algebra inner product, antisymmetry of $F_{\nu\mu}$, and an exchange of ν,μ. Consider the gauge-fixed form

$$\mathfrak{j}_R^{\mathrm{gf}}[a,b] = \langle \nabla_{A_{\eta,R}} a, \nabla_{A_{\eta,R}} b\rangle_{L^2} + \langle V_R a, b\rangle_{L^2}, \qquad \operatorname{Dom}\mathfrak{j}_R^{\mathrm{gf}} = H^1\Omega^1. \tag{13.67}$$

Proposition 13.3 (Closed gauge-fixed realization)**.** *The form equation (13.67) is densely defined, closed, symmetric, and bounded below. Its self-adjoint operator $\mathcal{J}_R^{\mathrm{gf}}$ has compact resolvent, domain $H^2\Omega^1$, and expression*

$$(\mathcal{J}_R^{\mathrm{gf}} a)_\nu = -\sum_{\mu=1}^{4}\nabla_{A_{\eta,R},e_\mu}\nabla_{A_{\eta,R},e_\mu}a_\nu - 2\sum_{\mu=1}^{4}[F_{A_{\eta,R},\nu\mu}, a_\mu]. \tag{13.68}$$

It is exactly the Hessian operator with the background-gauge term, before the overall factor g_{YM}^{-2}.

Proof. The first summand in equation (13.67), together with the L^2-norm, is an equivalent H^1-norm. The smooth endomorphism V_R is bounded and symmetric. The bounded perturbation theorem for closed forms gives closedness and a lower bound, and the representation theorem gives a self-adjoint operator. Its principal symbol is $|\xi|^2$ times the identity. Elliptic regularity identifies its maximal L^2-domain with $H^2\Omega^1$; direct substitution supplies the reverse inclusion. The compact embedding $H^1 \hookrightarrow L^2$ gives compact resolvent.

For the final assertion, define $[a\wedge b]_{\mu\nu} = [a_\mu, b_\nu] - [a_\nu, b_\mu]$. The exact two-parameter curvature expansion is

$$F_{A_{\eta,R}+sa+tb} = F_{A_{\eta,R}} + sd_A a + td_A b + \frac{s^2}{2}[a\wedge a] + st[a\wedge b] + \frac{t^2}{2}[b\wedge b].$$

It yields

$$\mathrm{YM}''_R\big|_{A_{\eta,R}}(a,b) = \frac{1}{g_{\mathrm{YM}}^2}\left(\langle d_A a, d_A b\rangle + \langle F_A, [a\wedge b]\rangle\right). \tag{13.69}$$

On the flat torus, commuting the covariant derivatives with the retained curvature convention gives

$$\langle d_A a, d_A b\rangle + \langle d_A^* a, d_A^* b\rangle = \langle \nabla_A a, \nabla_A b\rangle - \sum_{\nu,\mu}\langle a_\nu, [F_{\nu\mu}, b_\mu]\rangle,$$
$$\langle F_A, [a\wedge b]\rangle = -\sum_{\nu,\mu}\langle a_\nu, [F_{\nu\mu}, b_\mu]\rangle.$$

The factor $1/2$ in the two-form inner product is included in the second identity. Adding $g_{\mathrm{YM}}^{-2}\langle d_A^* a, d_A^* b\rangle$ to equation (13.69) gives exactly $g_{\mathrm{YM}}^{-2}\mathfrak{j}_R^{\mathrm{gf}}[a,b]$, including the coefficient -2 in equation (13.66). □

Proposition 13.4 (Resolvent intertwining and the reducing Coulomb slice)**.** *On smooth sections,*

$$\mathcal{J}_R^{\mathrm{gf}} d_{A_{\eta,R}} = d_{A_{\eta,R}} \Delta^{(0)}_{A_{\eta,R}}, \qquad d^*_{A_{\eta,R}} \mathcal{J}_R^{\mathrm{gf}} = \Delta^{(0)}_{A_{\eta,R}} d^*_{A_{\eta,R}}. \tag{13.70}$$

For $z \in \mathbb{C} \setminus \mathbb{R}$ and $f \in \operatorname{Dom} D_R^{\max}$,

$$D_R^{\max} (\mathcal{J}_R^{\mathrm{gf}} - z)^{-1} f = (\Delta^{(0)}_{A_{\eta,R}} - z)^{-1} D_R^{\max} f. \tag{13.71}$$

Thus $\mathscr{S}_R$ reduces $\mathcal{J}_R^{\mathrm{gf}}$, and its part

$$\mathcal{J}_R = \mathcal{J}_R^{\mathrm{gf}}\big|_{\mathscr{S}_R}, \qquad \operatorname{Dom} \mathcal{J}_R = H^2\Omega^1 \cap \mathscr{S}_R \tag{13.72}$$

is self-adjoint. Its closed form is the restriction of $\mathfrak{j}_R^{\mathrm{gf}}$ to $H^1\Omega^1 \cap \mathscr{S}_R$, and the Yang–Mills Hessian on this slice is

$$\mathrm{YM}''_R\big|_{A_{\eta,R}} = g_{\mathrm{YM}}^{-2} \mathfrak{j}_R. \tag{13.73}$$

Proof. For a smooth adjoint-valued scalar ϕ, write $\nabla_\mu = \nabla_{A_{\eta,R}, e_\mu}$. Retaining both curvature terms when commuting three derivatives gives

$$\begin{aligned}
&(\mathcal{J}_R^{\mathrm{gf}} d_A \phi - d_A \Delta_A^{(0)} \phi)_\nu \\
&\quad = -\nabla_\mu \nabla_\mu \nabla_\nu \phi + \nabla_\nu \nabla_\mu \nabla_\mu \phi - 2[F_{\nu\mu}, \nabla_\mu \phi] \\
&\quad = -[\nabla_\mu F_{\mu\nu}, \phi] = [(d_A^* F_A)_\nu, \phi] = 0.
\end{aligned}$$

This proves the first identity in equation (13.70); formal adjunction gives the second.

Let $u = (\mathcal{J}_R^{\mathrm{gf}} - z)^{-1} f$. The operator domain gives $u \in H^2$, hence $D_R^{\max} u \in H^1$. Distributionally applying the second intertwining identity to $(\mathcal{J}_R^{\mathrm{gf}} - z)u = f$ yields

$$(\Delta_A^{(0)} - z) D_R^{\max} u = D_R^{\max} f.$$

The right side is in L^2; scalar elliptic regularity upgrades $D_R^{\max} u$ to H^2, and the scalar resolvent proves equation (13.71). Both resolvents of the self-adjoint gauge-fixed operator preserve $\mathscr{S}_R$; adjunction shows that they also preserve its orthogonal complement. Therefore the slice is reducing and its operator part is self-adjoint with the displayed domain. The reducing projection commutes with the spectral calculus, so the form domain is exactly $H^1\Omega^1 \cap \mathscr{S}_R$. The Hessian identity is the restriction of the last assertion of theorem 13.3. □

13.6 Charge two, all polarizations, and the Morse index

Choose root vectors with $[H_{\mathrm{mag}}, E_\pm] = \pm 2i E_\pm$. Then

$$(\operatorname{ad} \mathscr{E}_{\eta,R})_{\mathbb{C}} = \underline{\mathbb{C}} H_{\mathrm{mag}} \oplus \mathscr{L}_{\eta,R}^{\otimes 2} E_+ \oplus \mathscr{L}_{\eta,R}^{\otimes(-2)} E_-. \tag{13.74}$$

The E_+ root connection has curvature $-4\pi i \eta_R$. Replacing E_η by $2E_\eta$ gives $J_{2,R} = J_{\eta,R}$, doubles both frequencies, and changes the vacuum dimension from six to twenty-four by equation (13.36). The E_- root has $q = -2$, uses the conjugate auxiliary structure $J_{-2,R} = -J_{\eta,R}$, and has the same positive frequencies and the same dimension $6(-2)^2 = 24$. Repeating the complete ladder proof of theorem 13.1 therefore gives, for either root, the joint charge-two scalar levels

$$E^{(2)}_{R;p,q} = 2(2p+1)\mathcal{B}_{R,+} + 2(2q+1)\mathcal{B}_{R,-}, \qquad p, q \in \mathbb{Z}_{\geq 0}, \tag{13.75}$$

each of complex dimension

$$|\operatorname{Pf}(2E_\eta)| = 24. \tag{13.76}$$

Coincident joint pairs must again be summed.

Choose $O_R \in SO(2)$ with $O_R^{\mathsf{T}} K_R O_R = \mathrm{diag}(\kappa_{R,+}, \kappa_{R,-})$, and define the constant orthonormal coframe

$$\begin{pmatrix} \mathrm{d}X_+ \\ \mathrm{d}X_- \end{pmatrix} = O_R^{\mathsf{T}} \begin{pmatrix} \mathrm{d}x^1 \\ \mathrm{d}x^3 \end{pmatrix}, \qquad \begin{pmatrix} \mathrm{d}Y_+ \\ \mathrm{d}Y_- \end{pmatrix} = O_R^{\mathsf{T}} \begin{pmatrix} \mathrm{d}x^2 \\ \mathrm{d}x^4 \end{pmatrix}. \tag{13.77}$$

Then

$$\eta_R = \frac{\mathcal{B}_{R,+}}{2\pi} \mathrm{d}X_+ \wedge \mathrm{d}Y_+ - \frac{\mathcal{B}_{R,-}}{2\pi} \mathrm{d}X_- \wedge \mathrm{d}Y_-. \tag{13.78}$$

The odd grouping permutation in equation (13.77) reverses orientation; it is used only to expose the two blocks and never to infer a Chern sign.

On the E_+ root line, the four constant circular one-forms diagonalize the spin endomorphism. Direct substitution into equation (13.66), with the charge-two root normalization retained, gives

polarization	V_R-shift	$\mathcal{J}_R^{\mathrm{gf}}$-eigenvalue on (p,q)
$\mathrm{d}X_+ + i\mathrm{d}Y_+$	$-4\mathcal{B}_{R,+}$	$2[(2p-1)\mathcal{B}_{R,+} + (2q+1)\mathcal{B}_{R,-}]$
$\mathrm{d}X_+ - i\mathrm{d}Y_+$	$+4\mathcal{B}_{R,+}$	$2[(2p+3)\mathcal{B}_{R,+} + (2q+1)\mathcal{B}_{R,-}]$
$\mathrm{d}X_- - i\mathrm{d}Y_-$	$-4\mathcal{B}_{R,-}$	$2[(2p+1)\mathcal{B}_{R,+} + (2q-1)\mathcal{B}_{R,-}]$
$\mathrm{d}X_- + i\mathrm{d}Y_-$	$+4\mathcal{B}_{R,-}$	$2[(2p+1)\mathcal{B}_{R,+} + (2q+3)\mathcal{B}_{R,-}]$

The E_- table is its anti-linear conjugate. The Cartan sector is the ordinary nonnegative coefficientwise Laplacian on one-forms.

Theorem 13.5 (Exact negative spectrum, walls, and real Morse index)**.** *Put*

$$\mathcal{B}_{R,\max} = \max\{\mathcal{B}_{R,+}, \mathcal{B}_{R,-}\}, \quad \mathcal{B}_{R,\min} = \min\{\mathcal{B}_{R,+}, \mathcal{B}_{R,-}\}, \quad \rho_R^* = \frac{\mathcal{B}_{R,\max}}{\mathcal{B}_{R,\min}} = \frac{\Xi_R + |S_R|}{\Xi_R - |S_R|}. \tag{13.79}$$

For every regular R, the negative spectrum of the self-adjoint Coulomb operator is exactly

$$\left\{ \nu_{R,n} = 2[(2n+1)\mathcal{B}_{R,\min} - \mathcal{B}_{R,\max}] = \frac{4\pi}{D_R}[n\Xi_R - (n+1)|S_R|] : 2n+1 < \rho_R^* \right\}. \tag{13.80}$$

Every indicated band has real multiplicity 48*, and*

$$\mathrm{ind}_{\mathbb{R}}(\mathcal{J}_R) = 48 \left\lceil \frac{\rho_R^* - 1}{2} \right\rceil. \tag{13.81}$$

At the n-th wall the strict inequality becomes equality. Equivalently,

$$\rho_R^* = 2n+1 \iff n\Xi_R = (n+1)|S_R| \iff 24n^2 D_R = (2n+1)S_R^2. \tag{13.82}$$

The corresponding branch is then a zero branch and is not counted in the Morse index. If one removes the absolute value by choosing the $\mathcal{B}_{R,+} > \mathcal{B}_{R,-}$ side, the polynomial equation in equation (13.82) must be accompanied by $S_R > 0$; otherwise squaring introduces the opposite side as an extraneous solution.

On the sufficiently deep tail of equation (13.28), the negative branches are exactly $n = 0, 1, 2,$

$$\nu_{R,n} = \frac{4\pi}{D_R}[n\Xi_R - (n+1)S_R], \qquad n = 0, 1, 2, \tag{13.83}$$

each of real multiplicity 48*. Therefore*

$$\mathrm{ind}_{\mathbb{R}}(\mathcal{J}_R) = 144, \tag{13.84}$$

and

$$\nu_{R,0} = -\frac{20\pi}{R} + O(R^{-2}), \qquad \nu_{R,1} = -\frac{12\pi}{R} + O(R^{-2}),$$
$$\nu_{R,2} = -\frac{4\pi}{R} + O(R^{-2}), \qquad \nu_{R,3} = +\frac{4\pi}{R} + O(R^{-2}). \tag{13.85}$$

The Hessian operator for equation (13.56) has eigenvalues $\nu_{R,n}/g_{\mathrm{YM}}^2$ and the same index.

Proof. Assume first $\mathcal{B}_{R,+} > \mathcal{B}_{R,-}$. In the first row of the polarization table, $p \geq 1$ makes the eigenvalue positive. At $p = 0$ it is

$$2[-\mathcal{B}_{R,+} + (2q+1)\mathcal{B}_{R,-}],$$

which is negative exactly when $2q + 1 < \rho_R^*$. The second and fourth rows are positive, and the third has minimum $2(\mathcal{B}_{R,+} - \mathcal{B}_{R,-}) > 0$. When $\mathcal{B}_{R,-} > \mathcal{B}_{R,+}$, the identical argument with the two planes exchanged shows that only the $q = 0$ part of the third row can be negative and gives the same formula with the ordered frequencies. If the frequencies agree, both candidate lowest branches are zero and none is negative. This proves the exhaustive list for the gauge-fixed operator.

Every negative gauge-fixed eigenform is automatically Coulomb. Elliptic regularity makes it smooth; if $\mathcal{J}_R^{\mathrm{gf}} a = \lambda a$ with $\lambda < 0$, then equation (13.70) gives

$$\Delta^{(0)}_{A_{\eta,R}}(d^*_{A_{\eta,R}} a) = \lambda d^*_{A_{\eta,R}} a.$$

Nonnegativity of the scalar operator forces $d_A^* a = 0$. Hence the full and Coulomb negative spectral spaces coincide.

One joint charge-two scalar space contributes complex dimension 24 in the E_+ sector. Its conjugate contributes complex dimension 24 in the complexification of the real adjoint problem. A complexified real eigenspace has complex dimension equal to the original real dimension, so the real multiplicity is $24 + 24 = 48$, not 24 or 96.

The count of integers $n \geq 0$ satisfying $2n + 1 < \rho_R^*$ is $\lceil (\rho_R^* - 1)/2 \rceil$, including the correct exclusion at an odd integer wall. Substitution of the ordered frequency formulas gives equation (13.80). The wall equation is $n\Xi_R = (n+1)|S_R|$; squaring and using $\Xi_R^2 - S_R^2 = 24 D_R$ gives the final polynomial form, and nonnegativity of both sides gives the converse. Finally, equation (13.28) leaves precisely $n = 0, 1, 2$, and the leading frequency terms give $\nu_{R,n} = (8n - 20)\pi/R + O(R^{-2})$. □

The lowest branch and the action excess are tied by the exact identities

$$\nu_{R,0} = -2|\mathcal{B}_{R,+} - \mathcal{B}_{R,-}| = -\frac{4\pi|S_R|}{D_R}, \qquad \mathrm{YM}_R(A_{\eta,R}) - \frac{24\pi^2}{g_{\mathrm{YM}}^2} = \frac{D_R}{8g_{\mathrm{YM}}^2}\nu_{R,0}^2. \tag{13.86}$$

13.7 The fixed and varying radial direct integrals

Choose $R_1 > R_0$ so that the complete deep-tail inequalities in equation (13.28) hold on

$$I := (R_1, \infty).$$

Let $Y = \mathbb{R}^4/\mathbb{Z}^4$. Denote by $\mathscr{L}_\eta \to Y$ the fixed marked Hermitian line bundle defined by the factor of automorphy in equation (13.9). The marking diffeomorphism and its transported bundle isomorphism are

$$\Phi_R : Y \longrightarrow F_R, \qquad [y] \longmapsto [P_R y], \qquad \widehat{\Phi}_R : \mathscr{L}_\eta \longrightarrow \mathscr{L}_{\eta,R}.$$

They are fixed by requiring

$$\widehat{\Phi}_R^{*}\nabla_{\eta,R} = \mathrm{d} - 2\pi i(6y^1\,\mathrm{d}y^4 + y^2\,\mathrm{d}y^3),$$

so no gauge, metric, lattice, or orientation normalization is hidden in the identification. Put

$$\mathscr{K}_R := L^2(F_R, \mathscr{L}_{\eta,R}; \mathrm{d}\,\mathrm{vol}_{g_R}).$$

The pullback of the physical volume is

$$\Phi_R^{*}(\mathrm{d}\,\mathrm{vol}_{g_R}) = \sqrt{\det(P_R^{\mathsf{T}}P_R)}\,\mathrm{d}^4y = D_R\,\mathrm{d}^4y. \tag{13.87}$$

Put

$$\mathscr{H}_* := L^2(Y, \mathscr{L}_\eta; \mathrm{d}^4y), \qquad \mathscr{D}_* := H^2(Y; \mathscr{L}_\eta),$$

where the Sobolev space is defined using any one fixed smooth reference metric and connection on the fixed compact bundle. Define

$$U_R : \mathscr{K}_R \longrightarrow \mathscr{H}_*, \qquad U_R s := D_R^{1/2}\widehat{\Phi}_R^{*}s. \tag{13.88}$$

Then

$$\|U_R s\|_{\mathscr{H}_*}^2 = D_R \int_Y |\widehat{\Phi}_R^{*}s|^2\,\mathrm{d}^4y = \|s\|_{\mathscr{K}_R}^2,$$

and

$$U_R^{-1}v = D_R^{-1/2}(\widehat{\Phi}_R^{-1})^{*}v.$$

Thus U_R is an onto unitary, not merely an isometric injection.

Transport the fibre operator to the fixed space:

$$A_R := U_R\Delta_{\eta,R}U_R^{-1}, \qquad \mathrm{Dom}(A_R) = \mathscr{D}_*. \tag{13.89}$$

Its closed quadratic form is

$$\widetilde{\mathfrak{a}}_R[v,w] = \int_Y (G_R^{-1})^{jk}\langle\nabla_{\eta,j}v, \nabla_{\eta,k}w\rangle\,\mathrm{d}^4y, \qquad G_R = P_R^{\mathsf{T}}P_R. \tag{13.90}$$

Here $\nabla_{\eta,j}$ is the $\mathrm{d}y^j$-component of the fixed marked connection displayed above. The factor $D_R^{1/2}$ in U_R is constant in y, and it cancels the factor D_R in (13.87); hence no unrecorded factor appears in (13.90).

Lemma 13.6 (Common domain and measurable resolvents)**.** *For every compact interval $K \Subset I$, the family*

$$R \longmapsto A_R \in \mathcal{B}(\mathscr{D}_*, \mathscr{H}_*)$$

is norm-continuous, in fact real analytic. For every $z \in \mathbb{C}\setminus\mathbb{R}$, the resolvents $(A_R - z)^{-1}$ are locally norm-continuous as operators on $\mathscr{H}_*$, and hence form a measurable operator field.

Proof. On K, the matrices G_R^{-1} are uniformly positive definite, and their entries are real analytic in R. In fixed local bundle trivializations, (13.90) therefore gives a uniformly elliptic second-order expression whose coefficients depend real analytically on R. Elliptic regularity identifies every operator domain with the same set $\mathscr{D}_*$, and the standard local coordinate estimate for a differential operator of order two gives norm-continuity in $\mathcal{B}(\mathscr{D}_*, \mathscr{H}_*)$.

Uniform ellipticity on K gives

$$\|v\|_{\mathscr{D}_*} \le C_K(\|A_R v\|_{\mathscr{H}_*} + \|v\|_{\mathscr{H}_*}), \qquad R \in K.$$

For $v = (A_R - z)^{-1} f$, self-adjointness gives $\|v\| \le |\operatorname{Im} z|^{-1} \|f\|$, while $A_R v = f + zv$. Consequently

$$\sup_{R \in K} \|(A_R - z)^{-1}\|_{\mathscr{H}_* \to \mathscr{D}_*} < \infty. \tag{13.91}$$

The resolvent identity, with its exact sign, for $R, S \in K$,

$$(A_R - z)^{-1} - (A_S - z)^{-1} = -(A_R - z)^{-1}(A_R - A_S)(A_S - z)^{-1},$$

together with (13.91), proves local norm-resolvent continuity. A countable compact exhaustion of I gives measurability on all of I. □

We now define the varying measurable Hilbert field rather than presuppose one. A section $R \mapsto s_R \in \mathscr{K}_R$ is declared measurable precisely when

$$R \longmapsto U_R s_R \in \mathscr{H}_*$$

is strongly measurable. This is the measurable field pulled back from the constant separable field $I \times \mathscr{H}_*$. Its direct integral is

$$\mathscr{H}_{\mathrm{var}} := \int_I^{\oplus} \mathscr{K}_R \, \mathrm{d}R.$$

The fixed radial Hilbert space is

$$\mathscr{H}_{\mathrm{rad}} := L^2(I, \mathrm{d}R; \mathscr{H}_*).$$

By construction,

$$\mathscr{U} := \int_I^{\oplus} U_R \, \mathrm{d}R : \mathscr{H}_{\mathrm{var}} \longrightarrow \mathscr{H}_{\mathrm{rad}}, \qquad (\mathscr{U} s)(R) = U_R s_R, \tag{13.92}$$

is an isometry. It is onto: if $\Psi \in \mathscr{H}_{\mathrm{rad}}$, then $s_R = U_R^{-1} \Psi(R)$ is measurable by the very definition of the varying field and has the same squared norm integral.

On the fixed direct integral, define the maximal multiplication operator

$$(\widetilde{\mathscr{M}_\eta} \Psi)(R) := A_R \Psi(R) \tag{13.93}$$

with the exact domain

$$\begin{aligned} \operatorname{Dom}(\widetilde{\mathscr{M}_\eta}) := \{ \Psi \in \mathscr{H}_{\mathrm{rad}} : {} & \Psi(R) \in \mathscr{D}_* \text{ for almost every } R, \\ & R \mapsto A_R \Psi(R) \text{ is strongly measurable}, \\ & \int_I \|A_R \Psi(R)\|_{\mathscr{H}_*}^2 \, \mathrm{d}R < \infty \}. \end{aligned} \tag{13.94}$$

On the varying direct integral, define

$$\begin{aligned} \operatorname{Dom}(\mathscr{M}_\eta) := \{ s \in \mathscr{H}_{\mathrm{var}} : {} & s_R \in \operatorname{Dom}(\Delta_{\eta,R}) \text{ for almost every } R, \\ & R \mapsto U_R \Delta_{\eta,R} s_R \text{ is strongly measurable}, \\ & \int_I \|\Delta_{\eta,R} s_R\|_{\mathscr{K}_R}^2 \, \mathrm{d}R < \infty \}, \end{aligned} \tag{13.95}$$

and

$$(\mathscr{M}_\eta s)_R := \Delta_{\eta,R} s_R \quad \text{for almost every } R. \tag{13.96}$$

Every expression in (13.93) and (13.96) is now typed on its actual fibre. Equations (13.89), (13.92), (13.94), and (13.95) give both the domain equality and the exact intertwining identity

$$\mathscr{U} \operatorname{Dom}(\mathscr{M}_\eta) = \operatorname{Dom}(\widetilde{\mathscr{M}_\eta}), \qquad \mathscr{U} \mathscr{M}_\eta \mathscr{U}^{-1} = \widetilde{\mathscr{M}_\eta}. \tag{13.97}$$

13.7.1 Density and self-adjointness on the maximal domain

Proposition 13.7 (Maximal-domain realization)**.** *The operators $\widetilde{\mathscr{M}_\eta}$ and $\mathscr{M}_\eta$ are densely defined, nonnegative, and self-adjoint on the domains* (13.94) *and* (13.95)*, respectively.*

Proof. For $\chi \in C_c(I)$ and $v \in \mathscr{D}_*$, local continuity of $R \mapsto A_R v$ makes it bounded on $\operatorname{supp}\chi$. Thus $\chi v \in \operatorname{Dom}(\mathscr{M}_\eta)$. Since $\mathscr{D}_*$ is dense in $\mathscr{H}_*$, the algebraic tensor product

$$C_c(I) \odot \mathscr{D}_*$$

is dense in $L^2(I;\mathscr{H}_*)$. Hence the fixed operator is densely defined. Fibrewise self-adjointness and Cauchy–Schwarz give, for Ψ, Θ in its domain,

$$\langle \widetilde{\mathscr{M}_\eta}\Psi, \Theta\rangle = \int_I \langle A_R\Psi(R), \Theta(R)\rangle \, \mathrm{d}R = \langle \Psi, \widetilde{\mathscr{M}_\eta}\Theta\rangle,$$

so it is symmetric.

Fix $z \in \mathbb{C} \setminus \mathbb{R}$. For $F \in \mathscr{H}_{\mathrm{rad}}$, define pointwise

$$(\mathcal{R}_z F)(R) := (A_R - z)^{-1} F(R). \tag{13.98}$$

The locally norm-continuous resolvent field, approximation of F by simple functions, and

$$\|(A_R - z)^{-1}\|_{\mathscr{H}_*\to\mathscr{H}_*} \le |\operatorname{Im} z|^{-1}$$

show that $\mathcal{R}_z F$ is strongly measurable and

$$\|\mathcal{R}_z F\|_{\mathscr{H}_{\mathrm{rad}}} \le |\operatorname{Im} z|^{-1} \|F\|_{\mathscr{H}_{\mathrm{rad}}}. \tag{13.99}$$

Pointwise, $(A_R - z)^{-1} F(R) \in \mathscr{D}_*$. The exact identity

$$A_R(A_R - z)^{-1} = I + z(A_R - z)^{-1} \tag{13.100}$$

gives the global uniform bound

$$\|A_R(A_R - z)^{-1}\| \le 1 + \frac{|z|}{|\operatorname{Im} z|}.$$

The right-hand side of (13.100), applied to $F(R)$, is strongly measurable and square-integrable. Therefore $\mathcal{R}_z F$ belongs to the full maximal domain (13.94); this is the domain step that pointwise inversion alone does not prove.

Fibrewise identities now yield

$$(\widetilde{\mathscr{M}_\eta} - z)\mathcal{R}_z = I \quad \text{on } \mathscr{H}_{\mathrm{rad}}, \qquad \mathcal{R}_z(\widetilde{\mathscr{M}_\eta} - z) = I \quad \text{on } \operatorname{Dom}(\widetilde{\mathscr{M}_\eta}).$$

In particular, the ranges of $\widetilde{\mathscr{M}_\eta} - i$ and $\widetilde{\mathscr{M}_\eta} + i$ are both all of $\mathscr{H}_{\mathrm{rad}}$. The range criterion for a densely defined symmetric operator proves self-adjointness. Moreover,

$$\langle \widetilde{\mathscr{M}_\eta}\Psi, \Psi\rangle = \int_I \langle A_R\Psi(R), \Psi(R)\rangle \, \mathrm{d}R \ge 0,$$

so it is nonnegative. The unitary equivalence (13.97) transfers density, self-adjointness, and nonnegativity to the varying-fibre operator. □

13.7.2 Measurable eigenvectors, including at band crossings

Lemma 13.8 (Explicit measurable eigenvector selection)**.** *For every fixed* $\mathbf{n} = (n_+, n_-) \in \mathbb{Z}^2_{\geq 0}$*, there is a strongly measurable map*

$$I \longrightarrow \mathscr{D}_*, \qquad R \longmapsto \phi_{\mathbf{n}}(R),$$

such that

$$\|\phi_{\mathbf{n}}(R)\|_{\mathscr{H}_*} = 1, \qquad A_R\phi_{\mathbf{n}}(R) = E_{\mathbf{n}}(R)\phi_{\mathbf{n}}(R) \tag{13.101}$$

for every $R \in I$*. The construction remains valid when* $E_{\mathbf{n}}(R)$ *crosses one or more other bands.*

Proof. Set

$$C_{\mathbf{n},R} := A_R - E_{\mathbf{n}}(R), \qquad \mathrm{Dom}(C_{\mathbf{n},R}) = \mathscr{D}_*.$$

The band $E_{\mathbf{n}}$ is continuous. The proof of Lemma 13.6, applied also to the scalar shift, shows that $(C_{\mathbf{n},R} - z)^{-1}$ is a measurable resolvent field for every nonreal z.

Let

$$\mathsf{P}_{\mathbf{n}}(R) := \mathbf{1}_{\{0\}}(C_{\mathbf{n},R}) = \mathbf{1}_{\{E_{\mathbf{n}}(R)\}}(A_R).$$

This singleton projection is not assumed continuous at a crossing. Instead, the spectral theorem gives the pointwise strong limit

$$\mathsf{P}_{\mathbf{n}}(R) = \operatorname*{s-lim}_{k\to\infty}(I + kC^2_{\mathbf{n},R})^{-1}, \tag{13.102}$$

and the exact resolvent formula

$$(I + kC^2_{\mathbf{n},R})^{-1} = \frac{1}{2i\sqrt{k}}\left[(C_{\mathbf{n},R} - i/\sqrt{k})^{-1} - (C_{\mathbf{n},R} + i/\sqrt{k})^{-1}\right] \tag{13.103}$$

shows that every approximating field is strongly measurable. Therefore $R \mapsto \mathsf{P}_{\mathbf{n}}(R)v$ is strongly measurable for every $v \in \mathscr{H}_*$. Proposition 13.1 says that $\mathsf{P}_{\mathbf{n}}(R) \neq 0$ for every R, including crossings.

Choose a countable dense sequence $(v_k)_{k\geq 1}$ in the separable space $\mathscr{H}_*$, and define the measurable partition

$$X_k = \{R : \|\mathsf{P}_{\mathbf{n}}(R)v_k\| > 0\} \cap \bigcap_{\ell<k} \{R : \|\mathsf{P}_{\mathbf{n}}(R)v_\ell\| = 0\}.$$

The sets X_k are pairwise disjoint and cover I: a nonzero bounded projection cannot annihilate a dense set. On X_k, set

$$\phi_{\mathbf{n}}(R) = \frac{\mathsf{P}_{\mathbf{n}}(R)v_k}{\|\mathsf{P}_{\mathbf{n}}(R)v_k\|}.$$

Countable measurable patching gives a strongly measurable unit vector. The range of the singleton spectral projection lies in $\mathrm{Dom}(A_R)$ and satisfies $(A_R - E_{\mathbf{n}}(R))\phi_{\mathbf{n}}(R) = 0$, which proves (13.101). □

13.7.3 Full spectrum and spectral type

Theorem 13.9 (Purely continuous full half-line spectrum)**.** *For both the fixed-fibre and varying-fibre maximal radial multiplication operators,*

$$\begin{aligned} \sigma(\widetilde{\mathscr{M}_\eta}) = \sigma_{\mathrm{ess}}(\widetilde{\mathscr{M}_\eta}) = \sigma_{\mathrm{c}}(\widetilde{\mathscr{M}_\eta}) = [0,\infty), \\ \sigma_{\mathrm{p}}(\widetilde{\mathscr{M}_\eta}) = \varnothing, \qquad \ker \widetilde{\mathscr{M}_\eta} = \{0\}, \end{aligned} \tag{13.104}$$

and

$$\begin{aligned}\sigma(\mathscr{M}_\eta) = \sigma_{\mathrm{ess}}(\mathscr{M}_\eta) = \sigma_{\mathrm{c}}(\mathscr{M}_\eta) = [0,\infty),\\ \sigma_{\mathrm{p}}(\mathscr{M}_\eta) = \varnothing, \qquad \ker \mathscr{M}_\eta = \{0\}.\end{aligned} \tag{13.105}$$

Here σ_{ess} is the Weyl essential spectrum.

Proof. It is enough to prove (13.104), because (13.97) is a unitary equivalence. Proposition 13.7 and nonnegativity give

$$\sigma(\widetilde{\mathscr{M}_\eta}) \subset [0,\infty).$$

The zero-energy singular sequence. Let $a_j = R_1 + 2j + 1$. Using the measurable normalized ground eigenvector from Lemma 13.8, define

$$\Psi_j(R) := \mathbf{1}_{[a_j, a_j+1]}(R)\phi_{(0,0)}(R).$$

These vectors have norm one and have pairwise disjoint supports, hence are orthonormal and converge weakly to zero. They belong to the maximal domain because

$$A_R\Psi_j(R) = \mathbf{1}_{[a_j, a_j+1]}(R)E_0(R)\phi_{(0,0)}(R)$$

is strongly measurable and square-integrable. From (13.45),

$$\|\widetilde{\mathscr{M}_\eta}\Psi_j\|^2 = \int_{a_j}^{a_j+1} E_0(R)^2\,\mathrm{d}R = \frac{196\pi^2}{a_j^2} + O(a_j^{-3}) \longrightarrow 0.$$

Thus (Ψ_j) is a singular Weyl sequence at zero, and

$$0 \in \sigma_{\mathrm{ess}}(\widetilde{\mathscr{M}_\eta}).$$

Positive energy: escaping crossings before disjoint supports. Fix $\lambda > 0$, and fix once and for all $R_\bullet = R_1 + 1 \in I$. For $n \in \mathbb{Z}_{\geq 0}$, retain the specific canonical band

$$e_n(R) := E_{(0,n)}(R) = \mathcal{B}_{R,+} + (2n+1)\mathcal{B}_{R,-}. \tag{13.106}$$

Since $\mathcal{B}_{R_\bullet,-} > 0$,

$$e_n(R_\bullet) \longrightarrow \infty \quad (n \to \infty).$$

For each fixed n, equations (13.25) and (13.26) give

$$e_n(R) \longrightarrow 0 \quad (R \to \infty).$$

Hence, for every sufficiently large n, the intermediate value theorem gives at least one crossing parameter $r_n > R_\bullet$ satisfying

$$e_n(r_n) = \lambda. \tag{13.107}$$

We prove explicitly that every such choice of crossing parameters escapes to infinity. Fix $M > R_\bullet$. Positivity and continuity give

$$c_M := \min_{R\in[R_\bullet, M]} \mathcal{B}_{R,-} > 0.$$

For every $R \in [R_\bullet, M]$,

$$e_n(R) \geq (2n+1)c_M.$$

Thus, once $(2n+1)c_M > \lambda$, equation (13.107) has no solution in $[R_\bullet, M]$, and the chosen r_n must satisfy $r_n > M$. Since M was arbitrary,

$$r_n \longrightarrow \infty. \tag{13.108}$$

Only now, before choosing any support, pass to a subsequence $n_j \to \infty$ such that, with $r_j := r_{n_j}$,

$$r_{j+1} > r_j + 2. \tag{13.109}$$

Continuity of e_{n_j} at r_j permits the subsequent choice of $\varepsilon_j \in (0, \frac{1}{4})$ such that

$$J_j := (r_j - \varepsilon_j, r_j + \varepsilon_j) \Subset I, \qquad \sup_{R \in J_j} |e_{n_j}(R) - \lambda| < \frac{1}{j}.$$

The order of construction matters: the separation (13.109) makes the intervals J_j pairwise disjoint.

Take the measurable normalized eigenvector $\phi_{(0,n_j)}(R)$ from Lemma 13.8; the singleton projection used there remains valid if the selected band crosses another one inside J_j. Define

$$\Theta_j(R) := |J_j|^{-1/2} \mathbf{1}_{J_j}(R) \phi_{(0,n_j)}(R). \tag{13.110}$$

The sections Θ_j have norm one and are orthogonal by disjointness of their supports. They lie in the maximal domain because

$$A_R \Theta_j(R) = |J_j|^{-1/2} \mathbf{1}_{J_j}(R) e_{n_j}(R) \phi_{(0,n_j)}(R)$$

is strongly measurable and square-integrable. Moreover,

$$\|(\widetilde{\mathscr{M}_\eta} - \lambda)\Theta_j\|^2 = \frac{1}{|J_j|} \int_{J_j} |e_{n_j}(R) - \lambda|^2 \, \mathrm{d}R < \frac{1}{j^2}.$$

Every orthonormal sequence is weakly null, so (Θ_j) is a singular Weyl sequence at λ. Thus every $\lambda > 0$ belongs to the Weyl essential spectrum. Together with the zero sequence and nonnegativity, this proves

$$\sigma(\widetilde{\mathscr{M}_\eta}) = \sigma_{\mathrm{ess}}(\widetilde{\mathscr{M}_\eta}) = [0, \infty). \tag{13.111}$$

Real analyticity, absence of flat bands, and no eigenvalues. Equation (13.20) writes

$$\Xi_R = \sqrt{S_R^2 + 24 D_R}$$

with a strictly positive real-analytic radicand. The positive square-root function is real analytic on $(0, \infty)$. Hence Ξ_R, $\mathcal{B}_{R,+}$, $\mathcal{B}_{R,-}$, and every $E_{\mathbf{n}}(R)$ are real analytic on the connected interval I. For each fixed $\mathbf{n}$,

$$E_{\mathbf{n}}(R) > 0 \quad (R < \infty), \qquad \lim_{R \to \infty} E_{\mathbf{n}}(R) = 0.$$

It follows that no band is constant: a constant band would have constant value equal to its limit zero, contradicting strict positivity at every finite R. This proves the absence of flat bands, rather than assuming it.

Fix $\lambda \in \mathbb{R}$. For every $\mathbf{n}$, the real-analytic function $E_{\mathbf{n}} - \lambda$ is not identically zero, so its zero set in I is discrete and has Lebesgue measure zero. Since $\mathbb{Z}^2_{\geq 0}$ is countable,

$$Z_\lambda := \{R \in I : \lambda \in \sigma_{\mathrm{p}}(A_R)\} = \bigcup_{\mathbf{n} \in \mathbb{Z}^2_{\geq 0}} \{R : E_{\mathbf{n}}(R) = \lambda\} \tag{13.112}$$

is a null set.

If $\widetilde{\mathscr{M}_\eta}\Psi = \lambda\Psi$, then $(A_R - \lambda)\Psi(R) = 0$ for almost every R. At every point where $\Psi(R) \neq 0$, Proposition 13.1 forces $R \in Z_\lambda$. Thus the essential support of Ψ is contained in the null set (13.112), and $\Psi = 0$. A self-adjoint operator has only real eigenvalues, so

$$\sigma_{\mathrm{p}}(\widetilde{\mathscr{M}_\eta}) = \varnothing, \qquad \ker \widetilde{\mathscr{M}_\eta} = \{0\}.$$

Self-adjoint operators have empty residual spectrum. Combining this with (13.111) proves that the entire spectrum is continuous and gives (13.104). Unitary equivalence gives (13.105). □

13.7.4 Exact interpretive boundary

Remark 13.10 (No radial derivative and no interacting Yang–Mills claim). The variable R in (13.93) and (13.96) is a direct-integral parameter. Neither operator contains a R-derivative. In particular, there is no term such as $-\partial_R^2$, no radial Sobolev condition, no cusp boundary condition, and no interaction between distinct R-fibres. This is why characteristic functions of radial intervals legitimately occur in the maximal-domain Weyl sequences above.

The theorem is therefore a complete spectral statement for the specified auxiliary decomposable family of native full-flux magnetic Laplacians. It does not construct, identify, or determine the spectrum or mass gap of an interacting quantum Yang–Mills Hamiltonian, and it makes no Osterwalder–Schrader reconstruction claim.

14 The native full-flux split-Cartan $SU(r)$ extension

This section fixes an integer $r \geq 2$ and extends the full-flux construction of section 13 from its already proved split $SU(2)$ case to arbitrary split-Cartan $SU(r)$ backgrounds. Every object remains on the same regular flat four-torus $F_R = \mathbb{R}^4/P_R\mathbb{Z}^4$, with the unrescaled Euclidean quotient metric. In particular, the integral two-form η_R, the line bundle $\mathscr{L}_{\eta,R}$, the two orientations, and the exact quantities

$$L_R, \quad q_R, \quad m_R, \quad D_R = L_R q_R + 6m_R^2 > 0,$$
$$S_R = 6L_R - q_R, \qquad \Xi_R = \sqrt{(6L_R + q_R)^2 + 144m_R^2}.$$

are precisely those of equations (13.1) and (13.18). The charge-one frequencies are

$$\mathcal{B}_{R,+} = \frac{\pi(S_R + \Xi_R)}{D_R}, \qquad \mathcal{B}_{R,-} = \frac{\pi(\Xi_R - S_R)}{D_R}, \tag{14.1}$$

and we shall use the globally ordered notation

$$\begin{aligned} \mathcal{B}_{R,\max} &= \max\{\mathcal{B}_{R,+}, \mathcal{B}_{R,-}\}, \quad \mathcal{B}_{R,\min} = \min\{\mathcal{B}_{R,+}, \mathcal{B}_{R,-}\}, \\ \rho_R^* &= \frac{\mathcal{B}_{R,\max}}{\mathcal{B}_{R,\min}}. \end{aligned} \tag{14.2}$$

Thus every assertion below remains valid on either side of $S_R = 0$.

14.1 The split bundle, determinant, and Chern number

Fix a nonzero trace-zero integral weight vector

$$\mathbf{m} = (m_1, \ldots, m_r) \in \mathbb{Z}^r, \qquad \sum_{j=1}^r m_j = 0, \qquad s_2(\mathbf{m}) := \sum_{j=1}^r m_j^2. \tag{14.3}$$

Define

$$\mathscr{E}_{\mathbf{m},R} := \bigoplus_{j=1}^{r} \mathscr{L}_{\eta,R}^{\otimes m_j}, \qquad H_{\mathbf{m}} := i\,\mathrm{diag}(m_1,\dots,m_r) \in \mathfrak{su}(r). \tag{14.4}$$

Proposition 14.1 (Parallel determinant trivialization and exact c_2)**.** *The ordered splitting in equation (14.4) has a canonical parallel determinant trivialization. With $c = [\eta_R]$,*

$$c_1(\mathscr{E}_{\mathbf{m},R}) = 0, \qquad c_2(\mathscr{E}_{\mathbf{m},R}) = -\frac{s_2(\mathbf{m})}{2}c^2. \tag{14.5}$$

The marked and physical-complex Chern numbers are respectively

$$\langle c_2, [F_R]_{\mathrm{mark}}\rangle = -6s_2(\mathbf{m}), \qquad k_{\mathbf{m}} := \langle c_2, [F_R]_{x\text{-complex}}\rangle = 6s_2(\mathbf{m}) > 0. \tag{14.6}$$

Moreover $s_2(\mathbf{m})$ is even. Hence this direct-sum family realizes only $k_{\mathbf{m}} \in 12\mathbb{Z}_{>0}$, not all topological $SU(r)$ sectors.

Proof. Let $j_n^\eta(y)$ denote the factor of automorphy in equation (13.9). The transition matrix of the ordered sum is

$$\mathrm{diag}\big((j_n^\eta)^{m_1},\dots,(j_n^\eta)^{m_r}\big).$$

Its determinant is $(j_n^\eta)^{\sum_j m_j} = 1$ identically. The tensor product of the ordered unit frames therefore defines a global determinant section, and the determinant connection has connection form proportional to $\sum_j m_j = 0$; the section is parallel.

The splitting principle gives

$$c(\mathscr{E}_{\mathbf{m},R}) = \prod_{j=1}^{r}(1+m_j c), \qquad c_2 = \sum_{i<j} m_i m_j c^2.$$

Since

$$2\sum_{i<j} m_i m_j = \left(\sum_j m_j\right)^2 - \sum_j m_j^2 = -s_2(\mathbf{m}),$$

this proves equation (14.5). The two evaluations follow from equation (13.15); retaining its orientation reversal is essential for the sign. Finally

$$s_2(\mathbf{m}) = \sum_j m_j^2 \equiv \sum_j m_j \equiv 0 \pmod 2,$$

because $a^2 \equiv a \pmod 2$. This proves the divisibility statement. □

The diagonal connection induced by the summands is

$$\nabla^{\mathbf{m},R} = \mathrm{d} - 2\pi(6y^1\,\mathrm{d}y^4 + y^2\,\mathrm{d}y^3)H_{\mathbf{m}}, \qquad F_{\mathbf{m},R} = -2\pi H_{\mathbf{m}}\eta_R. \tag{14.7}$$

We retain the invariant inner product and functional normalization

$$\begin{aligned} \langle X, Y\rangle_{\mathfrak{su}(r)} &= -\frac{1}{2}\,\mathrm{Tr}(XY), \qquad \|H_{\mathbf{m}}\|^2 = \frac{1}{2}s_2(\mathbf{m}), \\ \mathrm{YM}_R(A) &= \frac{1}{2g_{\mathrm{YM}}^2}\int_{F_R} |F_A|^2\,\mathrm{d}\,\mathrm{vol}_R\,. \end{aligned} \tag{14.8}$$

Theorem 14.2 (Yang–Mills criticality and exact action)**.** *The connection $\nabla^{\mathbf{m},R}$ is non-flat, reducible, and Yang–Mills. Its action is*

$$\mathrm{YM}_R(\nabla^{\mathbf{m},R}) = \frac{\pi^2 s_2(\mathbf{m})}{g_{\mathrm{YM}}^2 D_R}(q_R^2 + 36L_R^2 + 72m_R^2) \tag{14.9}$$

$$= \frac{2\pi^2 k_{\mathbf{m}}}{g_{\mathrm{YM}}^2} + \frac{\pi^2 s_2(\mathbf{m}) S_R^2}{g_{\mathrm{YM}}^2 D_R}. \tag{14.10}$$

On the canonical cusp tail,

$$\mathrm{YM}_R(\nabla^{\mathbf{m},R}) = \frac{\pi^2 s_2(\mathbf{m})}{g_{\mathrm{YM}}^2}\left(37 + \frac{35B}{R} + O(R^{-2})\right). \tag{14.11}$$

Proof. Both $H_{\mathbf{m}}$ and η_R are parallel for the diagonal connection and flat quotient metric, and their brackets vanish. Hence $d^*_{\nabla^{\mathbf{m},R}} F_{\mathbf{m},R} = 0$. Non-flatness follows from $\mathbf{m} \neq 0$ and $\eta_R \neq 0$.

In the physical orthonormal frame used in equation (13.78),

$$|\eta_R|^2 = \frac{\mathcal{B}_{R,+}^2 + \mathcal{B}_{R,-}^2}{4\pi^2}.$$

Using equation (14.8), $\mathrm{vol}(F_R) = D_R$, and the exact frequency square sum from the native module yields equation (14.9). The identity

$$q_R^2 + 36L_R^2 + 72m_R^2 = 12(L_R q_R + 6m_R^2) + (6L_R - q_R)^2 = 12D_R + S_R^2$$

together with $k_{\mathbf{m}} = 6s_2(\mathbf{m})$, proves equation (14.10). Substituting the retained expansions of L_R, q_R, m_R, D_R, without suppressing H, B, m, gives

$$\frac{q_R^2 + 36L_R^2 + 72m_R^2}{D_R} = 37 + \frac{35B}{R} + O(R^{-2}),$$

which proves the last formula. □

14.2 Every nonzero integral root charge

The native module proves the charge-one ladder decomposition and the charge-two specialization used by its $SU(2)$ theorem. The following lemma records, and proves, the exact integer-charge extension needed here.

Lemma 14.3 (Charge-d Landau decomposition)**.** *Let $d \in \mathbb{Z} \setminus \{0\}$, put $q = |d|$, and equip $\mathscr{L}_{\eta,R}^{\otimes d}$ with its tensor-power connection. Its scalar covariant Laplacian has joint levels*

$$q\big((2a+1)\mathcal{B}_{R,+} + (2b+1)\mathcal{B}_{R,-}\big), \qquad a, b \in \mathbb{Z}_{\geq 0}, \tag{14.12}$$

and every joint space has complex dimension $6q^2$. If different pairs (a, b) give the same numerical value, their orthogonal joint spaces are to be summed.

Proof. The curvature is $-2\pi i d\eta_R$. In the polar construction of theorem 13.1, replace the auxiliary complex structure by $J_{d,R} = \mathrm{sgn}(d) J_{\eta,R}$. Then $d\eta_R$ is positive of type $(1,1)$, so the vacuum is the holomorphic-section space of the positive bundle $\mathscr{L}_{\eta,R}^{\otimes d}$ on that auxiliary complex torus. Its tangent and canonical bundles are trivial. Kodaira vanishing and Hirzebruch–Riemann–Roch give

$$\dim_{\mathbb{C}} H^0 = \frac{1}{2}\int_{[F_R]_{d\eta\text{-symp}}} (d\eta_R)^2 = \frac{1}{2}q^2 \int_{[F_R]_{\mathrm{mark}}} \eta_R^2 = 6q^2.$$

Here changing d to $-d$ does not change the four-dimensional symplectic orientation. Scaling the curvature by d scales both positive oscillator frequencies by q, while the two canonical commutation relations and the lowering argument in theorem 13.1 are unchanged after this rescaling. Thus every joint level is unitarily obtained from the vacuum and has the asserted energy and dimension. Orthogonal joint spaces remain a direct sum when their numerical energies coincide. □

14.3 Closed gauge-fixed and Coulomb operators

Write $A = \nabla^{\mathbf{m},R}$ in this subsection. Let

$$\mathscr{H}^1_{\mathbf{m},R} = L^2\Omega^1(F_R; \operatorname{ad}\mathscr{E}_{\mathbf{m},R}), \qquad \mathscr{H}^0_{\mathbf{m},R} = L^2\Omega^0(F_R; \operatorname{ad}\mathscr{E}_{\mathbf{m},R}) \tag{14.13}$$

as real Hilbert spaces. Define the maximal distributional codifferential by

$$\begin{aligned} \operatorname{Dom} D^{\max}_{\mathbf{m},R} = \{a \in \mathscr{H}^1_{\mathbf{m},R} : \exists u \in \mathscr{H}^0_{\mathbf{m},R} \text{ such that} \\ \langle a, d_A\phi\rangle_{L^2} = \langle u, \phi\rangle_{L^2} \quad \text{for every} \\ \phi \in C^\infty(F_R; \operatorname{ad}\mathscr{E}_{\mathbf{m},R})\}, \end{aligned} \tag{14.14}$$

$$D^{\max}_{\mathbf{m},R} a = u, \qquad \mathscr{S}_{\mathbf{m},R} := \ker D^{\max}_{\mathbf{m},R}. \tag{14.15}$$

Density of smooth test scalars makes u unique. If $a_j \to a$ and $D^{\max}_{\mathbf{m},R} a_j \to u$ in L^2, passage to the limit in the defining identity gives $D^{\max}_{\mathbf{m},R} a = u$. Thus $D^{\max}_{\mathbf{m},R}$ is closed and $\mathscr{S}_{\mathbf{m},R}$ is a closed subspace.

Let $\Delta^{(0)}_A$ be the nonnegative self-adjoint scalar operator of the closed form

$$\mathfrak{q}^{(0)}_{\mathbf{m},R}[\phi,\psi] = \langle d_A\phi, d_A\psi\rangle_{L^2}, \qquad \operatorname{Dom}\mathfrak{q}^{(0)}_{\mathbf{m},R} = H^1\Omega^0. \tag{14.16}$$

It has domain $H^2\Omega^0$ and equals $d^*_A d_A$. In a constant orthonormal frame set

$$(V_{\mathbf{m},R}a)_\nu = -2\sum_{\mu=1}^{4} [F_{\mathbf{m},R,\nu\mu}, a_\mu]. \tag{14.17}$$

Invariance of the compact Lie-algebra inner product and $F_{\nu\mu} = -F_{\mu\nu}$ show directly that $V_{\mathbf{m},R}$ is a bounded symmetric endomorphism. Define

$$\mathfrak{j}^{\mathrm{gf}}_{\mathbf{m},R}[a,b] = \langle \nabla_A a, \nabla_A b\rangle_{L^2} + \langle V_{\mathbf{m},R}a, b\rangle_{L^2}, \qquad \operatorname{Dom}\mathfrak{j}^{\mathrm{gf}}_{\mathbf{m},R} = H^1\Omega^1. \tag{14.18}$$

Proposition 14.4 (Closed realization and exact Coulomb reduction)**.** *The form in equation (14.18) is densely defined, closed, symmetric, and bounded below. Its self-adjoint operator has compact resolvent, domain $H^2\Omega^1$, and expression*

$$(\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},R}a)_\nu = -\sum_{\mu=1}^{4} \nabla_{A,e_\mu}\nabla_{A,e_\mu}a_\nu - 2\sum_{\mu=1}^{4}[F_{\mathbf{m},R,\nu\mu}, a_\mu]. \tag{14.19}$$

For every $z \in \mathbb{C}\setminus\mathbb{R}$ and $f \in \operatorname{Dom} D^{\max}_{\mathbf{m},R}$,

$$D^{\max}_{\mathbf{m},R}(\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},R} - z)^{-1} f = (\Delta^{(0)}_A - z)^{-1} D^{\max}_{\mathbf{m},R} f. \tag{14.20}$$

Consequently $\mathscr{S}_{\mathbf{m},R}$ reduces $\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},R}$, and

$$\mathcal{J}_{\mathbf{m},R} := \mathcal{J}^{\mathrm{gf}}_{\mathbf{m},R}|_{\mathscr{S}_{\mathbf{m},R}}, \qquad \operatorname{Dom}\mathcal{J}_{\mathbf{m},R} = H^2\Omega^1 \cap \mathscr{S}_{\mathbf{m},R} \tag{14.21}$$

is self-adjoint, with compact resolvent and closed form $\mathfrak{j}_{\mathbf{m},R}$ *equal to the restriction of* $\mathfrak{j}^{\mathrm{gf}}_{\mathbf{m},R}$ *to* $H^1\Omega^1 \cap \mathscr{S}_{\mathbf{m},R}$*. The physical Yang–Mills Hessian on this slice is*

$$\mathrm{YM}''_R\big|_A = g_{\mathrm{YM}}^{-2}\mathfrak{j}_{\mathbf{m},R}. \tag{14.22}$$

Thus an eigenvalue ν *of* $\mathcal{J}_{\mathbf{m},R}$ *is the physical Hessian eigenvalue* ν/g_{YM}^2*, while its sign and Morse-index contribution are unchanged.*

Proof. The first term of equation (14.18), after addition of the L^2-norm, is an equivalent H^1-norm. The second term is a bounded symmetric form. The bounded perturbation theorem for closed forms and the representation theorem therefore give a lower-bounded self-adjoint operator. Its principal symbol is $|\xi|^2$ times the identity, so elliptic regularity gives the H^2 domain. Rellich compactness on the compact torus gives compact resolvent.

The two-parameter curvature expansion and the flat Weitzenböck identity, with the inner product and functional normalization in equation (14.8), give

$$\mathrm{YM}''_R\big|_A(a,b) + g_{\mathrm{YM}}^{-2}\langle d_A^* a, d_A^* b\rangle = g_{\mathrm{YM}}^{-2}\mathfrak{j}^{\mathrm{gf}}_{\mathbf{m},R}[a,b].$$

This proves the Hessian normalization after restriction to Coulomb gauge.

It remains to prove that the restriction is an operator rather than a formal constraint. For a smooth adjoint scalar ϕ, write $\nabla_\mu = \nabla_{A,e_\mu}$. Commuting all three derivatives and using the Yang–Mills equation gives

$$\begin{aligned}
&(\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},R} d_A\phi - d_A\Delta_A^{(0)}\phi)_\nu \\
&\quad = -\nabla_\mu\nabla_\mu\nabla_\nu\phi + \nabla_\nu\nabla_\mu\nabla_\mu\phi - 2[F_{\nu\mu}, \nabla_\mu\phi] \\
&\quad = -[\nabla_\mu F_{\mu\nu}, \phi] = [(d_A^* F_A)_\nu, \phi] = 0.
\end{aligned}$$

Formal adjunction yields, on smooth one-forms,

$$\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},R} d_A = d_A\Delta_A^{(0)}, \qquad d_A^*\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},R} = \Delta_A^{(0)} d_A^*. \tag{14.23}$$

For $u = (\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},R} - z)^{-1} f$, the operator domain gives $u \in H^2$, hence $D^{\max}_{\mathbf{m},R} u \in H^1$. Applying the second identity distributionally to $(\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},R} - z)u = f$ gives

$$(\Delta_A^{(0)} - z) D^{\max}_{\mathbf{m},R} u = D^{\max}_{\mathbf{m},R} f.$$

Scalar elliptic regularity and the scalar resolvent give equation (14.20). The two conjugate resolvents preserve $\mathscr{S}_{\mathbf{m},R}$; adjunction then makes its orthogonal complement invariant as well. The slice is reducing, so the self-adjoint part has the operator and form domains displayed above. □

14.4 Rootwise spectrum, coincidences, index, and nullity

For $i \neq j$, the matrix unit E_{ij} obeys

$$[H_{\mathbf{m}}, E_{ij}] = i(m_i - m_j)E_{ij},$$

so its complex root line is

$$\mathrm{Hom}(\mathscr{L}_{\eta,R}^{m_j}, \mathscr{L}_{\eta,R}^{m_i}) = \mathscr{L}_{\eta,R}^{m_i - m_j}, \qquad q_{ij} := |m_i - m_j|. \tag{14.24}$$

The factor q_{ij} is the complete charge normalization; for the existing $SU(2)$ weights $(1,-1)$, it gives $q_{12}=2$.

Let the distinct numerical values among $m_1,\dots,m_r$ have multiplicities $r_1,\dots,r_\ell$. The neutral centralizer is

$$\mathfrak{z}_{\mathbf{m}}=\{X\in\mathfrak{su}(r):[H_{\mathbf{m}},X]=0\}\cong\mathfrak{s}\left(\bigoplus_{a=1}^{\ell}\mathfrak{u}(r_a)\right),\qquad \dim_{\mathbb{R}}\mathfrak{z}_{\mathbf{m}}=\sum_{a=1}^{\ell}r_a^2-1. \tag{14.25}$$

Theorem 14.5 (Complete negative spectrum and numerical multiplicities)**.** *Put*

$$N_-(R):=\#\{n\in\mathbb{Z}_{\geq 0}:2n+1<\rho_R^*\}=\left\lceil\frac{\rho_R^*-1}{2}\right\rceil. \tag{14.26}$$

For each positive root $i<j$ with $q_{ij}>0$, the complete negative spectrum of the normalized Coulomb operator is

$$\nu_{ij,n}(R)=q_{ij}\big((2n+1)\mathcal{B}_{R,\min}-\mathcal{B}_{R,\max}\big),\qquad 0\leq n<N_-(R). \tag{14.27}$$

The contribution of that positive-root pair to each displayed band has real dimension $12q_{ij}^2$. For an actual numerical eigenvalue $\lambda<0$, its full real multiplicity is therefore

$$\operatorname{mult}_{\mathbb{R}}(\lambda)=\sum_{\substack{i<j,\ q_{ij}>0,\ 0\leq n<N_-(R)\\ \nu_{ij,n}(R)=\lambda}}12q_{ij}^2. \tag{14.28}$$

In particular, coincident root charges or accidental coincidences between different q_{ij},n add rather than being counted as one band. The real Morse index is

$$\boxed{\operatorname{ind}_{\mathbb{R}}\mathcal{J}_{\mathbf{m},R}=12N_-(R)\sum_{i<j}(m_i-m_j)^2=12rN_-(R)s_2(\mathbf{m}).} \tag{14.29}$$

Every eigenvalue in equation (14.27) is divided by g_{YM}^2 for the physical Hessian.

Proof. On the E_{ij} line, the scalar oscillator is the charge-q_{ij} operator of theorem 14.3. In a skew-normal orthonormal coframe, the four circular one-form polarizations diagonalize equation (14.17); the two polarizations in a plane of charge-q_{ij} frequency $q_{ij}\mathcal{B}$ shift the scalar energy by $\pm 2q_{ij}\mathcal{B}$. The unfavorable polarizations are positive. For the favorable polarization in the high-frequency plane, high-plane oscillator number zero leaves

$$q_{ij}\big((2n+1)\mathcal{B}_{R,\min}-\mathcal{B}_{R,\max}\big).$$

Increasing its high-plane oscillator number makes it positive. The favorable low-frequency polarization has bottom $q_{ij}(\mathcal{B}_{R,\max}-\mathcal{B}_{R,\min})\geq 0$, and increasing either number cannot lower it. This proves that the list is exhaustive.

Every joint scalar space on one complex root line has dimension $6q_{ij}^2$. The opposite root is its anti-linear conjugate. The pair is the complexification of a real root subspace, so its real dimension is $12q_{ij}^2$. Root subspaces and joint oscillator spaces are orthogonal; hence equality of numerical eigenvalues produces precisely the direct sum in equation (14.28).

The strict inequality in equation (14.26) counts the integers correctly even when ρ_R^* is an odd integer. Summing dimensions gives the first index formula. Finally,

$$\sum_{i<j}(m_i-m_j)^2=r\sum_i m_i^2-\left(\sum_i m_i\right)^2=rs_2(\mathbf{m}),$$

which gives the second formula.

It remains to justify Coulomb gauge. If $\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},R}a = \lambda a$ with $\lambda < 0$, elliptic regularity and equation (14.23) give

$$\Delta^{(0)}_A d^*_A a = \lambda d^*_A a.$$

The scalar operator is nonnegative, so $d^*_A a = 0$. Thus the negative gauge-fixed and Coulomb eigenspaces coincide, completing the proof. □

Theorem 14.6 (Exact neutral and wall nullities)**.** *Away from an odd wall, the neutral Coulomb kernel has real dimension*

$$\mathrm{null}_{\mathrm{neutral}}\,\mathcal{J}_{\mathbf{m},R} = 4\left(\sum_{a=1}^{\ell} r_a^2 - 1\right). \tag{14.30}$$

At a wall $\rho^*_R = 2n+1 > 1$, the nonzero-root part adds

$$12\sum_{i<j} q_{ij}^2 = 12rs_2(\mathbf{m}) \tag{14.31}$$

real zero modes. At the equality wall $S_R = 0$, equivalently $\rho^*_R = 1$, both favorable plane polarizations vanish and the nonzero root contribution doubles to

$$24\sum_{i<j} q_{ij}^2 = 24rs_2(\mathbf{m}). \tag{14.32}$$

The total nullity at a wall is the sum of the neutral term and the stated root term.

Proof. On $\mathfrak{z}_{\mathbf{m}}$, both the induced connection commutator and the spin endomorphism vanish. The operator is therefore the ordinary coefficientwise Laplacian on one-forms. Its kernel on a flat four-torus is the space of constant one-forms, all of which are Coulomb. Tensoring the four-dimensional real harmonic-one-form space with $\mathfrak{z}_{\mathbf{m}}$ proves equation (14.30).

If $\rho^*_R = 2n+1 > 1$, the high-plane favorable polarization at high oscillator number zero and low oscillator number n is the unique zero branch for each nonzero root. Its positive-root-pair real dimension is $12q_{ij}^2$. If $S_R = 0$, the two frequencies agree and the lowest favorable polarization in each of the two planes is zero, so this dimension doubles. These charged zero modes are Coulomb: the intertwining identity shows that their divergences lie in the kernel of the charged scalar Laplacian, but that scalar Laplacian has strictly positive ground energy by equation (14.12). Summing over positive roots and using $\sum_{i<j} q_{ij}^2 = rs_2(\mathbf{m})$ proves the result. □

For $n \geq 1$, the n-th odd wall is equivalently

$$\rho^*_R = 2n+1 \iff n\Xi_R = (n+1)|S_R| \iff 24n^2 D_R = (2n+1)S_R^2. \tag{14.33}$$

The final polynomial equation must be accompanied by the unsigned ordering in the middle expression if one needs to distinguish the two sides $S_R > 0$ and $S_R < 0$. The $n = 0$ wall is exactly $S_R = 0$.

On the sufficiently deep cusp tail, $S_R > 0$, $5 < \rho^*_R < 7$, and $N_-(R) = 3$. Hence

$$\mathrm{ind}_{\mathbb{R}}\,\mathcal{J}_{\mathbf{m},R} = 36rs_2(\mathbf{m}) = 6rk_{\mathbf{m}}, \tag{14.34}$$

and the three negative branches have the retained asymptotics

$$\nu_{ij,n}(R) = \frac{q_{ij}\pi}{R}(4n-10) + O(R^{-2}), \qquad n = 0,1,2. \tag{14.35}$$

For example, the genuinely $SU(3)$ data are

weights	nonzero q_{ij}	$k_{\mathbf{m}}$	deep index	neutral nullity
$(1,0,-1)$	$1,1,2$	12	216	8
$(1,1,-2)$	$3,3$	36	648	16
$(2,1,-3)$	$1,4,5$	84	1512	8

The middle row exhibits both required edge phenomena: one neutral root from the repeated weight and coincident numerical bands from the two charge-three positive roots. The specialization $r = 2$, $\mathbf{m} = (1,-1)$ is exactly the already proved native $SU(2)$ theorem and is not reproduced here.

14.5 Stability boundary, possible wall bifurcation, and exact scope

Proposition 14.7 (Where the split background is, and is not, unstable)**.** *If $S_R \neq 0$, then $\rho_R^* > 1$, $N_-(R) \geq 1$, and every nonzero positive root contributes a negative Coulomb direction. Since a nonzero trace-zero weight vector has at least one nonzero root, $\nabla^{\mathbf{m},R}$ is then not a local minimum modulo gauge. This applies in particular throughout the sufficiently deep cusp tail.*

If $S_R = 0$, the negative root spectrum is empty. In that case equation (14.10) attains the topological value $2\pi^2 k_{\mathbf{m}}/g_{\mathrm{YM}}^2$, while equation (14.32) supplies root zero modes. No assertion of strict stability follows because the Hessian is degenerate.

Proof. The first paragraph follows immediately from equation (14.2) and theorem 14.5. For the second, $S_R = 0$ makes the two frequencies equal and hence $\rho_R^* = 1$; the strict negative-band inequality has no solution. The action and nullity statements are exactly equation (14.10) and theorem 14.6. □

A negative eigenspace is a space of descent directions, not a Lyapunov–Schmidt bifurcation kernel. A local bifurcation analysis must instead be posed at one of the zero-mode parameters in theorem 14.6. It must first quotient or slice the ever-present centralizer-valued harmonic kernel, then identify the remaining kernel and cokernel of the gauge-fixed nonlinear Yang–Mills map, and finally compute the parameter transversality and nonlinear obstruction coefficients. The present linear calculation supplies the exact candidate kernel at each wall; it does not assert existence of a nearby irreducible branch.

The theorem proved here is strictly lower-rung and fibrewise. It constructs explicit reducible split-Cartan $SU(r)$ Yang–Mills critical points on the already existing regular flat four-torus fibres and determines their topology, action, normalized Jacobi spectrum, physical Hessian scaling, Morse index, and nullity. It constructs no new base, spacetime, cone, S^{12}, S^{24}, transfer matrix, or interacting quantum Hamiltonian. The radial multiplication operator already treated in the canonical native module is not repeated here.

There is also an exact obstruction to interpreting the nonzero-flux bundle as the restriction of a bundle on a proposed higher carrier. If $i : F_R \to Y$ is any map and $H^4(Y;\mathbb{Z}) = 0$, then every complex bundle $\mathscr{V} \to Y$ has $c_2(\mathscr{V}) = 0$, so naturality gives

$$c_2(i^*\mathscr{V}) = i^* c_2(\mathscr{V}) = 0.$$

This contradicts equations (14.5) and (14.6) for $\mathscr{E}_{\mathbf{m},R}$. Therefore this nonzero-c_2 bundle cannot extend over any carrier with vanishing fourth integral cohomology, including a space diffeomorphic to S^6, S^{12}, or S^{24}. Changing the gauge group changes the root system on the fibre; it does not by itself produce the proposed higher rung or any conclusion about a physical mass gap.

15 The exceptional first-coordinate elliptic quotient and Landau sector

The four period generators in $\mathbb{C}^2$ are

$$v_\gamma = (6\mu, \beta), \quad v_u = (\tau, \mu), \quad v_w = (1, 0), \quad v_\delta = (0, 1). \tag{15.1}$$

Let $E_\tau = \mathbb{C}/(\mathbb{Z}\tau + \mathbb{Z})$.

Proposition 15.1 (Exact descent criterion)**.** *The first-coordinate projection* $\mathrm{pr}_1 : \mathbb{C}^2 \to \mathbb{C}$ *descends to a surjective holomorphic homomorphism*

$$\overline{\mathrm{pr}}_1 : F_z \longrightarrow E_\tau$$

if and only if

$$6\mu = a\tau + b \qquad \text{for some } (a, b) \in \mathbb{Z}^2. \tag{15.2}$$

The pair (a, b) *is necessarily unique. On this locus, put*

$$\tau_f = a\mu - \beta. \tag{15.3}$$

Then $\Im\tau_f = -D > 0$ *and the connected kernel is the elliptic curve* $E_{\tau_f} = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau_f)$*, up to the harmless reversal of one lattice generator.*

Proof. The first coordinates of (15.1) are $6\mu, \tau, 1, 0$. Descent to E_τ is therefore equivalent to $6\mu \in \mathbb{Z}\tau + \mathbb{Z}$, which is (15.2); the images of v_u, v_w prove surjectivity. Uniqueness follows from $\Im\tau > 0$. The unimodular replacement

$$v_\gamma \longmapsto v_\gamma - av_u - bv_w = (0, \beta - a\mu)$$

shows that the kernel lattice is generated by 1 and $\beta - a\mu = -\tau_f$. Taking imaginary parts in (15.2) gives $6m = aT$, and hence

$$D = p - \frac{6m^2}{T} = p - am = \Im(\beta - a\mu) = -\Im\tau_f.$$

Since $D < 0$, $\tau_f \in \mathfrak{h}$. □

Define the marked exceptional set

$$\mathcal{D} = \bigcup_{(a,b)\in\mathbb{Z}^2} \mathcal{D}_{a,b}, \qquad \mathcal{D}_{a,b} = \{z : 6\mu(z) - a\tau(z) - b = 0\}. \tag{15.4}$$

Theorem 15.2 (No cusp accumulation and finiteness on the orbifold base)**.** *Assume the transformation laws and extension properties in Section 1. The generator laws act on the integral labels by*

$$g_0 : (a, b) \longmapsto (a, a + b), \qquad g_2 : (a, b) \longmapsto (-b, 6 + a). \tag{15.5}$$

Consequently $\mathcal{D}$ *is invariant under the triangle group and has a well-defined image in the completed orbifold base.*

(i) There is a punctured neighborhood of the distinguished cusp containing no point of $\mathcal{D}$*.*

(ii) The image of $\mathcal{D}$ *in the completed triangle-orbifold base is finite.*

Proof. If $z \in \mathcal{D}_{a,b}$, then the g_0 laws give

$$6\mu(g_0 z) = a(\tau(g_0 z) + 1) + b = a\tau(g_0 z) + (a + b),$$

while the g_2 laws give

$$6\mu(g_2 z) = 6 + a + \frac{b}{\tau(z)} = -b\tau(g_2 z) + (6 + a).$$

Both label maps are bijections, and g_0, g_2 generate the triangle group. This proves (15.5) and invariance of $\mathcal{D}$.

Suppose first that $z_j \in \mathcal{D}_{a_j,b_j}$ approaches the cusp in a fixed cusp component. Taking imaginary parts gives

$$a_j = \frac{6\Im\mu(z_j)}{\Im\tau(z_j)} \longrightarrow 0. \tag{15.6}$$

Because $a_j \in \mathbb{Z}$, one has $a_j = 0$ eventually. Then $6\mu(z_j) = b_j$. The holomorphic cusp extension of μ makes it bounded, so the integers b_j are bounded. After passage to a subsequence, $b_j = b$ is constant. The bounded holomorphic function $6\mu - b$ extends across the cusp and has zeros accumulating at the added interior point, hence vanishes identically.

No identity $6\mu = a\tau + b$ with fixed integers a, b is possible. Indeed, applying g_0 and using (1.10) forces $a = 0$. The identity would then make $\mu = b/6$ constant, whereas the g_2 law would require

$$\frac{b}{6} = 1 + \frac{b}{6\tau}$$

for the nonconstant function τ, an impossibility. This contradiction proves (i).

For (ii), truncate a fundamental polygon by the cusp neighborhood supplied by (i). Its orbifold closure is compact, and the stated elliptic regularity makes τ and μ bounded there, while $\Im\tau$ has a positive lower bound. At a point of $\mathcal{D}_{a,b}$,

$$a = \frac{6\Im\mu}{\Im\tau}, \qquad b = 6\mu - a\tau.$$

Thus only finitely many integer pairs (a, b) can occur on the compact truncation. For each such pair, the holomorphic function $6\mu - a\tau - b$ is not identically zero by the preceding transformation-law argument, so it has only finitely many zeros on the compact orbifold chart system. The union is finite. The period monodromy merely permutes the marked pairs, so this is a statement on the quotient base. □

On a descent fibre, the degree-six holomorphic line bundle isolated in the supplied workbench has a unitary constant-curvature connection. For the Hamiltonian $H_{\phi,\psi} = \frac{1}{2}\nabla^*\nabla$ its spectrum is

$$\operatorname{Spec}(H_{\phi,\psi}) = \left\{ \frac{12\pi}{\Im\tau_f}\left(j + \frac{1}{2}\right) : j \in \mathbb{Z}_{\geq 0} \right\}, \tag{15.7}$$

with multiplicity six at each level. The ground-shifted operator has levels $12\pi j/\Im\tau_f$ and a six-dimensional kernel.

Corollary 15.3 (The first-coordinate Landau sector does not run into the cusp)**.** *There is no sequence of actual descent fibres in the proposed S^6 family approaching the cusp. In particular, the limit $\Im\tau_f \to \infty$ cannot be inferred from the S^6 cusp and cannot be used to close the Landau-level spacing in* (15.7) *inside this family.*

Remark 15.4 (Correction of an ambient-family argument). The pointwise matrices

$$(\tau_T, \mu_T, \beta_T) = (i/T, 0, -iT)$$

do satisfy a nonzero period determinant and make certain flat or magnetic gaps small. They are, however, an arbitrary path in the ambient coefficient space. They are not shown to belong to the image of the asserted period map, and this specific first-coordinate magnetic construction also requires the independent arithmetic condition (15.2). Theorem 3.1 supplies the correct scalar gap-closing statement on the actual cusp; Corollary 15.3 shows that this exceptional first-coordinate Landau sector does not realize it. The independent full-torus construction in Theorem 9.1 does give a magnetic gap-closing sequence for every fixed nondegenerate integral flux on the marked cusp fibres, without asserting a monodromy-invariant global flux class or classifying holomorphic elliptic quotients. The native full-flux theorem separately treats the explicit invariant class $u \wedge w + 6\gamma \wedge \delta$ and does not use this exceptional elliptic quotient.

The descent theorem classifies only the specified first-coordinate map to $\mathbb{C}/(\mathbb{Z}\tau + \mathbb{Z})$. It does not prove that F_z has no other elliptic quotient or elliptic subtorus, that its Néron–Severi group vanishes, or that it admits no smooth integral magnetic line bundle; those are different typed questions.

16 The compact total space is a different spectral problem

Proposition 16.1 (Compact total-space scalar gap). *Assume that the proposed completion is a smooth compact manifold $X \cong S^6$. For every smooth Riemannian metric g_X on X, the scalar Laplace–Beltrami operator has compact resolvent,*

$$0 = \lambda_0(X, g_X) < \lambda_1(X, g_X) \le \lambda_2(X, g_X) \le \cdots, \qquad \lambda_j \to \infty,$$

and $\lambda_1(X, g_X) > 0$ because X is connected.

Proof. This is the standard elliptic spectral theorem on a connected closed Riemannian manifold (Chavel, 1984). □

The proposition does not contradict Theorem 3.1. The latter concerns a singularly degenerating family of auxiliary vertical metrics on regular torus fibres. No argument in the period construction shows that these Euclidean fibre metrics are restrictions of one smooth metric on the completed total space. The same warning applies to the quotient metrics in Theorem 5.1: a pointed $\mathbb{R}^4$ limit of their regular-fibre targets is not a metric or operator on the compact total space. A complex structure alone determines neither a preferred Riemannian metric nor a Hamiltonian.

17 Metric indeterminacy and the physical boundary

The preceding results attach gaplessness only to explicitly displayed scalar, magnetic, classical Hessian, and finite Gaussian operators. The following example shows why the complex-geometric data alone cannot select any one of those metric spectra.

Proposition 17.1 (Fixed complex torus, fixed volume, arbitrarily small gap). *Let*

$$A = (\mathbb{C}/(\mathbb{Z} + i\mathbb{Z}))^2, \qquad z_j = x_j + iy_j.$$

For every $R \geq 1$,

$$g_R = R^2(\mathrm{d}x_1^2 + \mathrm{d}y_1^2) + R^{-2}(\mathrm{d}x_2^2 + \mathrm{d}y_2^2) \tag{17.1}$$

is a flat Kähler metric of volume one on this one fixed complex torus, and

$$\lambda_1(\Delta_{g_R}) = \frac{4\pi^2}{R^2} \longrightarrow 0. \tag{17.2}$$

Therefore the complex structure, Hodge decomposition, integral cohomology, and coherent direct-image data, even together with total volume, do not determine a positive lower bound for the scalar Laplacian gap.

Proof. The two-form

$$\omega_R = R^2\mathrm{d}x_1 \wedge \mathrm{d}y_1 + R^{-2}\mathrm{d}x_2 \wedge \mathrm{d}y_2$$

is closed, positive, and invariant under the fixed complex structure, so g_R is Kähler. Its real metric matrix is $\mathrm{diag}(R^2, R^2, R^{-2}, R^{-2})$, of determinant one, and the coordinate fundamental domain has volume one. The Fourier character

$$e_{a,b,c,d} = e^{2\pi i(ax_1+by_1+cx_2+dy_2)}$$

has eigenvalue

$$4\pi^2(R^{-2}(a^2+b^2) + R^2(c^2+d^2)).$$

For $R \geq 1$ its minimum over nonzero integer quadruples is $4\pi^2R^{-2}$, attained exactly at $(\pm1, 0, 0, 0)$ and $(0, \pm1, 0, 0)$. This proves the claim; none of the listed complex or cohomological data varies with R. □

The mathematical mass gap in the Clay formulation concerns the Hamiltonian of a constructed four-dimensional quantum Yang–Mills theory, not an internal scalar Laplacian or a classical Hessian (Jaffe and Witten, 2006). Euclidean data reach such a Hamiltonian only after a positive-metric reconstruction satisfying, in particular, the Osterwalder–Schrader requirements (Osterwalder and Schrader, 1973, 1975). The exact operator-level implication that a transfer argument would need is the next proposition.

Proposition 17.2 (Typed Rayleigh-sequence transfer)**.** *Let $H \geq 0$ be self-adjoint on a Hilbert space $\mathcal{H}_{\mathrm{phys}}$. Put*

$$\mathcal{V}_0 = \ker H, \qquad P_0 = \mathbf{1}_{\{0\}}(H), \qquad \mathcal{H}_{\mathrm{exc}} = (I - P_0)\mathcal{H}_{\mathrm{phys}}, \tag{17.3}$$

and let

$$\mathfrak{h}[\psi] = \|H^{1/2}\psi\|^2, \qquad \mathcal{Q}(H) = \mathrm{Dom}(H^{1/2}). \tag{17.4}$$

For each n, let $\mathcal{K}_n$ be a Hilbert space, $\mathfrak{q}_n$ a nonnegative quadratic form on $D(\mathfrak{q}_n) \subseteq \mathcal{K}_n$, and $0 \neq u_n \in D(\mathfrak{q}_n)$. Suppose that a linear map

$$J_n : \mathrm{span}\{u_n\} \longrightarrow \mathcal{Q}(H)$$

is given. Set $\psi_n = (I - P_0)J_nu_n$. If there are $c > 0$, $C_n \geq 0$, and $\varepsilon_n \geq 0$ such that

$$\|\psi_n\| \geq c\|u_n\|, \tag{17.5}$$

$$\mathfrak{h}[\psi_n] \leq C_n\mathfrak{q}_n[u_n] + \varepsilon_n\|u_n\|^2, \tag{17.6}$$

$$C_n\frac{\mathfrak{q}_n[u_n]}{\|u_n\|^2} + \varepsilon_n \longrightarrow 0, \tag{17.7}$$

then the reducing restriction

$$H_{\mathrm{exc}} := H\big|_{\mathrm{Dom}(H)\cap\mathcal{H}_{\mathrm{exc}}}$$

is self-adjoint and

$$\inf \mathrm{Spec}(H_{\mathrm{exc}}) = 0, \qquad 0 \in \sigma_{\mathrm{ess}}(H_{\mathrm{exc}}). \tag{17.8}$$

Thus H has no positive gap above its full zero-energy sector.

Proof. The spectral projection P_0 commutes with H and $H^{1/2}$, so $\mathcal{H}_{\mathrm{phys}} = \ker H \oplus \mathcal{H}_{\mathrm{exc}}$ is a reducing decomposition. Hence H_{exc} is self-adjoint, nonnegative, has form domain $\mathcal{Q}(H)\cap\mathcal{H}_{\mathrm{exc}}$, and has zero kernel. Normalize $w_n = \psi_n/\|\psi_n\|$. Then

$$\mathfrak{h}[w_n] \le c^{-2}\left(C_n \frac{\mathfrak{q}_n[u_n]}{\|u_n\|^2} + \varepsilon_n\right) \longrightarrow 0.$$

The variational characterization of the bottom of a nonnegative self-adjoint operator therefore gives $\inf \mathrm{Spec}(H_{\mathrm{exc}}) = 0$. Its spectrum is closed, so zero belongs to it; its kernel is zero, so zero is not an eigenvalue. A self-adjoint operator has no residual spectrum. Hence zero is a continuous spectral point and belongs to the essential spectrum under the standard convention that removes isolated eigenvalues of finite multiplicity. □

Remark 17.3 (The missing physical morphism). The native unscaled limit retains two compact directions, and the cusp parameter T has not been identified with a spacetime box size, lattice spacing, correlation length, Euclidean time, or renormalization scale. No reflection-positive measure, transfer matrix, Osterwalder–Schrader reconstruction, Gauss-law physical quotient, local observable algebra, thermodynamic limit, or continuum limit in fixed physical units has been constructed. An eigenvalue of a four-dimensional elliptic Hessian has units of inverse length squared; without a time reconstruction and a scale conversion it is not an eigenvalue of a physical Hamiltonian.

The cusp calculations supply many possible auxiliary soft states: the scalar characters in Theorem 3.1, the continuum transverse modes in Theorem 8.2, the finite transverse modes in Theorem 8.3, the native full-flux Landau bands and their radial multiplication operator, and the finite Gaussian oscillator modes in Theorem 11.5. The reducible full-flux critical connections additionally have negative Jacobi bands, so their Hessians are not positive candidate vacuum Hamiltonians. None of these constructions supplies one already constructed, gauge-invariant, positive-metric physical Hilbert space for four-dimensional quantum Yang–Mills theory, its Hamiltonian H, or maps J_n satisfying (17.5)–(17.7) with constants controlled in physical units after gauge reduction, reconstruction, renormalization, the infinite-volume limit, and the continuum limit. Orthogonality to one chosen vacuum vector would not suffice unless vacuum uniqueness were also proved: the projection $I - P_0$ must remove the full zero-energy sector. Consequently Proposition 17.2 is an exact transfer theorem, not an assertion that the required transfer data exist. No nonlinear or quantum mass-gap conclusion follows here.

18 Conclusions

Conditional on the manuscript's period, monodromy, and toric cusp formulae in the precise scopes stated above, the following hierarchy is exact.

1. Every fixed regular flat fibre has a positive scalar spectral gap.

2. The family has no uniform flat-fibre gap at the actual cusp. Its first eigenvalue is $4\pi^2T^{-2}(1 + 2\min\{b_{\mathrm{I},0}, 0\}T^{-1} + O(T^{-2}))$, and after unit-volume normalization it is $4\pi^2T^{-1}(1 + (-b_{\mathrm{I},0}/2 + 2\min\{b_{\mathrm{I},0}, 0\})T^{-1} + O(T^{-2}))$.

3. More than one eigenvalue becomes soft: if $0 < E < 4\pi^2$, the exact fixed-energy count with Fourier multiplicity is $N_T(E) = E\mathcal{D}_T/(4\pi) + O_E(T)$, where $\mathcal{D}_T = |\det \mathsf{P}_z| = T^2 - b_{\mathrm{I},0}T + O(1)$. Thus the low-energy population has a two-dimensional leading density. The unit-volume count is $E\mathcal{D}_T^{1/2}/(4\pi) + O_E(\mathcal{D}_T^{1/4} + 1)$, and at the collapsing T^{-2} scale the finite limiting count is the lattice-point count in a two-dimensional Euclidean disc, with the stated threshold refinement.

4. The native pointed smooth limit of the marked four-tori is $\mathbb{T}^2 \times \mathbb{R}^2$, not $\mathbb{R}^4$. The degree-$M(T)^2$ marked local fibre covers have pointed $\mathbb{R}^4$ limit precisely when $M(T) \to \infty$ in the unscaled metric, and precisely when $M(T)/\sqrt{T} \to \infty$ for the pullback of the base-fibre unit-volume metric. Normalizing every covering torus to its own volume one never gives a pointed smooth $\mathbb{R}^4$ limit.

5. After $t_c = v^M$, the cover sublattice is preserved by the pulled-back cusp monodromy. Its exact Fourier spectrum separates the deck-invariant pullback sector from the new momenta, and those momentum lattices become dense in $\mathbb{R}^4$ when $T, M \to \infty$. This punctured statement is distinct from the completed local toric theorem below.

6. Up to sign, the unique primitive full-monodromy-invariant integral character gives a real, non-holomorphic circle quotient of the regular group family. In this paper's convention its quotient circumference is exactly $-TD/\sqrt{T^2 + m^2}$ and is $T - b_{\mathrm{I},0} + O(T^{-1})$ on the positive radial marked lift. The character is global; for the chosen radial fibre metrics its Fourier mode gap-closes, but it need not be the exact shortest mode at every parameter.

7. Four synchronized copies, without a base change or fibre cover, give an all-expanding real four-torus quotient with pointed $\mathbb{R}^4$ radial limit. Four copies are minimal only within the stated fixed primitive full-monodromy-invariant kernel class with its induced quotient metric. The different two-copy expanding quotient of the native fibre is cusp-local, and no finite base cover globalizes its particular kernel. Neither quotient is the native fibre or its degree-M^2 cover.

8. The literal invariant-character map does not extend through the asserted toric cusp with its fixed circle target. Conditional on $X \cong_{\mathrm{diff}} S^6$, its fourfold product has no extension to the fixed target $(\mathbb{R}/\mathbb{Z})^4$ from X^4 with the stated regular-fibre restriction, and no such fixed-target extension factors through $X^{\wedge 4} \cong S^{24}$. This is not an obstruction to a construction that changes the codomain.

9. The exact local Kummer construction after $t = v^N$ requires the source action $\Psi_{\bar\lambda}^{(N)}(\xi, v) = (\exp(2\pi i C(v^N)\bar\lambda/N)v^{B_0\bar\lambda}\xi, v)$. It gives a finite flat degree-N^3 map to the original completed cusp family. The factor to the base-changed target has generic degree N^2 and, for $N > 1$, is quasi-etale but nonflat precisely over the double and triple central-fibre strata. These are local cusp statements, not a full-base cover.

10. Separately, the canonical workbench constructs $q_2^{\wedge 2}$ from the sphere source S^{12} and $q_1^{\wedge 4}$ from the sphere source S^{24} by changing to compact cone targets. This paper does not reproduce their proofs. Their radial target slices have intrinsic pointed $\mathbb{R}^4$ limits but collapse in every fixed compatible ambient target metric.

11. The associated mean-zero scalar vertical direct integral is injective and gapless, with zero in its essential continuous spectrum. This statement identifies the threshold but does

not assert the stronger full-half-line spectral classification proved below for the native magnetic multiplication operator.

12. The classical Yang–Mills Hessian at the product connection, after Coulomb reduction and removal of harmonic one-forms, has exact gap (8.11) and closes as $4\pi^2/(g_{\mathrm{YM}}^2 T^2)$. The exact finite cubical cochain Hessian closes uniformly as T^{-2} for every $N \geq 2$; resolving the varying continuum geometry is the separate condition $N/T \to \infty$.

13. Keeping the covering degree M_{cov}^2 separate from the mesh count N_{mesh}^4 gives an honest transported cubical complex, typed continuum holonomy, and an explicit periodic compact-link SU(2) flux. Its logarithmic action is exactly $2\pi^2 M_{\mathrm{cov}}^2/(g_{\mathrm{YM}}^2 D_R)$, and its displayed $(1,2)$-sector Wilson functional converges to that value as $N_{\mathrm{mesh}} \to \infty$.

14. On the reduced finite linear Coulomb space, the Gaussian Hamiltonian has a unique ground state and exact gap $\sqrt{\lambda_{\min}(g_{\mathrm{YM}}^{-2} d^* d)}$. The diagonal sequence $R_j = j^2$, $M_{\mathrm{cov},j} = j$, $N_{\mathrm{mesh},j} = j^3$ simultaneously has pointed locally isometric $\mathbb{R}^4$ cover limit, vanishing maximal physical edge length, vanishing Gaussian gap, and vanishing displayed logarithmic action.

15. For every fixed nondegenerate integral alternating form E on the marked four-torus, the full-torus constant-curvature magnetic Laplacian has frequency product $b_1 b_2 = 4\pi^2 |\operatorname{Pf}(\mathsf{E})|/\mathcal{D}_T$ and level spacing at most $4\pi\sqrt{|\operatorname{Pf}(\mathsf{E})|/\mathcal{D}_T} = O(T^{-1})$. This is a fibrewise marked-cusp result, not an assertion of monodromy invariance or a physical mass gap.

16. The distinguished class $\eta = u \wedge w + 6\gamma \wedge \delta$ is invariant under the asserted monodromy generators, has elementary divisors $(1, 6)$ and Pfaffian 6, and is represented by the explicit cocycle and constant-curvature connection given below. Its exact charge-one frequencies have product $24\pi^2/D_R$ and asymptotics $\mathcal{B}_{R,+} = 12\pi/R + O(R^{-2})$ and $\mathcal{B}_{R,-} = 2\pi/R + O(R^{-2})$. Every joint Landau level has complex multiplicity six, the ground energy is $14\pi/R + O(R^{-2})$, and the first distinct spacing is $4\pi/R + O(R^{-2})$.

17. After the fixed marking and volume-correcting unitary trivialize the varying fibre Hilbert spaces, the full-flux magnetic family has common H^2 operator domain and locally norm-continuous resolvents. Its exact maximal radial multiplication operators, in both fixed- and varying-fibre forms over (R_1, ∞) with Lebesgue measure, are self-adjoint and satisfy

$$\sigma = \sigma_{\mathrm{ess}} = \sigma_{\mathrm{c}} = [0, \infty), \qquad \sigma_{\mathrm{p}} = \varnothing, \qquad \ker = \{0\}.$$

    The parameter R is not differentiated.

18. The nontrivial split bundle $\mathscr{L}_{\eta,R} \oplus \mathscr{L}_{\eta,R}^{-1}$ carries a reducible SU(2) Yang–Mills critical connection. Its gauge-fixed Jacobi form is closed, its operator domain is $H^2\Omega^1$, the maximal Coulomb slice is reducing, and on a sufficiently deep cusp tail its complete negative spectrum comprises three bands of real multiplicity 48, giving real Morse index 144. Thus this Hessian is not a positive fluctuation operator.

19. More generally, every nonzero trace-zero integral weight vector $\mathbf{m} = (m_1, \ldots, m_r)$ gives a split-Cartan SU(r) critical connection with $k_{\mathbf{m}} = 6\sum_j m_j^2$ and exact rootwise Jacobi spectrum. On the deep tail its real Morse index is $36r\sum_j m_j^2 = 6rk_{\mathbf{m}}$; the neutral and odd-wall nullities are given exactly in Theorem 14.6. These nonzero-c_2 bundles cannot be restrictions of bundles on a carrier with vanishing $H^4(-;\mathbb{Z})$.

20. The specific degree-six Landau construction pulled back from the first-coordinate elliptic quotient is confined to a finite exceptional locus and cannot approach the cusp. This

does not contradict the preceding full-torus magnetic theorem, whose line bundle is defined directly from an integral alternating form and is not pulled back from that elliptic quotient.

21. None of the preceding statements determines the scalar spectrum of a smooth metric on the compact six-dimensional total space. Even on one fixed complex two-torus, unit-volume flat Kähler metrics can have arbitrarily small scalar gap.

22. The finite Gaussian oscillator is not the compact-link Haar Hilbert space, the exponential map from cochains is only a local chart, and the magnetic Landau operators and reducible Jacobi operators remain fibrewise. Transfer of any auxiliary soft sequence to a physical Hamiltonian would require the explicit norm-noncollapse and energy-comparison morphisms in Proposition 17.2. No such physical Hilbert space, Hamiltonian, or morphisms are constructed here, so no nonlinear or quantum Yang–Mills mass-gap conclusion follows.

The main structural lesson is that compactness and a positive gap at every fibre do not survive integration over a noncompact degenerating modulus without a uniform coercive estimate. Conversely, a gapless decomposable vertical operator or classical Hessian cannot be identified with a field-theoretic Hamiltonian without the explicit operator-level bridge. Likewise, the native $\mathbb{T}^2 \times \mathbb{R}^2$ limit, finite-cover $\mathbb{R}^4$ unspooling, fixed-target quotient limits, and cone-target constructions are four different operations. The fixed-target obstruction concerns only the displayed character map to $(\mathbb{R}/\mathbb{Z})^4$ with its stated fibre restriction; the cone constructions change the codomain and distinguish intrinsic expanding slice metrics from collapse in a fixed compact ambient metric.

A Reproducibility formulas

For a direct algebraic reproducibility check, expansion along the last two columns of (1.1) gives

$$\det \mathsf{P}_z = T\left(p - \frac{6m^2}{T}\right)$$

and multiplication of $(1.1)^{\mathsf{T}}$ by the vector in (2.5) returns $(r, s, n_1, n_2)^{\mathsf{T}}$ exactly. On the radial cusp, substitution of $p = B_R - T$ gives

$$-TD = T(T - B_R) + 6m^2,$$

while squaring the first two columns of the inverse transpose gives (3.1) and their subtraction gives (3.2). These identities are exact. The proof of Theorem 3.1 then uses the uniform singular-value estimate (3.11); no finite lattice search or numerical sampling is part of the proof.

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